



JDBC™ 3.0 Specification

Proposed Final Draft 4

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Preface

This document supersedes and consolidates the content of these predecessor specifications:

- “JDBC: A Java SQL API”
- “JDBC 2.1 API”
- “JDBC 2.0 Standard Extension API”

New content is summarized in an introductory chapter and then incorporated throughout this document. The main body of the specification describes the API at a conceptual level. More extensive details and examples are relegated to the appendices.

Readers can also download the API specification (Javadoc™ API and comments) for a complete and precise definition of JDBC classes and interfaces. This documentation is available from the download page at

<http://java.sun.com/products/jdbc/download.html>

or it can be browsed at

<http://java.sun.com/j2se/1.4/docs/api/java/sql/package-summary.html>

[currently 1.3 for kestrel; will be 1.4 for merlin]

Typographic Conventions

Typeface	Meaning	Examples
AaBbCc123	The names of commands, files, and directories; on-screen computer output	Edit your <code>.login</code> file. Use <code>ls -a</code> to list all files. % You have mail.
AaBbCc123	What you type, when contrasted with on-screen computer output	% su Password:
<i>AaBbCc123</i>	Book titles, new words or terms, words to be emphasized	Read Chapter 6 in the <i>User's Guide</i> . These are called <i>class</i> options. You <i>must</i> be superuser to do this.
	Command-line variable; replace with a real name or value	To delete a file, type <code>rm filename</code> .

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Introduction

1.1 The JDBC API

The JDBC™ API provides programmatic access to relational data from the Java™ programming language. Using the JDBC API, applications written in the Java programming language can execute SQL statements, retrieve results, and propagate changes back to an underlying data source. The JDBC API can also be used to interact with multiple data sources in a distributed, heterogenous environment.

The JDBC API is based on the X/Open SQL CLI, which is also the basis for ODBC. JDBC provides a natural and easy-to-use mapping from the Java programming language to the abstractions and concepts defined in the X/Open CLI and SQL standards.

Since its introduction in January 1997, the JDBC API has become widely accepted and implemented. The flexibility of the API allows for a broad range of implementations.

1.2 Platforms

The JDBC API is part of the Java platform, which includes the Java™ 2 Standard Edition (J2SE™) and the Java™ 2 Enterprise Edition (J2EE™). The JDBC 3.0 API is divided into two packages: `java.sql` and `javax.sql`. Both packages are included in the J2SE and J2EE platforms.

1.3 Target Audience

This specification is targeted primarily towards the vendors of these types of products:

- drivers that implement the JDBC API
- application servers providing middle-tier services above the driver layer
- tools that use the JDBC API to provide services such as application generation

This specification is also intended to serve the following purposes:

- an introduction for end-users whose applications use the JDBC API
- a starting point for developers of other APIs layered on top of the JDBC API

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Goals

The JDBC API is a mature technology, having first been specified in January 1997. In its initial release, the JDBC API focused on providing a basic call-level interface to SQL databases. The JDBC 2.1 specification and the 2.0 Optional Package specification then broadened the scope of the API to include support for more advanced applications and for the features required by application servers to manage use of the JDBC API on behalf of their applications.

The overall goal of the JDBC 3.0 specification is to “round out” the API by filling in smaller areas of missing functionality. The following list outlines the goals and design philosophy for the JDBC API in general and the JDBC 3.0 API in particular:

1. Fit into the J2EE and J2SE platforms

The JDBC API is a constituent technology of the Java platform. The JDBC 3.0 API should be aligned with the overall direction of the Java 2 Enterprise Edition and Java 2 Standard Edition platforms.

2. Be consistent with SQL99

The JDBC API provides programmatic access from applications written in the Java programming language to standard SQL. At the time the JDBC 2.0 API was in development, the SQL99 specification was a moving target. SQL99 is now a published standard and includes features that are widely supported among DBMS vendors as well as features that only a few vendors support. The intent of the JDBC 3.0 API is to provide access to the subset of SQL99 features that are likely to be widely supported within the next five years.

3. Consolidate predecessor specifications

This document incorporates content from three prior JDBC specifications to provide a single standalone specification of the JDBC API.

4. Offer vendor-neutral access to common features

The JDBC API strives to provide high-bandwidth access to features commonly supported across different vendor implementations. The goal is to provide a degree of feature access comparable to what can be achieved by native applications. However, the API must be general and flexible enough to allow for a wide range of implementations.

5. Maintain the focus on SQL

The focus of the JDBC API has always been on accessing relational data from the Java programming language. This continues to be true with the JDBC 3.0 API. The JDBC 3.0 API does not preclude interacting with other technologies, including XML, CORBA, or non-relational data, but the primary target will still be relational data and SQL.

6. Provide a foundation for tools and higher-level APIs

The JDBC API presents a standard API to access a wide range of underlying data sources or legacy systems. Implementation differences are made transparent through JDBC API abstractions, making it a valuable target platform for tools vendors who want to create portable tools and applications.

Because it is a “call-level” interface from the Java programming language to SQL, the JDBC API is also suitable as a base layer for higher-level facilities such as Enterprise JavaBeans (EJB) 2.0 container-managed persistence and SQLJ.

7. Keep it simple

The JDBC API is intended to be a simple-to-use, straightforward interface upon which more complex entities can be built. This goal is achieved by defining many compact, single-purpose methods instead of a smaller number of complex, multi-purpose ones with control flag parameters.

8. Enhance reliability, availability, and scalability

Reliability, availability, and scalability are the themes of the J2EE and J2SE platforms, as well as the direction for future Java platforms. The JDBC 3.0 API stays true to these themes by enhancing support in several areas, including resource management, the reuse of prepared statements across logical connections, and error handling.

9. Maintain backward compatibility with existing applications and drivers

Existing JDBC technology-enabled drivers (“JDBC drivers”) and the applications that use them must continue to work in an implementation of the Java virtual machine that supports the JDBC 3.0 API. Applications that use only features defined in earlier releases of the JDBC API, excluding those that were deprecated by JDBC 2.0, will not require changes to continue running. It should be straightforward for existing applications to migrate to JDBC 3.0 technology.

10. Allow forward compatibility with Connectors

The Connector architecture defines a standard way to package and deploy a resource adapter that allows a J2EE container to integrate its connection, transaction, and security management with those of an external resource.

The JDBC 3.0 API provides the migration path for JDBC drivers to the Connector architecture. It should be possible for vendors whose products use JDBC technology to move incrementally towards implementing the Connector API. The expectation is that these implementors will write “resource manager wrappers” around their existing data source implementations so that they can be reused in a Connector framework.

11. Specify requirements unambiguously

The requirements for JDBC compliance need to be unambiguous and easy to identify. The JDBC 3.0 specification and the API documentation (Javadoc documentation) will clarify which features are required and which are optional.

Summary of New Features

3.1 Overview of changes

The JDBC 3.0 API introduces new material and changes in these areas:

- **Savepoint support**

Added the `Savepoint` interface, which contains new methods to set, release, or roll back a transaction to designated savepoints.

- **Reuse of prepared statements by connection pools**

Added the ability for deployers to control how prepared statements are pooled and reused by connections.

- **Connection pool configuration**

Defined a number of properties for the `ConnectionPoolDataSource` interface. These properties can be used to describe how `PooledConnection` objects created by `DataSource` objects should be pooled.

- **Retrieval of parameter metadata**

Added the interface `ParameterMetaData`, which describes the number, type and properties of parameters to prepared statements.

- **Retrieval of auto-generated keys**

Added a means of retrieving values from columns containing automatically generated values.

- **Ability to have multiple open `ResultSet` objects**

Added the new method `getMoreResults(int)`, which takes an argument that specifies whether `ResultSet` objects returned by a `Statement` object should be closed before returning any subsequent `ResultSet` objects.

- Passing parameters to `CallableStatement` objects by name

Added methods to allow a string to identify the parameter to be set for a `CallableStatement` object.
- Holdable cursor support

Added the ability to specify the of holdability of a `ResultSet` object.
- `BOOLEAN` data type

Added the data type `java.sql.Types.BOOLEAN`. `BOOLEAN` is logically equivalent to `BIT`.
- Making internal updates to the data in `Blob` and `Clob` objects

Added methods to allow the data contained in `Blob` and `Clob` objects to be altered.
- Retrieving and updating the object referenced by a `Ref` object

Added methods to retrieve the object referenced by a `Ref` object. Also added the ability to update a referenced object through the `Ref` object.
- Updating of columns containing `BLOB`, `CLOB`, `ARRAY` and `REF` types

Added of the `updateBlob`, `updateClob`, `updateArray`, and `updateRef` methods to the `ResultSet` interface.
- `DATALINK/URL` data type

Added the data type `java.sql.Types.DATALINK`, allowing JDBC drivers to store and retrieve references to external data.
- Transform groups and type mapping

Described the effect of transform groups and how this is reflected in the meta-data.
- Relationship between the JDBC SPI (Service Provider Interface) and the `Connector` architecture

Described the relationship between the JDBC SPI and the connector architecture in Chapter 19 “Relationship to Connectors”.
- `DatabaseMetadata` APIs

Added metadata for retrieving SQL type hierarchies. See the JDBC API Specification for details.

See Chapter 5 “Classes and Interfaces” for a list of the classes and interfaces affected by these changes.

Overview

The JDBC API provides a way for Java programs to access one or more sources of data. In the majority of cases, the data source is a relational DBMS, and its data is accessed using SQL. However, it is also possible for JDBC technology-enabled drivers ("JDBC drivers") to be implemented on top of other data sources, including legacy file systems and object-oriented systems. A primary motivation for the JDBC API is to provide a standard API for applications to access a wide variety of data sources.

This chapter introduces some of the key concepts of the JDBC API. In addition, it describes two common environments for JDBC applications, with a discussion of how different functional roles are implemented in each one. The two-tier and three-tier models are logical configurations that can be implemented on a variety of physical configurations.

4.1 Establishing a Connection

The JDBC API defines the `Connection` interface to represent a connection to an underlying data source.

In a typical scenario, a JDBC application will connect to a target data source using one of two mechanisms:

- **DriverManager** — this fully implemented class was introduced in the original JDBC 1.0 API and requires the application to load a specific driver using a hard-coded URL.
- **DataSource** — this interface was introduced in the JDBC 2.0 Optional Package API. It is preferred over **DriverManager** because it allows details about the underlying data source to be transparent to the application. A **DataSource** object's properties are set so that it represents a particular data source. When its `getConnection` method is invoked, the **DataSource** instance will return a connection to that data source. An application can be directed to a different data

source by simply changing the `DataSource` object's properties; no change in application code is needed. Likewise, a `DataSource` implementation can be changed without changing the application code that uses it.

The JDBC API also defines two important extensions of the `DataSource` interface to support enterprise applications. These extensions are the following two interfaces:

- `ConnectionPoolDataSource` — supports caching and reusing of physical connections, which improves application performance and scalability
- `XADataSource` — provides connections that can participate in a distributed transaction

4.2 Executing SQL Statements and Manipulating Results

Once a connection has been established, an application using the JDBC API can execute queries and updates against the target data source. The JDBC 3.0 API provides access to the most commonly implemented features of SQL99. Because different vendors vary in their level of support for these features, the JDBC API includes the `DatabaseMetaData` interface. Applications can use this interface to determine whether a particular feature is supported by the data source they are using. The JDBC API also defines escape syntax to allow an application to access non-standard vendor-specific features. The use of escape syntax has the advantage of giving JDBC applications access to the same feature set as native applications and at the same time maintaining the portability of the application.

Applications use methods in the `Connection` interface to specify transaction attributes and create `Statement`, `PreparedStatement`, or `CallableStatement` objects. These statements are used to execute SQL statements and retrieve results. The `ResultSet` interface encapsulates the results of an SQL query. Statements may also be batched, allowing an application to submit multiple updates to a data source as a single unit of execution.

The JDBC API extends the `ResultSet` interface with the `RowSet` interface, thereby providing a container for tabular data that is much more versatile than a standard result set. A `RowSet` object is a `JavaBeans™` component, and it may operate without being connected to its data source. For example, a `RowSet` implementation can be serializable and therefore sent across a network, which is particularly useful for small-footprint clients that want to operate on tabular data without incurring the overhead of a JDBC driver and data source connection. Another feature of a `RowSet` implementation is that it can include a custom reader for accessing any data in tabular format, not just data in a relational database. Further, a `RowSet` object can

update its rows while it is disconnected from its data source, and its implementation can include a custom writer that writes those updates back to the underlying data source.

4.2.1 Support for SQL Advanced Data Types

The JDBC API defines standard mappings to convert SQL data types to JDBC data types and back. This includes support for SQL99 advanced data types such as BLOB, CLOB, ARRAY, REF, STRUCT, and DISTINCT. JDBC drivers may also implement one or more customized type mappings for user-defined types (UDTs), in which the UDT is mapped to a class in the Java programming language. The JDBC 3.0 API also adds support for externally managed data, for example, data in a file outside the data source.

4.3 Two-tier Model

A two-tier model divides functionality into a client layer and a server layer, as shown in FIGURE 4-1.

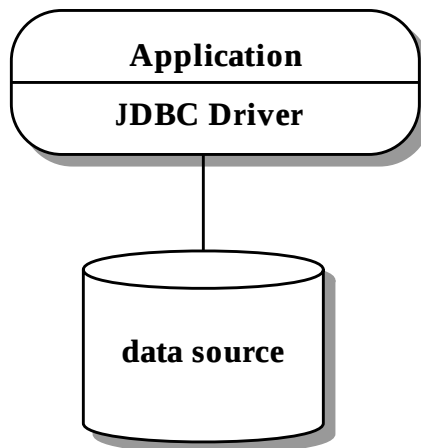


FIGURE 4-1 Two-tier Model

The client layer includes the application(s) and one or more JDBC drivers, with the application handling these areas of responsibility:

- presentation logic
- business logic
- transaction management for multiple-statement transactions or distributed transactions
- resource management

In this model, the application interacts directly with the JDBC driver(s), including establishing and managing the physical connection(s) and dealing with the details of specific underlying data source implementations. The application may use its knowledge of a specific implementation to take advantage of nonstandard features or do performance tuning.

Some drawbacks of this model include:

- mingling presentation and business logic with infrastructure and system-level functions. This presents an obstacle to producing maintainable code with a well-defined architecture.
- making applications less portable because they are tuned to a particular database implementation. Applications that require connections to multiple databases must be aware of the differences between the different vendors' implementations.
- limiting scalability. Typically, the application will hold onto one or more physical database connections until it terminates, limiting the number of concurrent applications that can be supported. In this model, issues of performance, scalability and availability are handled by the JDBC driver and the corresponding underlying data source. If an application deals with multiple drivers, it may also need to be aware of the different ways in which each driver/data source pair resolves these issues.

4.4 Three-tier Model

The three-tier model introduces a middle-tier server to house business logic and infrastructure, as shown in FIGURE 4-2.

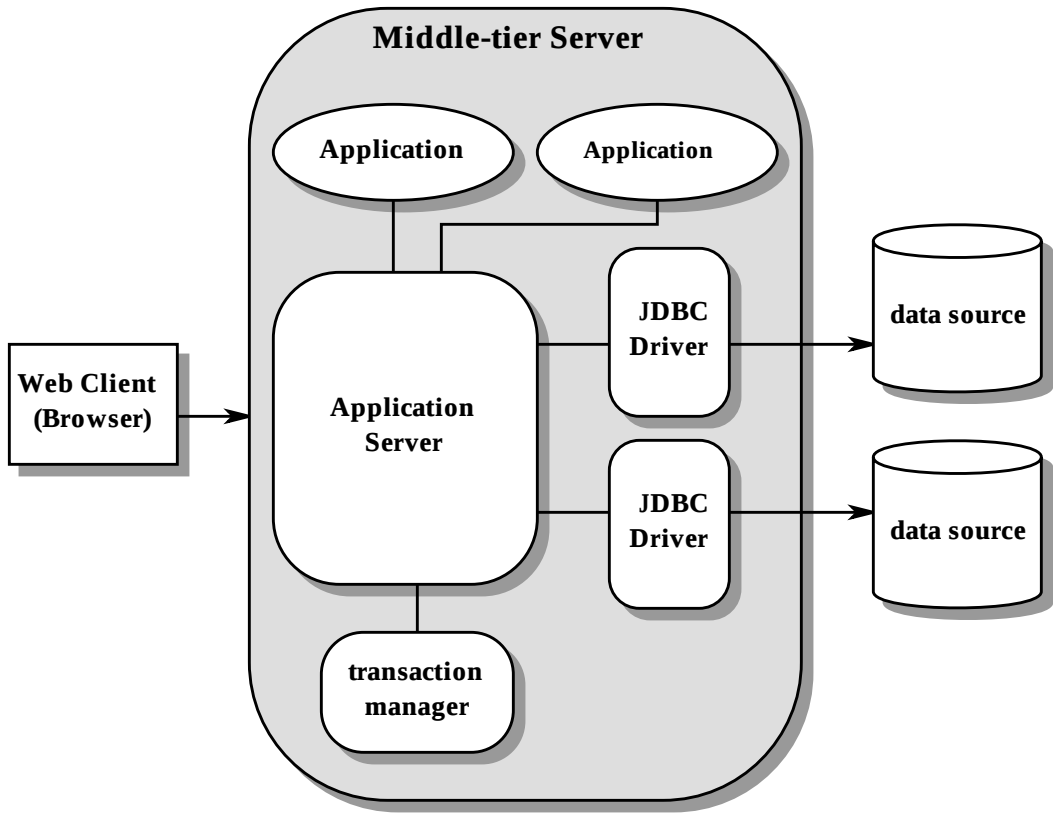


FIGURE 4-2 Three-tier Model

This architecture is designed to provide improved performance, scalability and availability for enterprise applications. Functionality is divided among the tiers as follows:

1. **Client tier** — a thin layer implementing presentation logic for human interaction. Java programs, web browsers and PDAs are typical client-tier implementations. The client interacts with the middle-tier application and does not need to include any knowledge of infrastructure or underlying data source functions.
2. **Middle-tier server** — a middle tier that includes:
 - *Applications* to interact with the client and implement business logic. If the application includes interaction with a data source, it will deal with higher-level abstractions, such as `DataSource` objects and logical connections rather than lower-level driver API.

- *An application server* to provide supporting infrastructure for a wide range of applications. This can include management and pooling of physical connections, transaction management, and the masking of differences between different JDBC drivers. This last point makes it easier to write portable applications. The application server role can be implemented by a J2EE server. Application servers implement the higher-level abstractions used by applications and interact directly with JDBC drivers.
 - *JDBC driver(s)* to provide connectivity to the underlying data sources. Each driver implements the standard JDBC API on top of whatever features are supported by its underlying data source. The driver layer may mask differences between standard SQL99 syntax and the native dialect supported by the data source. If the data source is not a relational DBMS, the driver implements the relational layer used by the application server.
- 3. Underlying data source** — the tier where the data resides. It can include relational DBMSs, legacy file systems, object-oriented DBMSs, data warehouses, spreadsheets, or other means of packaging and presenting data. The only requirement is a corresponding driver that supports the JDBC API.
-

4.5 JDBC in the J2EE Platform

J2EE components, such as JavaServer™ Pages, Servlets, and Enterprise Java Beans™ (EJB™) components, often require access to relational data and use the JDBC API for this access. When J2EE components use the JDBC API, the container manages their transactions and data sources. This means that J2EE component developers do not directly use the JDBC API's transaction and datasource management facilities. See the J2EE Platform Specification for further details.

Classes and Interfaces

The following classes and interfaces make up the JDBC API.

5.1 The `java.sql` Package

The core JDBC API is contained in the package `java.sql`. The classes and interfaces in `java.sql` are listed below. Classes are in bold type; interfaces are in standard type.

```
java.sql.Array  
java.sql.BatchUpdateException  
java.sql.Blob  
java.sql.CallableStatement  
java.sql.Clob  
java.sql.Connection  
java.sql.DataTruncation  
java.sql.DatabaseMetaData  
java.sql.Date  
java.sql.Driver  
java.sql.DriverManager  
java.sql.DriverPropertyInfo  
java.sql.ParameterMetaData  
java.sql.PreparedStatement  
java.sql.Ref  
java.sql.ResultSet
```

```
java.sql.ResultSetMetaData
java.sql.Savepoint
java.sql.SQLData
java.sql.SQLException
java.sql.SQLInput
java.sql.SQLOutput
java.sql.SQLPermission
java.sql.SQLWarning
java.sql.Statement
java.sql.Struct
java.sql.Time
java.sql.Timestamp
java.sql.Types
```

The following classes and interfaces are either new or updated in the JDBC 3.0 API. New classes and interfaces are in bold type.

```
java.sql.Blob
java.sql.CallableStatement
java.sql.Clob
java.sql.Connection
java.sql.DatabaseMetaData
java.sql.ParameterMetaData
java.sql.PreparedStatement
java.sql.Ref
Java.sql.ResultSet
java.sql.Savepoint
java.sql.SQLInput
java.sql.SQLOutput
java.sql.Statement
java.sql.Types
```

FIGURE 5-1 shows the interactions and relationships between the key classes and interfaces in the `java.sql` package. The methods involved in creating statements, setting parameters, and retrieving results are also shown.

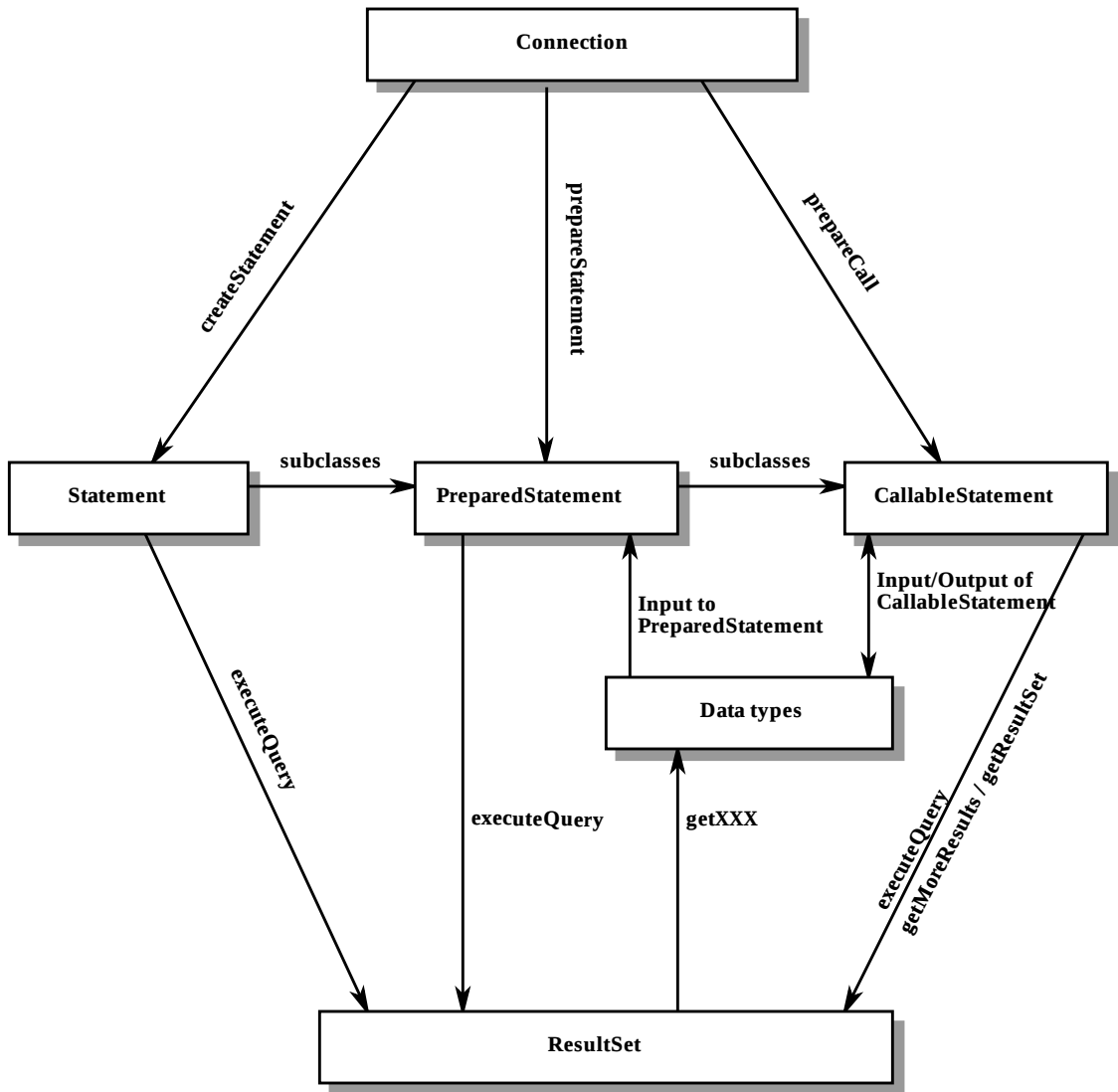


FIGURE 5-1 Relationships between major classes and interface in the `java.sql` package

5.2 The `javax.sql` Package

The following list contains the classes and interfaces that are contained in the `javax.sql` package. Classes are highlighted in bold; interfaces are in normal type.

`javax.sql.ConnectionEvent`

`javax.sql.ConnectionEventListener`

`javax.sql.ConnectionPoolDataSource`

`javax.sql.DataSource`

`javax.sql.PooledConnection`

`javax.sql.RowSet`

`javax.sql.RowSetEvent`

`javax.sql.RowSetInternal`

`javax.sql.RowSetListener`

`javax.sql.RowSetMetaData`

`javax.sql.RowSetReader`

`javax.sql.RowSetWriter`

`javax.sql.XAConnection`

`javax.sql.XADataSource`

Note – The classes and interfaces in the `javax.sql` package were first made available as the JDBC 2.0 Optional Package. This optional package was previously separate from the `java.sql` package, which was part of J2SE 1.2. Both packages (`java.sql` and `javax.sql`) are now part of J2SE 1.4.

FIGURE 5-2, FIGURE 5-3, FIGURE 5-4, and FIGURE 5-5 show the relationships between key classes and interfaces in these areas of functionality: `DataSource` objects, connection pooling, distributed transactions, and rowsets.

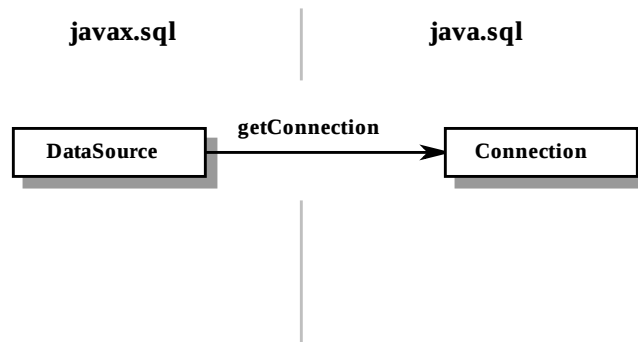


FIGURE 5-2 Relationship between `javax.sql.DataSource` and `java.sql.Connection`

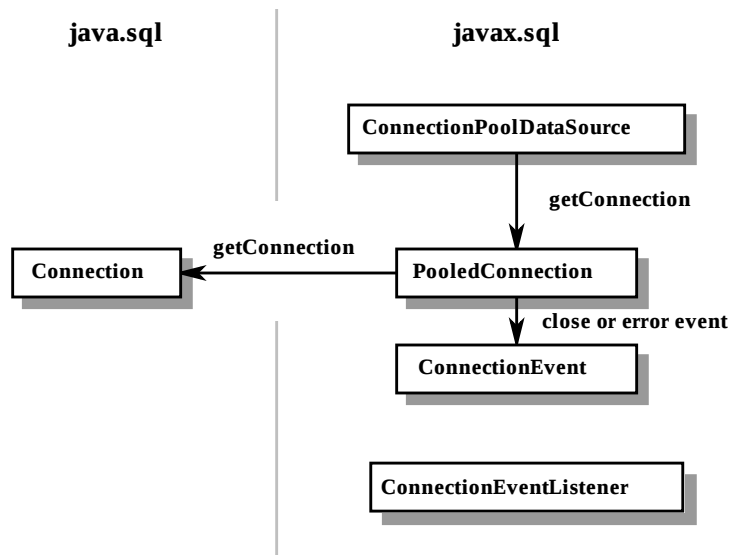


FIGURE 5-3 Relationships involved in connection pooling

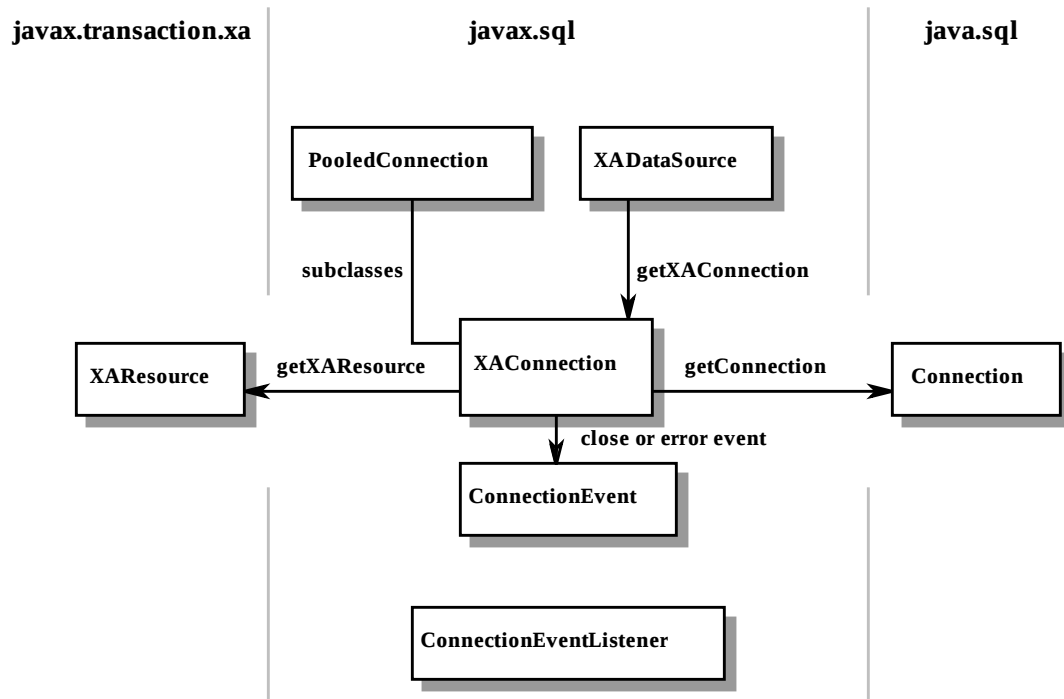


FIGURE 5-4 Relationships involved in distributed transaction support

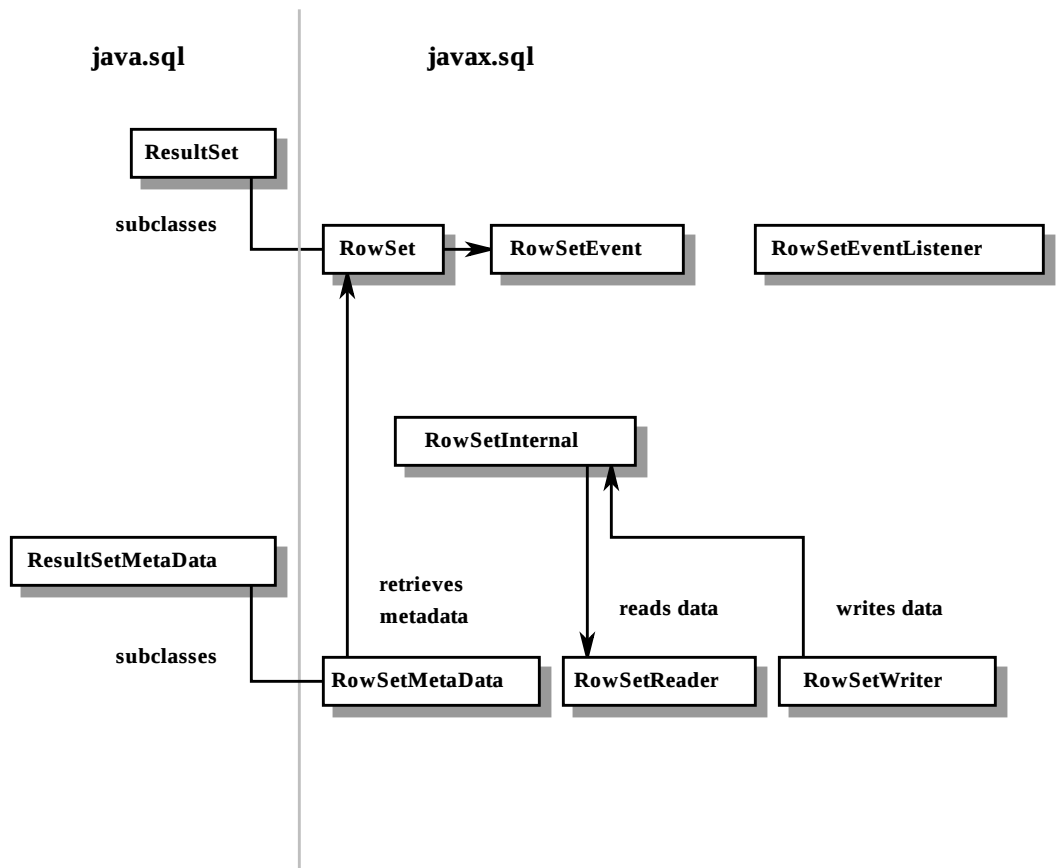


FIGURE 5-5 RowSet relationships

Compliance

This chapter identifies the features that a JDBC API implementation is required to support for each level of compliance. A JDBC API implementation includes a JDBC technology-enabled driver and its underlying data source. Therefore, compliance is defined in terms of what features are available above the driver layer.

Any features not identified here are optional. In general, a driver is not required to implement any feature that its underlying data source does not support.

6.1 Definitions

To avoid ambiguity, we will use these terms in our discussion of compliance:

- **JDBC API implementation** — a JDBC technology-enabled driver and its underlying data source. The driver may provide support for features that are not implemented by the underlying data source. It may also provide the mapping between standard syntax/semantics and the native API implemented by the data source.
- **Relevant specifications** — this document, the API specification, and the relevant SQL specification. This is also the order of precedence if a feature is described in more than one of these documents. For the JDBC 1.0 API, the relevant SQL specification is SQL92 and X/Open SQL CLI. For the JDBC 2.0 and 3.0 APIs, it is SQL92 plus the relevant sections of SQL99 and X/Open SQL CLI.
- **Supported feature** — a feature for which the JDBC API implementation supports standard syntax and semantics for that feature as defined in the relevant specifications.
- **Extension** — a feature that is not covered by any of the relevant specifications or a non-standard implementation of a feature that is covered.
- **Fully implemented** — a term applied to an interface that has all of its methods implemented to support the semantics defined in the relevant specifications.

- **Required interface** — an interface that must be included although it might not be fully implemented. Methods that are not implemented should throw an `SQLException` to indicate that the corresponding feature is not supported.

6.2 Guidelines and Requirements

The following guidelines apply to all levels of compliance:

- A JDBC API implementation must support Entry Level SQL92 plus the SQL command `DROP TABLE` (see note.)

Entry Level SQL92 represents a "floor" for the level of SQL that a JDBC API implementation must support. Access to features based on SQL99 should be provided in a way that is compatible with the relevant part of the SQL99 specification.

- Drivers must support escape syntax. Escape syntax is described in Chapter 13 "Statements".
- Drivers must support transactions. See Chapter 10 "Transactions" for details.
- Drivers should provide access to every feature implemented by the underlying data source, including features that extend the JDBC API. When a feature is not supported, the corresponding methods throw an `SQLException`. The intent is for applications using the JDBC API to have access to the same feature set as native applications.
- If a `DatabaseMetaData` method indicates that a given feature is supported, it must be supported via standard syntax and semantics as described in the relevant specifications. This may require the driver to provide the mapping to the data source's native API or SQL dialect if it differs from the standard.
- If a feature is supported, all of the relevant metadata methods must be implemented. For example, if a JDBC API implementation supports the `RowSet` interface, it must also implement the `RowSetMetaData` interface.
- If a feature is not supported, the corresponding `DatabaseMetaData` method must say so. Attempting to access the unsupported feature causes an `SQLException` to be thrown.

Note – A JDBC API implementation is required to support the `DROP TABLE` command as specified by SQL92, Transitional Level. However, support for the `CASCADE` and `RESTRICT` options of `DROP TABLE` is optional. In addition, the behaviour of `DROP TABLE` is implementation-defined when there are views or integrity constraints defined that reference the table being dropped.

6.3 JDBC 1.0 API Compliance

A driver that is compliant with the JDBC 1.0 API must do the following:

- Adhere to the preceding guidelines and requirements
- Fully implement the following interfaces:
 - `java.sql.Driver`
 - `java.sql.DatabaseMetaData` (excepting those methods introduced in the JDBC 2.0 API and the JDBC 3.0 API)
 - `java.sql.ResultSetMetaData` (excepting those methods introduced in the JDBC 2.0 API and the JDBC 3.0 API)
- Include the following required interfaces:
 - `java.sql.CallableStatement`
 - `java.sql.Connection`
 - `java.sql.PreparedStatement`
 - `java.sql.ResultSet`
 - `java.sql.Statement`

6.4 JDBC 2.0 API Compliance

A driver that is compliant with the JDBC 2.0 API must do the following:

- Comply with the JDBC 1.0 API requirements
- Fully implement the `DatabaseMetaData` interface, including the following methods added in the JDBC 2.0 API:
 - `deletesAreDetected`
 - `getConnection`
 - `getUDTs`
 - `insertsAreDetected`
 - `othersDeletesAreVisible`
 - `othersInsertsAreVisible`
 - `othersUpdatesAreVisible`
 - `ownDeletesAreVisible`
 - `ownInsertsAreVisible`

- `ownUpdatesAreVisible`
 - `supportsBatchUpdates`
 - `supportsResultSetConcurrency`
 - `supportsResultSetType`
 - `updatesAreDetected`
 - Implement the following additional `ResultSetMetaData` methods:
 - `getColumnClassName`
 - `getColumnType`
 - `getColumnTypeName`
-

6.5 JDBC 3.0 API Compliance

A driver that is compliant with the JDBC 3.0 API must do the following:

- Comply with the JDBC 2.0 API requirements
- Include the following required interfaces:
 - `java.sql.ParameterMetaData`
 - `java.sql.Savepoint`
- Fully implement the `DatabaseMetaData` interface, including the following methods added in the JDBC 3.0 API:
 - `supportsSavepoints`
 - `supportsNamedParameters`
 - `supportsMultipleOpenResults`
 - `supportsGetGeneratedKeys`
 - `getSuperTypes`
 - `getSuperTables`
 - `getAttributes`
 - `getResultSetHoldability`
 - `supportsResultSetHoldability`
 - `getSQLStateType`
 - `getDatabaseMajorVersion`
 - `getDatabaseMinorVersion`
 - `getJDBCMinorVersion`
 - `getJDBCMajorVersion`

6.6 Determining Compliance Level

The JDBC API is a constituent technology of the Java platform. Compliance with the JDBC API specification is determined as a subset of evaluating compliance with the overall platform.

Note – As of this writing, there is no separate evaluation of compliance level for the JDBC API.

6.7 Deprecated APIs

Deprecation refers to a class, interface, constructor, method or field that is no longer recommended and may cease to exist in a future version.

The following constructors and methods were deprecated in the JDBC 2.0 API:

```
java.sql.CallableStatement.getBigDecimal(int, int)
```

```
java.sql.Date(int, int, int)
```

```
java.sql.Date.getHours()
```

```
java.sql.Date.getMinutes()
```

```
java.sql.Date.getSeconds()
```

```
java.sql.Date.setHours(int)
```

```
java.sql.Date.setMinutes(int)
```

```
java.sql.Date.setSeconds(int)
```

```
java.sql.DriverManager.getLogStream()
```

```
java.sql.DriverManager.setLogStream(PrintStream)
```

```
java.sql.PreparedStatement.setUnicodeStream(int, InputStream,  
int)
```

```
java.sql.ResultSet.getBigDecimal(int, int)
```

```
java.sql.ResultSet.getBigDecimal(String, int)
```

```
java.sql.ResultSet.getUnicodeStream(int)
```

```
java.sql.ResultSet.getUnicodeStream(String)
```

```
java.sql.Time(int, int, int)
```

```
java.sql.Time.getDate()
```

```
java.sql.Time.getDay()
```

```
java.sql.Time.getMonth()
```

```
java.sql.Time.getYear()
```

```
java.sql.Time.setDate(int)
```

```
java.sql.Time.setMonth(int)
```

```
java.sql.Time.setYear(int)
```

```
java.sql.Timestamp(int, int, int, int, int, int, int)
```

Database Metadata

The `DatabaseMetaData` interface is implemented by JDBC drivers to provide information about their underlying data sources. It is used primarily by application servers and tools to determine how to interact with a given data source. Applications may also use `DatabaseMetaData` methods to get information about a data source, but this is less typical.

The `DatabaseMetaData` interface includes over 150 methods, which can be categorized according to the types of information they provide:

- general information about the data source
- whether or not the data source supports a given feature or capability
- data source limits
- what SQL objects the data source contains and attributes of those objects
- transaction support offered by the data source

The `DatabaseMetaData` interface also contains over 40 fields, which are constants used as return values for various `DatabaseMetaData` methods.

This chapter presents an overview of the `DatabaseMetaData` interface, gives examples to illustrate the categories of metadata methods, and introduces some new methods. For a comprehensive listing, however, the reader should consult the JDBC 3.0 API specification.

JDBC also defines the `ResultSetMetaData` interface, which is discussed in Chapter 14 “Result Sets”.

7.1 Creating a DatabaseMetadata Object

A `DatabaseMetaData` object is created with the `Connection` method `getMetaData`. Once created, it can be used to dynamically discover information about the underlying data source. CODE EXAMPLE 7-1 creates a `DatabaseMetadata` object and uses it to determine the maximum number of characters allowed for a table name.

```
// con is a Connection object
DatabaseMetaData dbmd = con.getMetaData();
int maxLen = dbmd.getMaxTableNameLength();
```

CODE EXAMPLE 7-1 Creating and using a `DatabaseMetadata` object

7.2 Retrieving General Information

Some `DatabaseMetaData` methods are used to dynamically discover general information about a data source as well as some details about its implementation. Some of the methods in this category are:

- `getURL`
- `getUserName`
- `getDatabaseProductVersion`, `getDriverMajorVersion` and `getDriverMinorVersion`
- `getSchemaTerm`, `getCatalogTerm` and `getProcedureTerm`
- `nullsAreSortedHigh` and `nullsAreSortedLow`
- `usesLocalFiles` and `usesLocalFilePerTable`
- `getSQLKeywords`

7.3 Determining Feature Support

A large group of `DatabaseMetaData` methods can be used to determine whether a given feature or set of features is supported by the driver or underlying data source. Beyond this, some of the methods describe what *level* of support is provided. Some of the methods that describe support for individual features are:

- `supportsAlterTableWithDropColumn`
- `supportsBatchUpdates`
- `supportsTableCorrelationNames`
- `supportsPositionedDelete`
- `supportsFullOuterJoins`
- `supportsStoredProcedures`
- `supportsMixedCaseQuotedIdentifiers`

Methods to describe a level of feature support include:

- `supportsANSI92EntryLevelSQL`
- `supportsCoreSQLGrammar`

7.4 Data Source Limits

Another group of methods provides the limits imposed by a given data source. Some of the methods in this category are:

- `getMaxRowSize`
- `getMaxStatementLength`
- `getMaxTablesInSelect`
- `getMaxConnections`
- `getMaxCharLiteralLength`
- `getMaxColumnsInTable`

Methods in this group return the limit as an `int`. A return value of zero means that there is no limit or the limit is unknown.

7.5 SQL Objects and Their Attributes

Some `DatabaseMetaData` methods provide information about the SQL objects that populate a given data source. This group also includes methods to determine the attributes of those objects. Methods in this group return `ResultSet` objects in which each row describes a particular object. For example, the method `getUDTs` returns a `ResultSet` object in which there is a row for each UDT that has been defined in the data source. Examples of this category are:

- `getSchemas` and `getCatalogs`
- `getTables`
- `getPrimaryKeys`
- `getProcedures` and `getProcedureColumns`
- `getUDTs`

7.6 Transaction Support

A small group of methods provides information about the transaction semantics supported by the data source. Examples of this category include:

- `supportsMultipleTransactions`
- `getDefaultTransactionIsolation`

7.7 New Methods

The JDBC 3.0 API introduces the following new `DatabaseMetaData` methods:

- `getSuperTypes` — returns a description of the user-defined type hierarchies defined in a given schema in the underlying data source
- `getSuperTables` — returns a description of the table hierarchies defined in a given schema in the underlying data source
- `getAttributes` — returns a description of user-defined type attributes available from a given catalog in the underlying data source
- `getSQLStateType` — returns the type of `SQLStates` that will be returned by the method `SQLException.getSQLState`, described in “`SQLException`” on page 49.

- `supportsSavepoints` — returns `true` if the JDBC API implementation supports savepoints, described in “Savepoints” on page 65.
- `supportsNamedParameters` — returns `true` if the JDBC API implementation supports named parameters for `CallableStatement` objects, described in “Setting Parameters” on page 98
- `supportsMultipleOpenResults` — returns `true` if the JDBC API implementation supports multiple open result sets for `CallableStatement` objects, described in “Returning Unknown or Multiple Results” on page 107
- `supportsGetGeneratedKeys` — returns `true` if the JDBC API implementation supports the retrieval of automatically generated keys, described in “Retrieving Auto Generated Keys” on page 112
- `getResultSetHoldability` — returns the default holdability of `ResultSet` objects returned by the driver

A complete definition of these methods may be found in the JDBC 3.0 API specification (Javadoc comments).

7.8 Modified Methods

The JDBC 3.0 API modifies the definitions of these existing `DatabaseMetaData` methods, adding support for type hierarchies:

- `getTables` — returns descriptions of the tables that match the given catalog, schema, table name, and type criteria
- `getColumns` — returns descriptions of the columns that match the given catalog, schema, table name, and column name criteria
- `getUDTs` — returns descriptions of the user-defined types that match the given catalog, schema, type name, and type criteria
- `getSchemas` — now returns the catalog for each schema as well as the schemata.

The JDBC 3.0 API specification includes updated definitions of these methods.

Exceptions

The `SQLException` class and its subtypes provide information about errors and warnings that occur while a data source is being accessed.

8.1 SQLException

An instance of `SQLException` is thrown when an error occurs during an interaction with a data source. The exception contains the following information:

- a textual description of the error. The `String` containing the description can be retrieved by calling the method `SQLException.getMessage`.
- a `SQLState`. The `String` containing the `SQLState` can be retrieved by calling the method `SQLException.getSQLState`.

The value of the `SQLState` string will depend on the underlying data source setting the value. Both X/Open and SQL99 define `SQLState` values and the conditions in which they should be set. Although the sets of values overlap, the values defined by SQL99 are not a superset of X/Open.

The `DatabaseMetaData` method `getSQLStateType` allows an application to determine if the `SQLStates` being returned by a data source are X/Open or SQL99.

- an error code. This is an integer value identifying the error that caused the `SQLException` to be thrown. Its value and meaning are implementation-specific and may be the actual error code returned by the underlying data source. The error code can be retrieved using the `SQLException.getErrorCode` method.
- a reference to any "chained" exceptions. If more than one error occurs or the event leading up to the exception being thrown can be described as a chain of events, the exceptions are referenced via this chain. A chained exception can be retrieved by calling the `SQLException.getNextException` method on the exception that was thrown. If no more exceptions are chained, the `getNextException` method returns `null`.

`SQLWarning`, `DataTruncation` and `BatchUpdateException` are the three subclasses that extend `SQLException`. These subclasses are described in the following sections.

8.2 SQLWarning

Methods in the following interfaces will generate an `SQLWarning` object if they cause a database access warning:

- `Connection`
- `Statement` and its subtypes, `PreparedStatement` and `CallableStatement`
- `ResultSet`

When a method generates an `SQLWarning` object, the caller is not informed that a data access warning has occurred. The method `getWarnings` must be called on the appropriate object to retrieve the `SQLWarning` object. However, the `DataTruncation` sub-class of `SQLWarning` may be thrown in some circumstances. See “`DataTruncation`” on page 50 for more details.

If multiple data access warnings occur, they are chained to the first one and can be retrieved by calling the `SQLWarning.getNextWarning` method. If there are no more warnings in the chain, `getNextWarning` returns null.

Subsequent `SQLWarning` objects continue to be added to the chain until the next statement is executed or, in the case of a `ResultSet` object, when the cursor is repositioned, at which point all `SQLWarning` objects in the chain are removed.

8.3 DataTruncation

The `DataTruncation` class, a subclass of `SQLWarning`, provides information when data is truncated. When data truncation occurs on a write to the data source, a `DataTruncation` object is thrown. The data value that has been truncated may have been written to the data source even if a warning has been generated. When data truncation occurs on a read from the data source, a `SQLWarning` is reported.

A `DataTruncation` object contains the following information:

- the descriptive String “Data truncation”
- the `SQLState` “01004”

- a boolean to indicate whether a column value or a parameter was truncated. The method `DataTruncation.getParameter` returns `true` if a parameter was truncated and `false` if a column value was truncated.
- an `int` giving the index of the column or parameter that was truncated. If the index of the column or parameter is unknown, the method `DataTruncation.getIndex` returns `-1`. If the index is unknown, the values returned by the methods `DataTruncation.getParameter` and `DataTruncation.getRead` are undefined.
- a boolean to indicate whether the truncation occurred on a read or a write operation. The method `DataTruncation.getRead` returns `true` if the truncation occurred on a read and `false` if the truncation occurred on a write.
- an `int` indicating the size of the target field in bytes. The method `DataTruncation.getDataSize` returns the number of bytes of data that could have been transferred or `-1` if the number of bytes is unknown.
- an `int` indicating the actual number of bytes that were transferred. The method `DataTruncation.getTransferSize` returns the number of bytes actually transferred or `-1` if the number of bytes is unknown.

8.3.1 Silent Truncation

The `Statement.setMaxFieldSize` method allows a maximum size (in bytes) to be set. This limit applies only to the `BINARY`, `VARBINARY`, `LONGVARBINARY`, `CHAR`, `VARCHAR` and `LONGVARCHAR` data types.

If a limit has been set using `setMaxFieldSize` and there is an attempt to read or write data that exceeds the limit, any truncation that occurs as a result of exceeding the set limit will *not* be reported.

8.4 BatchUpdateException

A `BatchUpdateException` object provides information about errors that occur while a batch of statements is being executed. This exception's behavior is described in Chapter 15 "Batch Updates".

Connections

A `Connection` object represents a connection to a data source via a JDBC technology-enabled driver. The data source can be a DBMS, a legacy file system, or some other source of data with a corresponding JDBC driver. A single application using the JDBC API may maintain multiple connections. These connections may access multiple data sources, or they may all access a single data source.

From the JDBC driver perspective, a `Connection` object represents a client session. It has associated state information such as user ID, a set of SQL statements and result sets being used in that session, and what transaction semantics are in effect.

To obtain a connection, the application may interact with either:

- the `DriverManager` class working with one or more `Driver` implementations

OR

- a `DataSource` implementation

Using a `DataSource` object is the preferred method because it enhances application portability, it makes code maintenance easier, and it makes it possible for an application to transparently make use of connection pooling and distributed transactions. All J2EE components that establish a connection to a data source use a `DataSource` object to get a connection.

This chapter describes the various types of JDBC drivers and the use of the `Driver` interface, the `DriverManager` class, and the basic `DataSource` interface. `DataSource` implementations that support connection pooling and distributed transactions are discussed in Chapter 11 “Connection Pooling” and Chapter 12 “Distributed Transactions”.

9.1 Types of Drivers

There are many possible implementations of JDBC drivers. These implementations are categorized as follows:

- Type 1 — drivers that implement the JDBC API as a mapping to another data access API, such as ODBC. Drivers of this type are generally dependent on a native library, which limits their portability. The JDBC-ODBC Bridge driver is an example of a Type 1 driver.
- Type 2 — drivers that are written partly in the Java programming language and partly in native code. These drivers use a native client library specific to the data source to which they connect. Again, because of the native code, their portability is limited.
- Type 3 — drivers that use a pure Java client and communicate with a middleware server using a database-independent protocol. The middleware server then communicates the client's requests to the data source.
- Type 4 — drivers that are pure Java and implement the network protocol for a specific data source. The client connects directly to the data source.

9.2 The Driver Interface

JDBC drivers must implement the `Driver` interface, and the implementation must contain a static initializer that will be called when the driver is loaded. This initializer registers a new instance of itself with the `DriverManager`, as shown in CODE EXAMPLE 9-1.

```
public class AcmeJdbcDriver implements java.sql.Driver {
    static {
        java.sql.DriverManager.registerDriver(new AcmeJdbcDriver());
    }
    ...
}
```

CODE EXAMPLE 9-1 Example static initializer for a driver implementing `java.sql.Driver`

When an application loads a `Driver` implementation, which is shown in CODE EXAMPLE 9-2, the static initializer will automatically register an instance of the driver.

```
Class.forName("com.acme.jdbc.AcmeJdbcDriver");
```

CODE EXAMPLE 9-2 Loading a driver that implements `java.sql.Driver`

To insure that drivers can be loaded using this mechanism, drivers are required to provide a no-argument constructor, that is, a constructor that takes no arguments.

The `DriverManager` class invokes `Driver` methods when it wishes to interact with a registered driver. The `Driver` interface also includes the method `acceptsURL`. The `DriverManager` can use this method to determine which of its registered drivers it should use for a given Universal Resource Locator (URL).

When the `DriverManager` is trying to establish a connection, it calls that driver's `connect` method and passes the driver the URL. If the `Driver` implementation understands the URL, it will return a `Connection` object; otherwise it returns `null`.

9.2.1 URL Syntax

The recommended JDBC URL syntax is structured as follows:

```
jdbc:<subprotocol>:<subname>
```

where a subprotocol names a particular kind of database connectivity mechanism that may be supported by one or more drivers. The contents of the subname will depend on the subprotocol.

The recommended syntax for a network address specified as part of a subname follows the standard URL naming convention for the subname:

```
//hostname:port/subsubname
```

The subsubname can have arbitrary internal syntax.

9.2.2 Registering Subprotocol names

Sun Microsystems, Java Software Division, will act as an informal registry for JDBC subprotocol names. Send mail to jdbc@eng.sun.com to reserve a subprotocol name.

9.3 The DriverManager Class

The `DriverManager` class works with the `Driver` interface to manage the set of drivers available to a JDBC client. When the client requests a connection and provides a URL, the `DriverManager` is responsible for finding a driver that recognizes the URL and for using it to connect to the corresponding data source.

Key `DriverManager` methods include:

- `registerDriver` — this method adds a driver to the set of available drivers and is invoked implicitly when the driver is loaded. The `registerDriver` method is typically called by the static initializer provided by each driver.
- `getConnection` — the method the JDBC client invokes to establish a connection. The invocation includes a JDBC URL, which the `DriverManager` passes to each driver in its list until it finds one whose `Driver.connect` method recognizes the URL. That driver returns a `Connection` object to the `DriverManager`, which in turn passes it to the application.

CODE EXAMPLE 9-3 illustrates how a JDBC client obtains a connection from the `DriverManager`.

```
// Load the driver. This creates an instance of the driver
// and calls the registerDriver method to make acme.db.Driver
// available to clients.

Class.forName("acme.db.Driver");

// Set up arguments for the call to the getConnection method.
// The sub-protocol "odbc" in the driver URL indicates the
// use of the JDBC-ODBC bridge.
String url = "jdbc:odbc:DSN";
String user = "SomeUser";
String passwd = "SomePwd";

// Get a connection from the first driver in the DriverManager
// list that recognizes the URL "jdbc:odbc:DSN".
Connection con = DriverManager.getConnection(url, user, passwd);
```

CODE EXAMPLE 9-3 Loading a driver and getting a connection using the `DriverManager`

The `DriverManager` class also provides two other `getConnection` methods:

- `getConnection(String url)` for connecting to data sources that do not use a username and password.
- `getConnection(String url, java.util.Properties prop)`, which allows the client to connect using a set of properties describing the user name and password along with any additional information that may be required.

The `DriverPropertyInfo` class provides information on the properties that the JDBC driver can understand.

See the JDBC 3.0 API Specification for more details.

9.3.1 The `SQLPermission` Class

The `SQLPermission` class represents a set of permissions that a codebase may be granted.

Currently the only permission defined is `setLog`. The `SecurityManager` will check for the `setLog` permission when an Applet calls either the `DriverManager` method `setLogWriter` or `setLogStream`. If the codebase does not have the `setLog` permission, a `java.lang.SecurityException` exception will be thrown.

See the JDBC 3.0 API Specification for more details.

9.4 The `DataSource` Interface

The `DataSource` interface, introduced in JDBC 2.0 Optional Package, is the preferred approach to obtaining data source connections. A JDBC driver that implements the `DataSource` interface returns connections that implement the same interface, `Connection`, as those returned by a `DriverManager` using the `Driver` interface. Using a `DataSource` object increases application portability by making it possible for an application to use a logical name for a data source instead of having to supply information specific to a particular driver. A logical name is mapped to a `DataSource` object via a naming service that uses the Java Naming and Directory Interface™ (JNDI). The `DataSource` object represents a physical data source and provides connections to that data source. If the data source or information about it changes, the properties of the `DataSource` object can simply be modified to reflect the changes; no change in application code is necessary.

The `DataSource` interface can be implemented so that it transparently provides the following:

- Increased performance and scalability through connection pooling
- Support for distributed transactions through the `XADataSource` interface

The next three sections discuss (1) basic `DataSource` properties, (2) how logical naming using the JNDI API improves an application’s portability and makes it easier to maintain, and (3) how to obtain a connection.

Connection pooling and distributed transactions will be discussed in Chapter 11 “Connection Pooling” and Chapter 12 “Distributed Transactions”.

9.4.1 DataSource Properties

The JDBC API defines a set of properties to identify and describe a `DataSource` implementation. The actual set required for a specific implementation depends on the type of `DataSource` object, that is, whether it is a basic `DataSource` object, a `ConnectionPoolDataSource` object, or an `XADataSource` object. The only property required for all `DataSource` implementations is `description`.

The following table describes the standard `DataSource` properties:

TABLE 9-1 Standard Data Source Properties

Property Name	Type	Description
databaseName	String	name of a particular database on a server
dataSourceName	String	a data source name; used to name an underlying <code>XADataSource</code> object or <code>ConnectionPoolDataSource</code> object when pooling of connections is done
description	String	description of this data source
networkProtocol	String	network protocol used to communicate with the server
password	String	a database password
portNumber	int	port number where a server is listening for requests
roleName	String	the initial SQL rolename
serverName	String	database server name
user	String	user’s account name

`DataSource` properties follow the convention specified for properties of `JavaBeans`TM components in the `JavaBeans 1.01 Specification`. `DataSource` implementations may augment this set with implementation-specific properties. If new properties are added, they must be given names that do not conflict with the standard property names.

`DataSource` implementations must provide “getter” and “setter” methods for each property they support. These properties typically are initialized when the `DataSource` object is deployed, as in `CODE EXAMPLE 9-4`, in which a `VendorDataSource` object implements the `DataSource` interface.

```
VendorDataSource vds = new VendorDataSource();  
vds.setServerName("my_database_server");  
String name = vds.getServerName();
```

CODE EXAMPLE 9-4 Setting and getting a `DataSource` property

`DataSource` properties are not intended to be directly accessible by JDBC clients. This design is reinforced by defining the access methods on the implementation class rather than on the public `DataSource` interface used by applications. Furthermore, the object that the client manipulates can be a wrapper that only implements the `DataSource` interface. The setter and getter methods for the properties need not be exposed to the client.

Management tools that need to manipulate the properties of a `DataSource` implementation can access those properties using introspection.

9.4.2 The JNDI API and Application Portability

The Java Naming and Directory Interface (JNDI) API provides a uniform way for applications to access remote services over the network. This section describes how it is used to register and access a JDBC `DataSource` object. See the JNDI specification for a complete description of this interface.

Using the JNDI API, applications can access a `DataSource` object by specifying its logical name. A naming service using the JNDI API maps this logical name to a corresponding data source. This scheme greatly enhances portability because any of the `DataSource` properties, such as `portNumber` or `serverName`, can be changed without impacting the JDBC client code. In fact, the application can be re-directed to a different underlying data source in a completely transparent fashion. This is particularly useful in the three-tier environment, where an application server hides the details of accessing different data sources.

`CODE EXAMPLE 9-5` illustrates the use of a JNDI-based naming service to deploy a new `VendorDataSource` object.

```
// Create a VendorDataSource object and set some properties
VendorDataSource vds = new VendorDataSource();
vds.setServerName("my_database_server");
vds.setDatabaseName("my_database");
vds.setDescription("data source for inventory and personnel");

// Use the JNDI API to register the new VendorDataSource object.
// Reference the root JNDI naming context and then bind the
// logical name "jdbc/AcmeDB" to the new VendorDataSource object.
Context ctx = new InitialContext();
ctx.bind("jdbc/AcmeDB", vds);
```

CODE EXAMPLE 9-5 Registering a DataSource object with a JNDI-based naming service

Note – J2EE components use a special convention for naming their data sources — see Chapter 5 "Naming" in the J2EE platform specification for more details.

9.4.3 Getting a Connection with a DataSource Object

Once a DataSource object has been registered with a JNDI-based naming service, an application can use it to obtain a connection to the physical data source that it represents, as is done in CODE EXAMPLE 9-6.

```
// Get the initial JNDI naming context
Context ctx = new InitialContext();

// Get the DataSource object associated with the logical name
// "jdbc/AcmeDB" and use it to obtain a database connection
DataSource ds = (DataSource)ctx.lookup("jdbc/AcmeDB");
Connection con = ds.getConnection("user", "pwd");
```

CODE EXAMPLE 9-6 Getting a Connection object using a DataSource object

The DataSource implementation bound to the name “jdbc/AcmeDB” can be modified or replaced without affecting the application code.

Transactions

Transactions are used to provide data integrity, correct application semantics, and a consistent view of data during concurrent access. All JDBC compliant drivers are required to provide transaction support. Transaction management in the JDBC API mirrors the SQL99 specification and includes these concepts:

- Auto-commit mode
- Transaction isolation levels
- Savepoints

This chapter describes transaction semantics associated with a single `Connection` object. Transactions involving multiple `Connection` objects are discussed in Chapter 12 “Distributed Transactions”.

10.1 Transaction Boundaries and Auto-commit

When to start a new transaction is a decision made implicitly by either the JDBC driver or the underlying data source. Although some data sources implement an explicit “begin transaction” statement, there is no JDBC API to do so. Typically, a new transaction is started when the current SQL statement requires one and there is no transaction already in place. Whether or not a given SQL statement requires a transaction is also specified by SQL99.

The `Connection` attribute *auto-commit* specifies when to end transactions. Enabling auto-commit causes the JDBC driver to do a transaction commit after each individual SQL statement as soon as it is complete. The point at which a statement is considered to be “complete” depends on the type of SQL statement as well as what the application does after executing it:

- For Insert, Update, Delete, and DDL statements, the statement is complete as soon as it has finished executing.
- For Select statements, the statement is complete when the associated result set is closed. The result set is closed as soon as one of the following occurs:
 - all of the rows have been retrieved
 - the associated `Statement` object is re-executed
 - another `Statement` object is executed on the same connection
- For `CallableStatement` objects, the statement is complete when *all* of the associated result sets have been closed.

10.1.1 Disabling Auto-commit Mode

CODE EXAMPLE 10-1 shows how to disable auto-commit mode.

```
// Assume con is a Connection object
con.setAutoCommit(false);
```

CODE EXAMPLE 10-1 Setting auto-commit off

When auto-commit is disabled, each transaction must be explicitly committed by calling the `Connection` method `commit` or else explicitly rolled back by calling the `Connection` method `rollback`. This is appropriate for cases where transaction management is being done in a layer above the driver, such as:

- when the application needs to group multiple SQL statements into a single transaction
- when the transaction is being managed by the application server

The default is for auto-commit mode to be enabled when the `Connection` object is created. If the value of auto-commit is changed in the middle of a transaction, the current transaction is committed. It is an error to enable auto-commit for a connection participating in a distributed transaction, as described in Chapter 12 “Distributed Transactions”.

10.2 Transaction Isolation Levels

Transaction isolation levels specify what data is “visible” to the statements within a transaction. They directly impact the level of concurrent access by defining what interaction, if any, is possible between transactions against the same target data source. Possible interaction between concurrent transactions is categorized as follows:

- *dirty reads* occur when transactions are allowed to see uncommitted changes to the data. In other words, changes made inside a transaction are visible outside the transaction before they are committed. If the changes are rolled back instead of being committed, it is possible for other transactions to have done work based on incorrect, transient data.
- *nonrepeatable reads* occur when:
 - a. Transaction A reads a row
 - b. Transaction B changes the row
 - c. Transaction A reads the same row a second time and gets different results
- *phantom reads* occur when:
 - a. Transaction A reads all rows that satisfy a **WHERE** condition
 - b. Transaction B inserts an additional row that satisfies the same condition
 - c. Transaction A reevaluates the **WHERE** condition and picks up the additional “phantom” row

JDBC augments the four levels of transaction isolation defined by SQL99, by adding **TRANSACTION_NONE**. From least restrictive to most restrictive, the transaction isolation levels are:

1. **TRANSACTION_NONE** — indicates that the driver does not support transactions, which means that it is not a JDBC compliant driver.
2. **TRANSACTION_READ_UNCOMMITTED** — allows transactions to see uncommitted changes to the data. This means that dirty reads, nonrepeatable reads, and phantom reads are possible.
3. **TRANSACTION_READ_COMMITTED** — means that any changes made inside a transaction are not visible outside the transaction until the transaction is committed. This prevents dirty reads, but nonrepeatable reads and phantom reads are still possible.
4. **TRANSACTION_REPEATABLE_READ** — disallows dirty reads and nonrepeatable reads. Phantom read are still possible.

5. `TRANSACTION_SERIALIZABLE` — specifies that dirty reads, nonrepeatable reads, and phantom reads are prevented.

10.2.1 Using the `setTransactionIsolation` Method

The default transaction level for a `Connection` object is determined by the driver supplying the connection. Typically, it is the default transaction level supported by the underlying data source.

The `Connection` method `setTransactionIsolation` is provided to allow clients to change the transaction isolation level for a given `Connection` object. The new isolation level remains in effect for the remainder of the session or until the next invocation of the `setTransactionIsolation` method.

The result of invoking the method `setTransactionIsolation` in the middle of a transaction is implementation-defined.

The return value of the method `getTransactionIsolation` should reflect the change in isolation level when it actually occurs. It is recommended that drivers implement the `setTransactionIsolation` method to change the isolation level starting with the next transaction. Committing the current transaction to make the effect immediate is also a valid implementation.

It is possible for a given JDBC driver to not support all four transaction isolation levels (not counting `TRANSACTION_NONE`). If a driver does not support the isolation level specified in an invocation of `setTransactionIsolation`, it is allowed to substitute a higher, more restrictive transaction isolation level. If a driver is unable to substitute a higher transaction level, it throws an `SQLException`. The `DatabaseMetaData` method `supportsTransactionIsolationLevel` may be used to determine whether or not the driver supports a given level.

10.2.2 Performance Considerations

As the transaction isolation level increases, more locking and other DBMS overhead is required to ensure the correct semantics. This in turn lowers the degree of concurrent access that can be supported. As a result, applications may see decreased performance when they use a higher transaction isolation level. For this reason, the transaction manager, whether it is the application itself or part of the application server, should weigh the need for data consistency against the requirements for performance when determining which transaction isolation level is appropriate.

10.3 Savepoints

Savepoints provide finer-grained control of transactions by marking intermediate points within a transaction. Once a savepoint has been set, the transaction can be rolled back to that savepoint without affecting preceding work.

The `DatabaseMetaData.supportsSavepoints` method can be used to determine whether a JDBC API implementation supports savepoints.

10.3.1 Setting and Rolling Back to a Savepoint

The JDBC 3.0 API adds the method `Connection.setSavepoint`, which sets a savepoint within the current transaction. The `Connection.rollback` method has been overloaded to take a savepoint argument.

CODE EXAMPLE 10-2 inserts a row into a table, sets the savepoint *svpt1*, and then inserts a second row. When the transaction is later rolled back to *svpt1*, the second insertion is undone, but the first insertion remains intact. In other words, when the transaction is committed, only the row containing 'FIRST' will be added to TAB1.

```
Statement stmt = conn.createStatement();
int rows = stmt.executeUpdate("INSERT INTO TAB1 (COL1) VALUES " +
                              "'('FIRST')'");

// set savepoint
Savepoint svpt1 = conn.setSavepoint("SAVEPOINT_1");
rows = stmt.executeUpdate("INSERT INTO TAB1 (COL1) " +
                          "VALUES ('SECOND')");

...
conn.rollback(svpt1);
...
conn.commit();
```

CODE EXAMPLE 10-2 Rolling back a transaction to a savepoint

10.3.2 Releasing a Savepoint

The method `Connection.releaseSavepoint` takes a `Savepoint` object as a parameter and removes it from the current transaction.

Once a savepoint has been released, attempting to reference it in a rollback operation will cause an `SQLException` to be thrown.

Any savepoints that have been created in a transaction are automatically released and become invalid when the transaction is committed or when the entire transaction is rolled back.

Rolling a transaction back to a savepoint automatically releases and makes invalid any other savepoints that were created after the savepoint in question.

Connection Pooling

In a basic `DataSource` implementation, there is a 1:1 correspondence between the client's `Connection` object and the physical database connection. When the `Connection` object is closed, the physical connection is dropped. Thus, the overhead of opening, initializing, and closing the physical connection is incurred for each client session.

A *connection pool* solves this problem by maintaining a cache of physical database connections that can be reused across client sessions. Connection pooling greatly improves performance and scalability, particularly in a three-tier environment where multiple clients can share a smaller number of physical database connections. In FIGURE 11-1, the JDBC driver provides an implementation of `ConnectionPoolDataSource` that the application server uses to build and manage the connection pool.

The algorithm used to manage the connection pool is implementation-specific and varies with application servers. The application server provides its clients with an implementation of the `DataSource` interface that makes connection pooling transparent to the client. As a result, the client gets better performance and scalability while using the same JNDI and `DataSource` APIs as before.

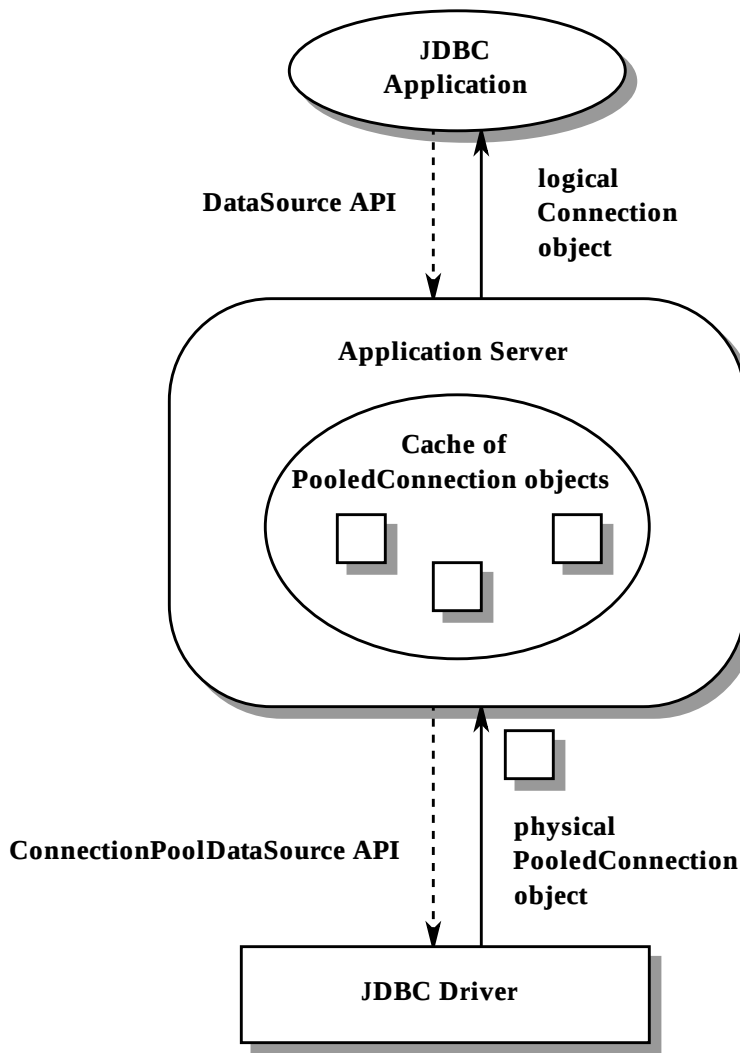


FIGURE 11-1 Connection pooling

The following sections introduce the `ConnectionPoolDataSource` interface, the `PooledConnection` interface, and the `ConnectionEvent` class. These pieces, which operate beneath the `DataSource` and `Connection` interfaces used by the client, are incorporated into a step-by-step description of a typical connection pooling implementation. This chapter also describes some important differences

between a basic `DataSource` object and one that implements connection pooling. In addition, it discusses how a pooled connection can maintain a pool of reusable `PreparedStatement` objects.

Although much of the discussion in this chapter assumes a three-tier environment, connection pooling is also relevant in a two-tier environment. In a two-tier environment, the JDBC driver implements both the `DataSource` and `ConnectionPoolDataSource` interfaces. This implementation allows an application that opens and closes multiple connections to benefit from connection pooling.

11.1 ConnectionPoolDataSource and PooledConnection

Typically, a JDBC driver implements the `ConnectionPoolDataSource` interface, and the application server uses it to obtain `PooledConnection` objects. CODE EXAMPLE 11-1 shows the signatures for the two versions of the `getPooledConnection` method.

```
public interface ConnectionPoolDataSource {  
    PooledConnection getPooledConnection() throws SQLException;  
    PooledConnection getPooledConnection(String user,  
                                         String password) throws SQLException;  
    ...  
}
```

CODE EXAMPLE 11-1 The `ConnectionPoolDataSource` interface

A `PooledConnection` object represents a physical connection to a data source. The JDBC driver's implementation of `PooledConnection` encapsulates all of the details of maintaining that connection.

An application server caches and reuses `PooledConnection` objects within its implementation of the `DataSource` interface. When a client calls the method `DataSource.getConnection`, the application server uses the physical `PooledConnection` object to obtain a logical `Connection` object. CODE EXAMPLE 11-2 shows the `PooledConnection` interface definition.

```

public interface PooledConnection {
    Connection getConnection() throws SQLException;
    void close() throws SQLException;
    void addConnectionEventListener(
        ConnectionEventListener listener);
    void removeConnectionEventListener(
        ConnectionEventListener listener);
}

```

CODE EXAMPLE 11-2 The `PooledConnection` interface

When an application is finished using a connection, it closes the logical connection using the method `Connection.close`. This closes the logical connection but does not close the physical connection. Instead, the physical connection is returned to the pool so that it can be reused.

Connection pooling is completely transparent to the client: A client obtains a pooled connection and uses it just the same way it obtains and uses a nonpooled connection.

11.2 Connection Events

Recall that when an application calls the method `Connection.close`, the underlying physical connection—the `PooledConnection` object—is available for reuse. JavaBeans-style events are used to notify the connection pool manager (the application server) that a `PooledConnection` object can be recycled.

In order to be notified of an event on a `PooledConnection` object, the connection pool manager must implement the `ConnectionEventListener` interface and then be registered as a listener by that `PooledConnection` object. The `ConnectionEventListener` interface defines the following two methods, which correspond to the two kinds of events that can occur on a `PooledConnection` object:

- `connectionClosed` — triggered when the logical `Connection` object associated with this `PooledConnection` object is closed, that is, the application called the method `Connection.close`
- `connectionErrorOccurred` — triggered when a fatal error, such as the server crashing, causes the connection to be lost

A connection pool manager registers itself as a listener for a `PooledConnection` object using the `PooledConnection.addConnectionEventListener` method. Typically, a connection pool manager registers itself as a `ConnectionEventListener` before returning a `Connection` object to an application.

The driver invokes the `ConnectionEventListener` methods `connectionClosed` and `connectionErrorOccurred` when the corresponding events occur. Both methods take a `ConnectionEvent` object as a parameter, which can be used to determine which `PooledConnection` object was closed or had an error. When the JDBC application closes its logical connection, the JDBC driver notifies the connection pool manager (the listener) by calling the listener's implementation of the method `connectionClosed`. At this point, the connection pool manager can return the `PooledConnection` object to the pool for reuse.

When an error occurs, the JDBC driver notifies the listener by calling its `connectionErrorOccurred` method and then throws an `SQLException` object to the application to notify it of the same error. In the event of a fatal error, the bad `PooledConnection` object is not returned to the pool. Instead, the connection pool manager calls the `PooledConnection.close` method on the `PooledConnection` object to close the physical connection.

11.3 Connection Pooling in a Three-tier Environment

The following sequence of steps outlines what happens when a JDBC client requests a connection from a `DataSource` object that implements connection pooling:

- The client calls `DataSource.getConnection`.
- The application server providing the `DataSource` implementation looks in its connection pool to see if there is a suitable `PooledConnection` object—a physical database connection—available. Determining the suitability of a given `PooledConnection` object may include matching the client's user authentication information or application type as well as using other implementation-specific criteria. The lookup method and other methods associated with managing the connection pool are specific to the application server.
- If there are no suitable `PooledConnection` objects available, the application server calls the `ConnectionPoolDataSource.getPooledConnection` method to get a new physical connection. The JDBC driver implementing `ConnectionPoolDataSource` creates a new `PooledConnection` object and returns it to the application server.

- Regardless of whether the `PooledConnection` was retrieved from the pool or was newly created, the application server does some internal bookkeeping to indicate that the physical connection is now in use.
- The application server calls the method `PooledConnection.getConnection` to get a logical `Connection` object. This logical `Connection` object is actually a “handle” to a physical `PooledConnection` object, and it is this handle that is returned by the `DataSource.getConnection` method when connection pooling is in effect.
- The application server registers itself as a `ConnectionEventListener` by calling the method `PooledConnection.addConnectionEventListener`. This is done so that the application server will be notified when the physical connection is available for reuse.
- The logical `Connection` object is returned to the JDBC client, which uses the same `Connection` API as in the basic `DataSource` case. Note that the underlying physical connection cannot be reused until the client calls the method `Connection.close`.

Connection pooling can also be implemented in a two-tier environment where there is no application server. In this case, the JDBC driver provides both the implementation of `DataSource` which is visible to the client and the underlying `ConnectionPoolDataSource` implementation.

11.4 DataSource Implementations and Connection Pooling

Aside from improved performance and scalability, a JDBC application should not see any difference between accessing a `DataSource` object that implements connection pooling and one that does not. However, there are some important differences in the application server and driver level implementations.

A basic `DataSource` implementation, that is, one that does not implement connection pooling, is typically provided by a JDBC driver vendor. In a basic `DataSource` implementation, the following are true:

- The `DataSource.getConnection` method creates a new `Connection` object that represents a physical connection and encapsulates all of the work to set up and manage that connection.
- The `Connection.close` method shuts down the physical connection and frees the associated resources.

In a `DataSource` implementation that includes connection pooling, a great deal happens behind the scenes. In such an implementation, the following are true:

- The `DataSource` implementation includes an implementation-specific connection pooling module that manages a cache of `PooledConnection` objects. The `DataSource` object is typically implemented by the application server as a layer on top of the driver's implementations of the `ConnectionPoolDataSource` and `PooledConnection` interfaces.
- The `DataSource.getConnection` method calls `PooledConnection.getConnection` to get a logical handle to an underlying physical connection. The overhead of setting up a new physical connection is incurred only if there are no existing connections available in the connection pool. When a new physical connection is needed, the connection pool manager will call the `ConnectionPoolDataSource` method `getPooledConnection` to create one. The work to manage the physical connection is delegated to the `PooledConnection` object.
- The `Connection.close` method closes the logical handle, but the physical connection is maintained. The connection pool manager is notified that the underlying `PooledConnection` object is now available for reuse. If the application attempts to reuse the logical handle, the `Connection` implementation throws an `SQLException`.
- A single physical `PooledConnection` object may generate many logical `Connection` objects during its lifetime. For a given `PooledConnection` object, only the most recently produced logical `Connection` object will be valid. Any previously existing `Connection` object is automatically closed when the associated `PooledConnection.getConnection` method is called. Listeners (connection pool managers) are not notified in this case.

This gives the application server a way to take a connection away from a client. This is an unlikely scenario but may be useful if the application server is trying to force an orderly shutdown.

- A connection pool manager shuts down a physical connection by calling the method `PooledConnection.close`. This method is typically called only in certain circumstances: when the application server is undergoing an orderly shutdown, when the connection cache is being reinitialized, or when the application server receives an event indicating that an unrecoverable error has occurred on the connection.

11.5 Deployment

Deploying a `DataSource` object that implements connection pooling requires that both a client-visible `DataSource` object and an underlying `ConnectionPoolDataSource` object be registered with a JNDI-based naming service.

The first step is to deploy the `ConnectionPoolDataSource` implementation, as is done in CODE EXAMPLE 11-3.

```
// ConnectionPoolDS implements the ConnectionPoolDataSource
// interface. Create an instance and set properties.
com.acme.jdbc.ConnectionPoolDS cpds =
    new com.acme.jdbc.ConnectionPoolDS();
cpds.setServerName("bookserver");
cpds.setDatabaseName("booklist");
cpds.setPortNumber(9040);
cpds.setDescription("Connection pooling for bookserver");

// Register the ConnectionPoolDS with JNDI, using the logical name
// "jdbc/pool/bookserver_pool"
Context ctx = new InitialContext();
ctx.bind("jdbc/pool/bookserver_pool", cpds);
```

CODE EXAMPLE 11-3 Deploying a `ConnectionPoolDataSource` object

Once this step is complete, the `ConnectionPoolDataSource` implementation is available as a foundation for the client-visible `DataSource` implementation. The `DataSource` implementation is deployed such that it references the `ConnectionPoolDataSource` implementation, as shown in CODE EXAMPLE 11-4.

```
// PooledDataSource implements the DataSource interface.
// Create an instance and set properties.
com.acme.appserver.PooledDataSource ds =
    new com.acme.appserver.PooledDataSource();
ds.setDescription("Datasource with connection pooling");

// Reference the previously registered ConnectionPoolDataSource
ds.setDataSourceName("jdbc/pool/bookserver_pool");

// Register the DataSource implementation with JNDI, using the logical
// name "jdbc/bookserver".
Context ctx = new InitialContext();
ctx.bind("jdbc/bookserver", ds);
```

CODE EXAMPLE 11-4 Deploying a `DataSource` object backed by a `ConnectionPoolDataSource` object

The `DataSource` object is now available for use in an application.

11.6 Reuse of Statements by Pooled Connections

The JDBC 3.0 specification introduces the feature of statement pooling. This feature, which allows an application to reuse a `PreparedStatement` object in much the same way it can reuse a connection, is made available through a pooled connection.

FIGURE 11-2 provides a logical view of how a pool of `PreparedStatement` objects can be associated with a `PooledConnection` object. As with the `PooledConnection` object itself, the `PreparedStatement` objects can be reused by multiple logical connections in a transparent manner.

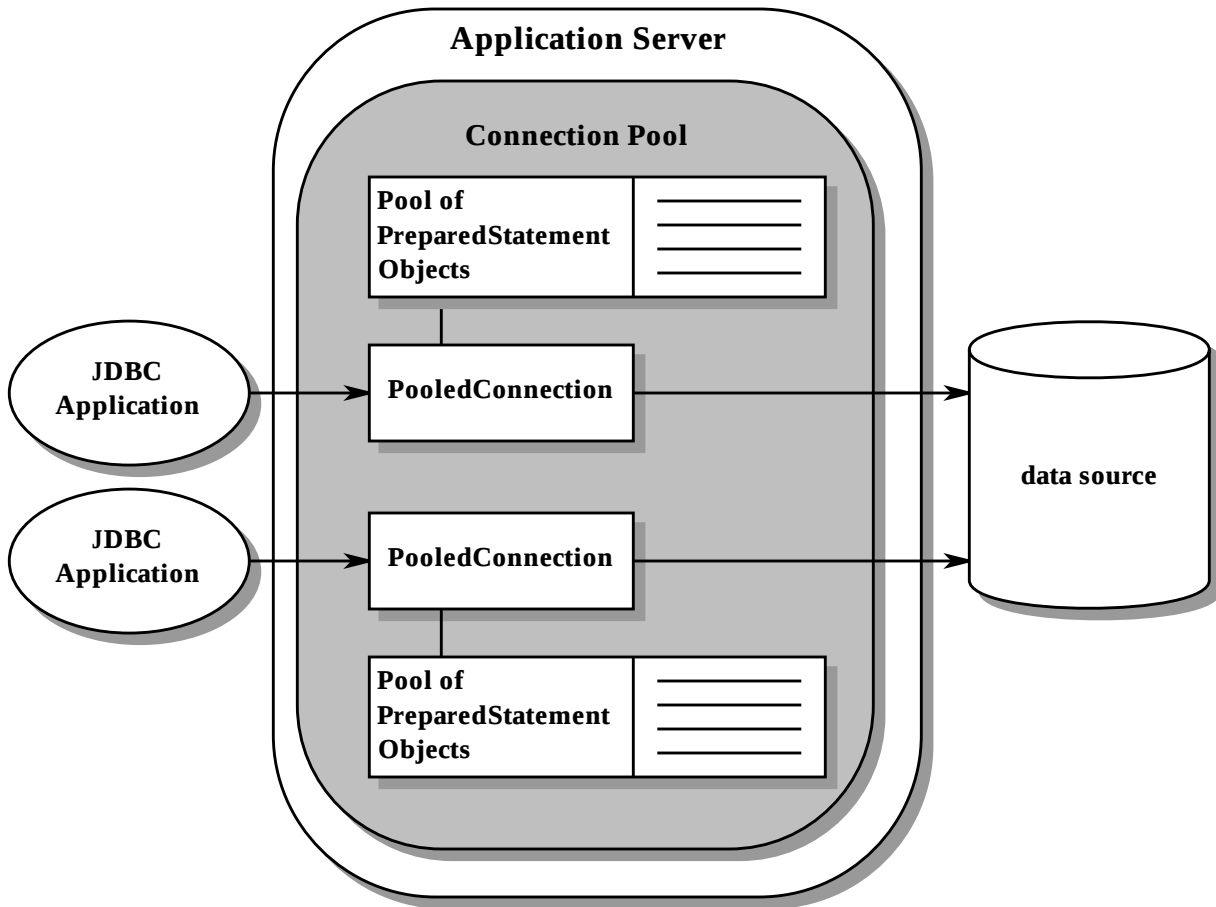


FIGURE 11-2 Logical view of prepared statements reused by pooled connections

In FIGURE 11-2, the connection pool and statement pool are implemented by the application server. However, this functionality could also be implemented by the driver or underlying data source. This discussion of statement pooling is meant to allow for any of these implementations.

11.6.1 Using a Pooled Statement

If a pooled connection reuses statements, the reuse must be completely transparent to an application. In other words, from the application's point of view, using a `PreparedStatement` object that participates in statement pooling is exactly the

same as using one that does not. Statements are kept open for reuse entirely under the covers, so there is no change in application code. If an application closes a `PreparedStatement` object, it must still call `Connection.prepareStatement` in order to use it again. The only visible effect of statement pooling is a possible improvement in performance.

An application may find out whether a data source supports statement pooling by calling the `DatabaseMetaData` method `supportsStatementPooling`. If the return value is `true`, the application can then choose to use `PreparedStatement` objects knowing that they are being pooled.

In many cases, reusing statements is a significant optimization. This is especially true for complex prepared statements. However, it should also be noted that leaving large numbers of statements open may have an adverse impact on the use of resources.

11.6.2 Closing a Pooled Statement

An application closes a pooled statement exactly the same way it closes a nonpooled statement. Whether it is pooled or not, a statement that has been closed is no longer available for use by the application, and an attempt to reuse it will cause an exception to be thrown.

The following methods can close a pooled statement:

- `Statement.close` — called by an application; if the statement is being pooled, closes the logical statement used by the application but does not close the physical statement being pooled
- `Connection.close` — called by an application
 - Nonpooled connection — closes the physical connection and all statements created by that connection. This is necessary because the garbage collection mechanism is unable to detect when externally managed resources can be released.
 - Pooled connection — closes the logical connection and the logical statements it returned but leaves open the underlying `PooledConnection` object and any associated pooled statements

An application cannot directly close a physical statement that is being pooled; instead, this is done by the connection pool manager.

An application also has no direct control over how statements are pooled. A pool of statements is associated with a `PooledConnection` object, whose behaviour is determined by the properties of the `ConnectionPoolDataSource` object that produced it. Section 11.7 “`ConnectionPoolDataSource` Properties” discusses these properties.

11.7 **ConnectionPoolDataSource Properties**

As with the `DataSource` interface, the JDBC API defines a set of properties that can be used to configure the behaviour of connection pools. These are shown in TABLE 11-1:

TABLE 11-1 Standard Connection Pool Properties

Property Name	Type	Description
maxStatements	int	The total number of statements that the pool should keep open. 0 (zero) indicates that caching of statements is disabled.
initialPoolSize	int	The number of physical connections the pool should contain when it is created
minPoolSize	int	The number of physical connections the pool should keep available at all times. 0 (zero) indicates that connections should be created as needed.
maxPoolSize	int	The maximum number of physical connections that the pool should contain. 0 (zero) indicates no maximum size.
maxIdleTime	int	The number of seconds that a physical connection should remain unused in the pool before the connection is closed. 0 (zero) indicates no limit.
propertyCycle	int	The interval, in seconds, that the pool should wait before enforcing the current policy defined by the values of the above connection pool properties

Connection pool configuration properties follow the convention specified for JavaBeans components in the JavaBeans specification. Driver vendors may choose to augment this set with implementation-specific properties. If so, the additional properties must be given names that do not conflict with the standard property names.

Like `DataSource` implementations, `ConnectionPoolDataSource` implementations must provide “getter” and “setter” methods for each property they support. These properties are typically initialized when the `ConnectionPoolDataSource` object is deployed. CODE EXAMPLE 11-5 illustrates setting properties in a vendor’s implementation of the `ConnectionPoolDataSource` interface.

```
VendorConnectionPoolDS vcp = new VendorConnectionPoolDS();  
vcp.setMaxStatements(25);  
vcp.setInitialPoolSize(10);  
vcp.setMinPoolSize(1);  
vcp.setMaxPoolSize(0);  
vcp.setMaxIdleTime(0);  
vcp.setPropertyCycle(300);
```

CODE EXAMPLE 11-5 Setting connection pool configuration properties

The properties set on a `ConnectionPoolDataSource` object apply to the `PooledConnection` objects that it creates. An application server managing a pool of `PooledConnection` objects uses these properties to determine how to manage its pool.

`ConnectionPoolDataSource` configuration properties are not intended to be directly accessible by JDBC clients. Management tools that need to manipulate the properties of a `ConnectionPoolDataSource` implementation can access those properties using introspection.

Distributed Transactions

Up to this point, the discussion of transactions has focused on the local case—transactions involving a single data source. This chapter introduces the distributed case where a single transaction involves multiple connections to one or more underlying data sources.

The following discussion includes these topics:

- distributed transaction infrastructure
- transaction managers and resource managers
- the `XDataSource`, `XAConnection`, and `XAResource` interfaces
- two-phase commit

Transaction management in the JDBC API is designed to fit with the Java Transaction API™ (JTA™). The examples presented here are high-level; the JTA specification should be consulted for a more substantial discussion.

12.1 Infrastructure

Distributed transactions require an infrastructure that provides these roles:

- Transaction manager — controls transaction boundaries and manages the two-phase commit protocol. This typically will be an implementation of JTA.
- JDBC drivers that implement the `XDataSource`, `XAConnection`, and `XAResource` interfaces. These are described in the next section.
- An application-visible implementation of `DataSource` to “sit on top of” each `XDataSource` object and interact with the transaction manager. The `DataSource` implementation is typically provided by an application server.

- Resource manager(s) to manage the underlying data. In the context of the JDBC API, a resource manager is a DBMS server. The term “resource manager” is borrowed from JTA to emphasize the point that distributed transactions using the JDBC API follow the architecture specified in that document.

This infrastructure is most often implemented in a three-tier architecture that includes the following:

1. A client tier
2. A middle tier that includes applications, an EJB server working with an external transaction manager, and a set of JDBC drivers
3. Multiple resource managers

Distributed transactions can also be implemented in two tiers. In a two-tier architecture, the application itself acts as the transaction manager and interacts directly with the JDBC drivers' `XDataSource` implementations.

The following diagram illustrates the distributed transaction infrastructure:

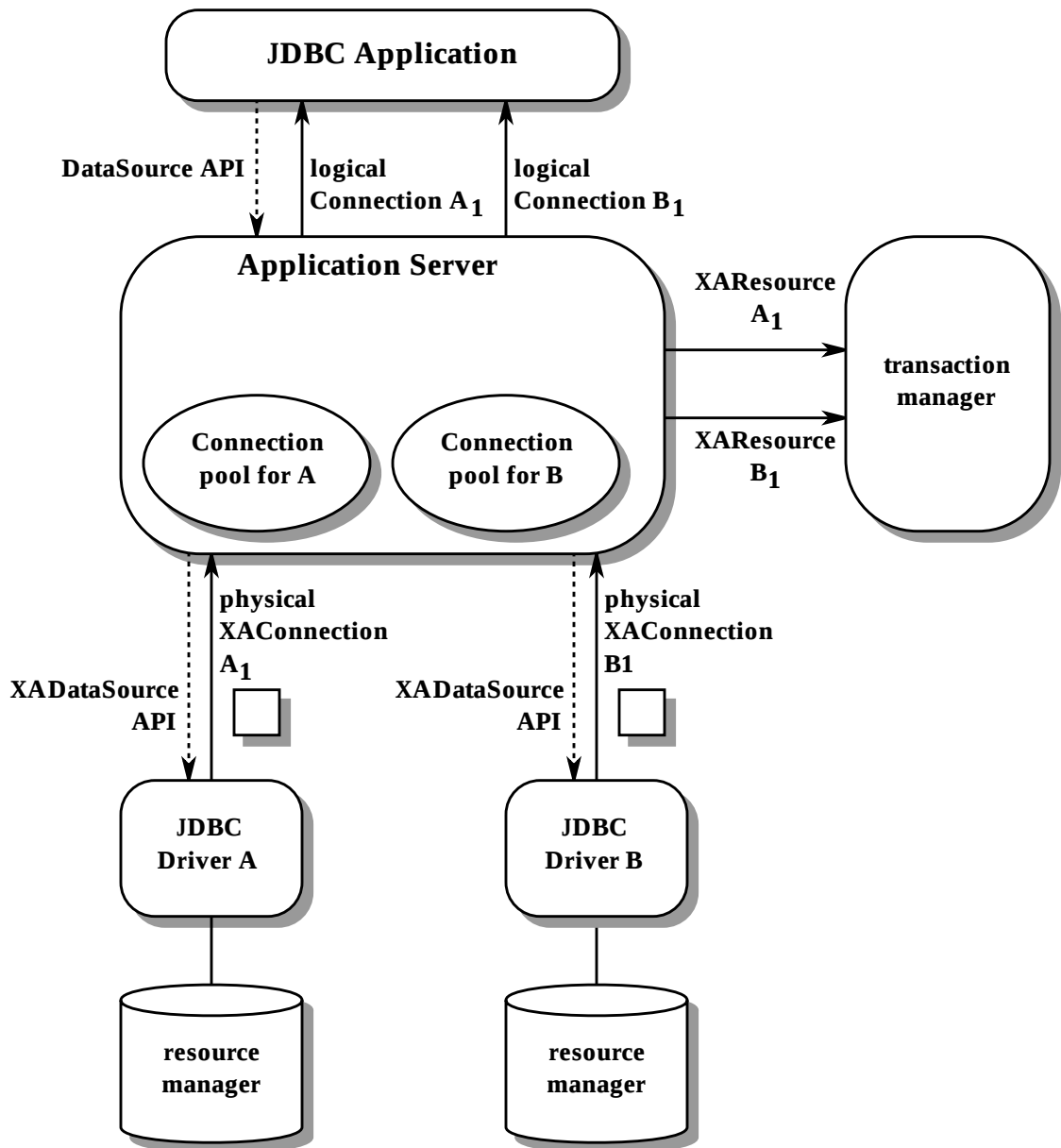


FIGURE 12-1 Infrastructure for distributed transactions

The following sections provide more detail on the components of this architecture.

12.2 XADataSource and XAConnection

The `XADataSource` and `XAConnection` interfaces, which are defined in the package `javax.sql`, are implemented by JDBC drivers that support distributed transactions. An `XAConnection` object is a `PooledConnection` object that can participate in a distributed transaction. More precisely, `XAConnection` extends the `PooledConnection` interface by adding the method `getXAResource`. This method produces an `XAResource` object that can be used by a transaction manager to coordinate the work done on this connection with the other participants in the distributed transaction. CODE EXAMPLE 12-1 gives the definition of the `XAConnection` interface.

```
public interface XAConnection extends PooledConnection {
    javax.transaction.xa.XAResource getXAResource()
        throws SQLException;
}
```

CODE EXAMPLE 12-1 The `XAConnection` interface

Because they extend the `PooledConnection` interface, `XAConnection` objects support all the methods of `PooledConnection` objects. They are reusable physical connections to an underlying data source and produce logical connection handles that can be passed back to a JDBC application.

`XAConnection` objects are produced by an `XADataSource` object. There is some similarity between `ConnectionPoolDataSource` objects and `XADataSource` objects in that they are both implemented below a `DataSource` layer that is visible to the JDBC application. This architecture allows JDBC drivers to support distributed transactions in a way that is transparent to the application. CODE EXAMPLE 12-2 shows the signatures for the two `getXAConnection` methods defined in `XADataSource`.

```
public interface XADataSource {
    XAConnection getXAConnection() throws SQLException;
    XAConnection getXAConnection(String user,
        String password) throws SQLException;
    ...
}
```

```
}
```

CODE EXAMPLE 12-2 The XDataSource interface

Typically, `DataSource` implementations built on top of an `XDataSource` implementation will also include a connection pooling module.

12.2.1 Deploying an XDataSource Object

Deploying an `XDataSource` object is done in exactly the same manner as previously described for `ConnectionPoolDataSource` objects. The two-step process includes deploying the `XDataSource` object and the application-visible `DataSource` object, as is done in CODE EXAMPLE 12-3.

```
// com.acme.jdbc.XDataSource implements the
// XDataSource interface.

// Create an instance and set properties.
com.acme.jdbc.XDataSource xads = new com.acme.jdbc.XDataSource();
xads.setServerName("bookstore");
xads.setDatabaseName("bookinventory");
xads.setPortNumber(9040);
xads.setDescription("XDataSource for inventory");

// First register xads with a JNDI naming service, using the
// logical name "jdbc/xa/inventory_xa"
Context ctx = new InitialContext();
ctx.bind("jdbc/xa/inventory_xa", xads);

// Next register the overlying DataSource object for application
// access. com.acme.appserver.DataSource is an implementation of
// the DataSource interface.

// Create an instance and set properties.
com.acme.appserver.DataSource ds =
    new com.acme.appserver.DataSource();
ds.setDescription("Datasource supporting distributed transactions");
```

```
// Reference the previously registered XADataSource
ds.setDataSourceName("jdbc/xa/inventory_xa");

// Register the DataSource implementation with a JNDI naming service,
// using the logical name "jdbc/inventory".
ctx.bind("jdbc/inventory", ds);
```

CODE EXAMPLE 12-3 Deploying a DataSource object backed by an XADataSource object

12.2.2 Getting a Connection

As in the connection pooling case, the application call to the method `DataSource.getConnection` returns a logical handle produced by the physical `XAConnection` object. The application code to get a logical connection is shown in CODE EXAMPLE 12-4.

```
Context ctx = new InitialContext();
DataSource ds = (DataSource)ctx.lookup("jdbc/inventory");
Connection con = ds.getConnection("myID", "mypasswd");
```

CODE EXAMPLE 12-4 Application code to get a logical connection

CODE EXAMPLE 12-5 is an example of code from the middle-tier server's implementation of the method `DataSource.getConnection`.

```
// Assume xads is a driver's implementation of XADataSource
XADataSource xads = (XADataSource)ctx.lookup("jdbc/xa/" +
                                             "inventory_xa");

// xacon implements XAConnection
XAConnection xacon = xads.getXAConnection("myID", "mypasswd");
// Get a logical connection to pass back up to the application
Connection con = xacon.getConnection();
```

CODE EXAMPLE 12-5 Getting a logical connection from an XAConnection object

12.3 XAResource

The XAResource interface is defined in the JTA specification and is the mapping in the Java programming language of the X/Open Group XA interface. An XAResource object is produced by calling the `XAConnection.getXAResource` method and is used to associate an `XAConnection` object with a distributed transaction. A given `XAConnection` object may be associated with at most one transaction at a time. The JDBC driver maintains a one-to-one correspondence between an XAResource object and its associated `XAConnection` object; that is, multiple calls to the `getXAResource` method must all return the same object.

In a typical scenario, the middle-tier application server calls the method `XAConnection.getXAResource` and passes the returned object to an external transaction manager. The transaction manager uses the XAResource object exclusively—it does not access an `XAConnection` object directly.

The transaction manager coordinates the work of multiple XAResource objects, each of which represents a resource manager participating in the distributed transaction. Note that two XAResource objects may “point” to the same resource manager, that is, they may be associated with `XAConnection` objects that were produced by the same `XADatasource` object.

The following XAResource methods are used by the transaction manager to implement a two-phase commit protocol. Each method takes an `xid` parameter that identifies the distributed transaction:

- **start** — tells the resource manager that the subsequent operations are part of the distributed transaction.
- **end** — marks the end of this resource manager’s part of the distributed transaction.
- **prepare** — gets the resource manager’s vote on whether to commit or roll back the distributed transaction.
- **commit** — tells the resource manager to commit its part of the distributed transaction. This method is invoked only if all the participating resource managers voted to commit the transaction.
- **rollback** — tells the resource manager to roll back its part of the distributed transaction. This method is invoked if one or more of the participating resource managers voted to roll back the transaction.

See the JTA specification for a complete description of the XAResource interface.

12.4 Transaction Management

Participation in a distributed transaction is defined as the work done between invocations of the methods `XAResource.start` and `XAResource.end`. Outside these boundaries, the transaction mode is local, and a connection behaves exactly like a local connection.

With one exception, there is no difference in how an application participating in a distributed transaction is coded. In contrast to the local case, the boundaries of a distributed transaction must be controlled by an external transaction manager that is coordinating the work of multiple connections. For this reason, it is an error for applications to call any of the following `Connection` methods while they are participating in a distributed transaction:

- `setAutoCommit(true)`
- `commit`
- `rollback`
- `setSavepoint`

The JDBC driver throws an `SQLException` if one of these operations is attempted on a connection that is participating in a distributed transaction. If the connection is later used for a local transaction, these operations are legal at that point.

Applications should also refrain from calling `Connection.setTransactionIsolation` within the bounds of a distributed transaction. The resulting behavior is implementation-defined.

If a connection has auto-commit mode already enabled at the time it joins a global transaction, the attribute will be ignored. The auto-commit behavior will resume when the connection returns to local transaction mode.

12.4.1 Two-phase Commit

The following steps outline how a transaction manager uses `XAResource` objects to implement the two-phase commit protocol. These steps assume a three-tier architecture where an application server is working with an external transaction manager:

1. The application server gets `XAResource` objects from two different connections:

```
// XAConA connects to resource manager A
javax.transaction.xa.XAResource resourceA = XAConA.getXAResource();
```

```
// XAConB connects to resource manager B
javax.transaction.xa.XAResource resourceB = XAConB.getXAResource();
```

CODE EXAMPLE 12-6 Getting the XAResource object from an XAConnection object

2. The application server passes the XAResource objects to the transaction manager. The transaction manager does not access the associated XAConnection objects directly.
3. The transaction manager uses the XAResource objects to assign a piece of the transaction to each of the associated resource managers. The transaction is identified by `xid`, which represents the identifier generated by the transaction manager when the transaction is created.

```
// Send work to resource manager A. The TMNOFLAGS argument indicates
// we are starting a new branch of the transaction, not joining or
// resuming an existing branch.
resourceA.start(xid, javax.transaction.xa.TMNOFLAGS);
// do work with resource manager A
...
// tell resource manager A that it's done, and no errors have occurred
resourceA.end(xid, javax.transaction.xa.TMSUCCESS);

// do work with resource manager B.
resourceB.start(xid, javax.transaction.xa.TMNOFLAGS);
// B's part of the distributed transaction
...
resourceB.end(xid, javax.transaction.xa.TMSUCCESS);
```

CODE EXAMPLE 12-7 Starting and ending transaction branches using the XAResource interface

4. The transaction manager initiates the two-phase commit protocol by asking each participant to vote:

```
resourceA.prepare(xid);
resourceB.prepare(xid);
```

CODE EXAMPLE 12-8 Initiating two-phase commit

A participating resource manager can vote to roll back the transaction by throwing a `javax.transaction.xa.XAException`.

5. If both participants vote to commit, the transaction manager tells each one to commit its piece of the distributed transaction (the second parameter tells the resource manager not to use a one phase commit protocol on behalf of the xid):

```
resourceA.commit(xid, false);  
resourceB.commit(xid, false);
```

CODE EXAMPLE 12-9 Committing the distributed transaction

6. If either resource manager votes to roll back, the transaction manager tells each one to roll back its piece of the transaction:

```
resourceA.rollback(xid);  
resourceB.rollback(xid);
```

CODE EXAMPLE 12-10 Rolling back the distributed transaction

The transaction manager is not required to use the same `XAResource` object to commit/rollback a transaction branch as was used to execute the branch. If different `XAResource` objects are used, however, they must be associated with `XAConnection` objects that connect to the same resource manager.

Note – Steps 1-6 also apply to the case where `XAConA` and `XAConB` are two physical connections to the same resource manager.

12.5 Closing the Connection

In a typical distributed transaction environment, the middle-tier server needs to be notified when an application has finished using a connection. As in the earlier discussion of `PooledConnection` objects, the middle-tier server will add itself as a `ConnectionEventListener` so that it will be notified when the application calls the method `Connection.close`. At this point, the server will notify the transaction manager so that it can end the transaction branch for the corresponding `XAResource` object. If the server's `DataSource` implementation includes connection pooling, the connection pooling module will be notified that it can return the physical `XAConnection` object to the pool.

Note – A distributed transaction may still be active after a participating `Connection` object is closed. This is not true for local transactions.

12.6 Limitations of the XAResource Interface

The `javax.transaction.xa.XAResource` interface is limited to defining only the set of methods needed to join and participate in global transactions, as defined by the X/Open XA standard. This allows any resource manager that implements the interface to participate with any other resource manager or transaction manager that has the same level of support.

Functionality that is not defined in the X/Open standard is correspondingly not defined in the `XAResource` interface. Resource managers that provide for support of features not defined in the X/Open XA standard, such as setting isolation levels, cursor holdability, and savepoints in global transactions, will have to do so in an implementation-defined way.

Users who use implementation-defined features should be aware that they will limit the portability of their applications.

Statements

This section describes the `Statement` interface and its subclasses, `PreparedStatement` and `CallableStatement`. It also describes related topics, including escape syntax, performance hints, and auto-generated keys.

13.1 The Statement Interface

The `Statement` interface defines methods for executing SQL statements that do not contain parameter markers. The `PreparedStatement` interface adds methods for setting input parameters, and the `CallableStatement` interface adds methods for retrieving output parameter values returned from stored procedures.

13.1.1 Creating Statements

`Statement` objects are created by `Connection` objects, as is done in CODE EXAMPLE 13-1.

```
Connection conn = dataSource.getConnection(user, passwd);  
Statement stmt = conn.createStatement();
```

CODE EXAMPLE 13-1 Creating a `Statement` object

Each `Connection` object can create multiple `Statement` objects that may be used concurrently by the program. This is demonstrated in CODE EXAMPLE 13-2.

```
// get a connection from the DataSource object ds  
Connection conn = ds.getConnection(user, passwd);  
// create two instances of Statement
```

```
Statement stmt1 = conn.createStatement();
Statement stmt2 = conn.createStatement();
```

CODE EXAMPLE 13-2 Creating multiple `Statement` objects from a single connection

13.1.1.1 Setting `ResultSet` Characteristics

Additional constructors may be used to set the type and concurrency or the type, concurrency, and holdability of any result sets produced by a statement. See Chapter 14 “Result Sets” for more on the `ResultSet` interface.

CODE EXAMPLE 13-3 creates a `Statement` object that returns result sets that are scrollable, that are insensitive to changes made while the `ResultSet` object is open, that can be updated, and that do not close the `ResultSet` objects when a commit operation is implicitly or explicitly performed.

```
Connection conn = ds.getConnection(user, passwd);
Statement stmt = conn.createStatement(
    ResultSet.TYPE_SCROLL_INSENSITIVE,
    ResultSet.CONCUR_UPDATABLE,
    ResultSet.HOLD_CURSOR_OVER_COMMIT);
```

CODE EXAMPLE 13-3 Creating a scrollable, insensitive, updatable result set that stays open after the method `commit` is called

See Chapter 14 “Result Sets” for more information on `ResultSet` types.

13.1.2 Executing Statement Objects

The method used to execute a `Statement` object depends on the type of SQL statement being executed. If the `Statement` object represents an SQL query returning a `ResultSet` object, the method `executeQuery` should be used. If the SQL is known to be a DDL statement or a DML statement returning an update count, the method `executeUpdate` should be used. If the type of the SQL statement is not known, the method `execute` should be used.

13.1.2.1 Returning a `ResultSet` object

CODE EXAMPLE 13-4 shows the execution of an SQL string returning a `ResultSet` object.

```
Statement stmt = conn.createStatement();
ResultSet rs = stmt.executeQuery("select TITLE, AUTHOR, ISBN " +
                                "from BOOKLIST");

while (rs.next()){
    ...
}
```

CODE EXAMPLE 13-4 Executing a `Statement` object that returns a `ResultSet` object

If the SQL string being executed does not return a `ResultSet` object, the method `executeQuery` throws an `SQLException`.

13.1.2.2 Returning an Update Count

In **CODE EXAMPLE 13-5**, the SQL statement being executed returns the number of rows affected by the update.

```
Statement stmt = conn.createStatement();
int rows = stmt.executeUpdate("update STOCK set ORDER = 'Y' " +
                              "where SUPPLY = 0");

if (rows > 0) {
    ...
}
```

CODE EXAMPLE 13-5 Executing a `Statement` object that returns an update count

The method `executeUpdate` throws an `SQLException` if the SQL string being executed does not return an update count.

13.1.2.3 Using the Method `execute`

The method `execute` should be used only when the SQL string being executed could return an update count, a `ResultSet` object, multiple `ResultSet` objects, or the type of the statement is not known. The `execute` method returns `true` if the first result is a `ResultSet` object and `false` if it is an update count. Additional methods must be called to retrieve the `ResultSet` object or update count or to retrieve additional results, if any.

```
String sql;
...
```

```

Statement stmt = conn.createStatement();
boolean b = stmt.execute(sql);
if (b == true) {
    // b is true if a ResultSet is returned
    ResultSet rs;
    rs = stmt.getResultSet();
    while (rs.next()) {
        ...
    }
} else {
    // b is false if an update count is returned
    int rows = stmt.getUpdateCount();
    if (rows > 0) {
        ...
    }
}

```

CODE EXAMPLE 13-6 Executing a `Statement` object that may return an update count or a `ResultSet` object

When the SQL string being executed returns a `ResultSet` object, the method `getUpdateCount` returns `-1`. If the SQL string being executed returns an update count, the method `getResultSet` returns `null`.

13.1.3 Closing Statement Objects

An application calls the method `Statement.close` to indicate that it has finished processing a statement. All `Statement` objects will be closed when the connection that created them is closed. However, it is good coding practice for applications to close statements as soon as they have finished processing them. This allows any external resources that the statement is using to be released immediately.

Closing a `Statement` object will close and invalidate any instances of `ResultSet` produced by that `Statement` object. The resources held by the `ResultSet` object may not be released until garbage collection runs again, so it is a good practice to explicitly close `ResultSet` objects when they are no longer needed.

These comments about closing `Statement` objects apply to `PreparedStatement` and `CallableStatement` objects as well.

13.2 The PreparedStatement Interface

The `PreparedStatement` interface extends `Statement`, adding the ability to set values for parameter markers contained within the statement.

`PreparedStatement` objects represent SQL statements that can be prepared, or precompiled, for execution once and then executed multiple times. Parameter markers, represented by “?” in the SQL string, are used to specify input values to the statement that may vary at runtime.

13.2.1 Creating a PreparedStatement Object

An instance of `PreparedStatement` is created in the same manner as a `Statement` object, except that the SQL command is supplied when the statement is created:

```
Connection conn = ds.getConnection(user, passwd);
PreparedStatement ps = conn.prepareStatement("INSERT INTO BOOKLIST" +
                                           "(AUTHOR, TITLE, ISBN) VALUES (?, ?, ?)");
```

CODE EXAMPLE 13-7 Creating a `PreparedStatement` object with three placeholder markers

13.2.1.1 Setting ResultSet Characteristics

As with `createStatement`, the method `prepareStatement` defines a constructor that can be used to specify the characteristics of result sets produced by that prepared statement.

```
Connection conn = ds.getConnection(user, passwd);
PreparedStatement ps = conn.prepareStatement(
    "SELECT AUTHOR, TITLE FROM BOOKLIST WHERE ISBN = ?",
    ResultSet.TYPE_FORWARD_ONLY,
    ResultSet.CONCUR_UPDATABLE);
```

CODE EXAMPLE 13-8 Creating a `PreparedStatement` object that returns forward only, updatable result sets

13.2.2 Setting Parameters

The `PreparedStatement` interface defines setter methods that are used to substitute values for each of the parameter markers in the precompiled SQL string. The names of the methods follow the pattern "set<Type>".

For example, the method `setString` is used to specify a value for a parameter marker that expects a string. Each of these setter methods takes at least two parameters. The first is always an `int` equal to the ordinal position of the parameter to be set, starting at 1. The second and any remaining parameters specify the value to be assigned to the parameter.

```
PreparedStatement ps = conn.prepareStatement("INSERT INTO BOOKLIST" +
                                             "(AUTHOR, TITLE, ISBN) VALUES (?, ?, ?)");
ps.setString(1, "Zamiatin, Evgenii");
ps.setString(2, "We");
ps.setLong(3, 0140185852);
```

CODE EXAMPLE 13-9 Setting parameters in a `PreparedStatement` object

A value must be provided for each parameter marker in the `PreparedStatement` object before it can be executed. The methods used to execute a `PreparedStatement` object (`executeQuery`, `executeUpdate` and `execute`) will throw an `SQLException` if a value is not supplied for a parameter marker.

The values set for the parameter markers of a `PreparedStatement` object are not reset when it is executed. The method `clearParameters` can be called to explicitly clear the values that have been set. Setting a parameter with a different value will replace the previous value with the new one.

13.2.2.1 Type Conversions

The data type specified in a `PreparedStatement` setter method is a data type in the Java programming language. The JDBC driver is responsible for mapping this to the corresponding JDBC type (one of the SQL types defined in `java.sql.Types`) so that it is the appropriate type to be sent to the data source. The default mapping is specified in Appendix B TABLE B-2.

13.2.2.2 Type Conversions Using the Method `setObject`

The method `setObject` can be used to convert an object in the Java programming language to a JDBC type.

The conversion is explicit when `setObject` is passed a Java `Object` and a JDBC data type. The driver will attempt to convert the `Object` to the specified JDBC type before passing it to the data source. If the object cannot be converted to the target type, an `SQLException` object is thrown. In CODE EXAMPLE 13-10, a Java `Object` of type `Integer` is being converted to the JDBC type `SHORT`.

```
Integer value = new Integer(15);
ps.setObject(1, value, java.sql.Types.SHORT);
```

CODE EXAMPLE 13-10 Converting an `Integer` object to an SQL `SHORT`

If `setObject` is called without a type parameter, the Java `Object` is implicitly mapped using the default mapping for that object type.

```
Integer value = new Integer(15);
// value is mapped to java.sql.Types.INTEGER
ps.setObject(1, value);
```

CODE EXAMPLE 13-11 The method `setObject` using the default mapping

The default mapping is described in Appendix B TABLE B-4

Note – The method `setObject` will do custom mapping for SQL UDTs that have a custom mapping. See Chapter 17 “Customized Type Mapping” for more information.

13.2.2.3 Setting NULL Parameters

The method `setNull` can be used to set any parameter to JDBC `NULL`. It takes two parameters, the ordinal position of the parameter marker, and the JDBC type of the parameter.

```
ps.setNull(2, java.sql.Types.VARCHAR);
```

CODE EXAMPLE 13-12 Setting a `String` parameter to JDBC `NULL`

If a Java `null` is passed to any of the setter methods that take a Java object, the parameter will be set to JDBC `NULL`.

13.2.3 Describing Outputs and Inputs of a PreparedStatement Object

The method `PreparedStatement.getMetaData` retrieves a `ResultSetMetaData` object containing a description of the columns that will be returned by a prepared statement when it is executed. The `ResultSetMetaData` object contains a record for each column being returned. Methods in the `ResultSetMetaData` interface provide information about the number of columns being returned and the characteristics of each column.

```
PreparedStatement pstmt = conn.prepareStatement(
    "SELECT * FROM CATALOG");
ResultSetMetaData rsmd = pstmt.getMetaData();
int colCount = rsmd.getColumnCount();
int colType;
String colLabel;
for (int i = 1; i <= colCount; i++) {
    colType = rsmd.getColumnType(i);
    colLabel = rsmd.getColumnLabel(i);
    ...
}
```

CODE EXAMPLE 13-13 Creating a `ResultSetMetaData` object and retrieving column information from it

The method `PreparedStatement.getParameterMetaData` returns a `ParameterMetaData` object describing the parameter markers that appear in the `PreparedStatement` object. Methods in the `ParameterMetaData` interface provide information about the number of parameters and their characteristics.

```
PreparedStatement pstmt = conn.prepareStatement(
    "SELECT * FROM BOOKLIST WHERE ISBN = ?");
...
ParameterMetaData pmd = pstmt.getParameterMetaData();
int colType = pmd.getParameterType(1);
...
```

CODE EXAMPLE 13-14 Creating a `ParameterMetaData` object and retrieving parameter information from it

See the API specification for more details.

13.2.4 Executing a PreparedStatement Object

As with `Statement` objects, the method used to execute a `PreparedStatement` object depends on the type of SQL statement being executed. If the `PreparedStatement` object is a query returning a `ResultSet` object, it should be executed with the method `executeQuery`. If it is a DML statement returning a row count, it should be executed with the method `executeUpdate`. The method `execute` should be used only if the return type of the statement is unknown.

If any of the `PreparedStatement` execute methods is called with an SQL string as a parameter, an `SQLException` is thrown.

13.2.4.1 Returning a ResultSet Object

CODE EXAMPLE 13-15 shows a query being prepared and then executed multiple times.

```
PreparedStatement pstmt = conn.prepareStatement("SELECT AUTHOR, " +
                                                "TITLE FROM BOOKLIST WHERE SECTION = ?");
for (int i = 1; i <= maxSectionNumber; i++) {
    pstmt.setInt(1, i);
    ResultSet rs = pstmt.executeQuery();
    while (rs.next()) {
        // process the record
    }
    rs.close();
}
pstmt.close();
```

CODE EXAMPLE 13-15 Preparing and executing a statement returning a result set

If the statement being executed does not return a `ResultSet` object, the method `executeQuery` throws an `SQLException`.

13.2.4.2 Returning a Row Count

If the statement being prepared and executed is a DML or DDL operation, it should be executed using the method `executeUpdate`. This method returns the number of rows that the statement affected.

```

PreparedStatement pstmt = conn.prepareStatement(
    "update stock set reorder = 'Y' where stock < ?");
pstmt.setInt(1, 5);
int num = pstmt.executeUpdate();

```

CODE EXAMPLE 13-16 Preparing and executing a statement returning an update count

If the statement being executed returns a `ResultSet` object, an `SQLException` is thrown.

13.2.4.3 Using the Method `execute`

If the return type of a `PreparedStatement` object is not known or may return multiple `ResultSet` objects, it should be executed with the `execute` method. As is true with `Statement` objects, the methods `getResultSet` and `getUpdateCount` can be used to retrieve a result.

```

PreparedStatement pstmt = conn.prepareStatement(sqlStatement);
// set any parameters the user passes
...

boolean b = pstmt.execute();
if (b == true) {
    ResultSet rs = pstmt.getResultSet();
    // process a ResultSet
    ...
} else {
    int rowCount = pstmt.getUpdateCount();
    // process row count
    ...
}

```

CODE EXAMPLE 13-17 Preparing and executing a statement that may return a result set or an update count

13.3 The CallableStatement Interface

The `CallableStatement` interface extends `PreparedStatement` with methods for executing and retrieving results from stored procedures.

13.3.1 Creating a CallableStatement Object

As with `Statement` and `PreparedStatement` objects, `CallableStatement` objects are created by `Connection` objects. CODE EXAMPLE 13-18 shows the creation of a `CallableStatement` object for calling the stored procedure ‘validate’, which has a return parameter and two other parameters.

```
CallableStatement cstmt = conn.prepareCall(
    "{? = call validate(?, ?)}");
```

CODE EXAMPLE 13-18 Creating a `CallableStatement` object

All the examples in this chapter use the escape syntax for calling stored procedures. See “Stored Procedures” on page 111.

13.3.2 Setting Parameters

`CallableStatement` objects may take three types of parameters: IN, OUT, and INOUT. The parameter can be specified as either an ordinal parameter or a named parameter. A value must be set for each parameter marker in the statement.

The number, type, and attributes of parameters to a stored procedure can be determined using the `DatabaseMetaData` method `getProcedureColumns`.

Parameter ordinals, which are integers passed to the appropriate setter method, refer to the parameter markers (“?”) in the statement, starting at one. Literal parameter values in the statement do not increment the ordinal value of the parameter markers. In CODE EXAMPLE 13-19, the two parameter markers have the ordinal values 1 and 2.

```
CallableStatement cstmt = con.prepareCall(
    "{CALL PROC(?, \"Literal_Value\", ?)}");
cstmt.setString(1, "First");
cstmt.setString(2, "Third");
```

CODE EXAMPLE 13-19 Specifying ordinal parameters

Named parameters can also be used to specify specific parameters. This is especially useful when a procedure has many parameters with default values. Named parameters can be used to specify only the values that have no default value. The name of a parameter corresponds to the `COLUMN_NAME` field returned by `DatabaseMetaData.getProcedureColumns`.

In CODE EXAMPLE 13-20, the procedure `COMPLEX_PROC` takes ten parameters, but only the first and fifth parameters, `PARAM_1` and `PARAM_5`, are required.

```
CallableStatement cstmt = con.prepareCall(
    "{CALL COMPLEX_PROC(?, ?)}";
cstmt.setString("PARAM_1", "Price");
cstmt.setFloat("PARAM_5", 150.25);
```

CODE EXAMPLE 13-20 Specifying two input parameters to a stored procedure

Additional methods in the `CallableStatement` interface allow parameters to be registered and retrieved by name.

The `DatabaseMetaData.supportsNamedParameters` method can be called to determine if a JDBC driver and underlying data source support specifying named parameters.

It is not possible to combine setting parameters with ordinals and with names in the same statement. If ordinals and names are used for parameters in the same statement, an `SQLException` is thrown.

Note – In some cases it may not be possible to provide only some of the parameters for a procedure. For example, if the procedure name is overloaded, the data source determines which procedure to call based on the number of parameters. Enough parameters must be provided to allow the data source to resolve any ambiguity.

13.3.2.1 IN Parameters

IN parameters are assigned values using the setter methods as described in “Setting Parameters” on page 98. In CODE EXAMPLE 13-21, a string parameter and a date parameter are set.

```
cstmt.setString(1, "October");
cstmt.setDate(2, date);
```

CODE EXAMPLE 13-21 Setting IN parameters

13.3.2.2 OUT Parameters

The method `registerOutParameter` must be called to set the type for each OUT parameter before a `CallableStatement` object is executed. When the stored procedure returns from execution, it will use these types to set the values for any OUT parameters.

The values of OUT parameters can be retrieved using the appropriate getter methods defined in the `CallableStatement` interface. CODE EXAMPLE 13-22 shows the execution of a stored procedure with two OUT parameters, a string and float, and the retrieval of the OUT parameter values.

```
CallableStatement cstmt = conn.prepareCall(
    "{CALL GET_NAME_AND_NUMBER(?, ?)}");
cstmt.registerOutParameter(1, java.sql.Types.STRING);
cstmt.registerOutParameter(2, java.sql.Types.FLOAT);
cstmt.execute();
// Retrieve OUT parameters
String name = cstmt.getString(1);
float number = cstmt.getFloat(2);
```

CODE EXAMPLE 13-22 Registering and retrieving OUT parameters

13.3.2.3 INOUT Parameters

Parameters that are both input and output parameters must be both set by using the appropriate setter method and also registered by calling the `registerOutParameter` method. The type implied by the setter method (see TABLE B-1 in Appendix B “Data Type Conversion Tables”) and the type supplied to the method `registerOutParameter` must be the same.

CODE EXAMPLE 13-23 shows the stored procedure `calc`, which takes one INOUT float parameter.

```
CallableStatement cstmt = conn.prepareCall("{CALL CALC(?)}");
cstmt.setFloat(1, 1237.98f);
cstmt.registerOutParameter(1, java.sql.Types.FLOAT);
cstmt.execute();
float f = cstmt.getFloat(1);
```

CODE EXAMPLE 13-23 Executing a `CallableStatement` object with an INOUT parameter

13.3.3 Executing a CallableStatement Object

As with `Statement` and `PreparedStatement` objects, the method used to execute a `CallableStatement` object depends on whether it returns a single `ResultSet` object, an update count, or multiple mixed results.

13.3.3.1 Returning a Single ResultSet Object

CODE EXAMPLE 13-24 shows the execution of a `CallableStatement` object that takes one input parameter and returns a single `ResultSet` object.

```
CallableStatement cstmt = conn.prepareCall("{CALL GETINFO(?)}");
cstmt.setLong(1, 1309944422);
ResultSet rs = cstmt.executeQuery();
// process the results
while (rs.next()) {
    ...
}
rs.close();
cstmt.close();
```

CODE EXAMPLE 13-24 Executing a `CallableStatement` object that returns a single result set

The method `executeQuery` throws an `SQLException` if the stored procedure does not return a `ResultSet` object.

13.3.3.2 Returning a Row Count

CODE EXAMPLE 13-25 shows the execution of a `CallableStatement` object that returns a row count.

```
CallableStatement cstmt = conn.prepareCall("{call GETCOUNT(?)}");
cstmt.setString(1, "Smith");
int count = cstmt.executeUpdate();
cstmt.close();
```

CODE EXAMPLE 13-25 Executing a `CallableStatement` object returning an update count

If the stored procedure does not return a row count, the method `executeUpdate` throws an `SQLException`.

13.3.3.3 Returning Unknown or Multiple Results

If the type or number of results returned by a `CallableStatement` object are not known until run time, the `CallableStatement` object should be executed with the method `execute`. The methods `getMoreResults`, `getUpdateCount`, and `getResultSet` can be used to retrieve all the results.

The method `execute` returns `true` if the first result is a `ResultSet` object and `false` if it is an update count.

When the method `execute` returns `true`, the method `getResultSet` is called to retrieve the `ResultSet` object. When `execute` returns `false`, the method `getUpdateCount` returns an `int`. If this number is greater than or equal to zero, it indicates the number of rows that were affected by the statement. If it is `-1`, it indicates that there are no more results.

If multiple results are being returned, the method `getMoreResults` can be called to get the next result. As with the method `execute`, `getMoreResults` will return `true` if the next result is a `ResultSet` object and `false` if the next result is a row count or no more result are available.

CODE EXAMPLE 13-26 shows how to retrieve all the results from a `CallableStatement` object.

```
CallableStatement cstmt = conn.prepareCall(procCall);
boolean retval = cstmt.execute();
ResultSet rs;
int count;
do {
    if (retval == false) {
        count = cstmt.getUpdateCount();
        if (count == -1) {
            // no more results
            break;
        } else {
            // process row count
        }
    } else { // ResultSet
        rs = cstmt.getResultSet();
        // process ResultSet
    }
    retval = cstmt.getMoreResults();
}
```

```
while (true);
```

CODE EXAMPLE 13-26 Executing a callable statement that returns multiple results

By default, each call to the method `getMoreResults` closes any previous `ResultSet` object returned by the method `getResultSet`. However, the method `getMoreResults` may take a parameter that specifies whether a `ResultSet` object returned by `getResultSet` should be closed. The `Statement` interface defines three constants that can be supplied to the method `getMoreResults`:

- `CLOSE_CURRENT_RESULT` — indicates that the current `ResultSet` object should be closed when the next `ResultSet` object is returned
- `KEEP_CURRENT_RESULT` — indicates that the current `ResultSet` object should not be closed when the next `ResultSet` object is returned
- `CLOSE_ALL_RESULTS` — indicates that any `ResultSet` objects that have been kept open should be closed when the next result is returned

If the current result is an update count and not a `ResultSet` object, any parameter passed to `getMoreResults` is ignored.

To determine whether a driver implements this feature, an application can call the `DatabaseMetaData` method `supportsMultipleOpenResults`.

```
ResultSet rs1 = cstmt.getResultSet();
rs1.next();
...
retval = cstmt.getMoreResults(Statement.KEEP_CURRENT_RESULT);
if (retval == true) {
    ResultSet rs2 = cstmt.getResultSet();
    rs2.next();
    ...
    rs1.next();
}
retval = cstmt.getMoreResults(Statement.CLOSE_ALL_RESULTS);
...
```

CODE EXAMPLE 13-27 Keeping multiple results from a `CallableStatement` object open

13.4 Escape Syntax

The SQL string used in a `Statement` object may include JDBC *escape syntax*. Escape syntax allows the driver to more easily scan for syntax that requires special processing. Implementing this special processing in the driver layer improves application portability.

Special escape processing might be needed for the following:

- commonly used features that do not have standard syntax defined by SQL, or where the native syntax supported by the underlying data source varies widely among vendors. The driver may translate the escape syntax to a specific native syntax in this case.
- features that are not supported by the underlying data source but are implemented by the driver.

Escape processing for a `Statement` object is turned on or off using the method `setEscapeProcessing`, with the default being on. The `RowSet` interface also includes a `setEscapeProcessing` method. The `RowSet` method applies to the SQL string used to populate a `RowSet` object. The `setEscapeProcessing` method does not work for a `PreparedStatement` object because its SQL string may have been precompiled when the `PreparedStatement` object was created.

JDBC defines escape syntax for the following:

- scalar functions
- date and time literals
- outer joins
- calling stored procedures
- escape characters for LIKE clauses

13.4.1 Scalar Functions

Almost all underlying data sources support numeric, string, time, date, system, and conversion functions on scalar values. The escape syntax to access a scalar function is:

```
{fn <function-name> (argument list)}
```

For example, the following code calls the function `concat` with two arguments to be concatenated:

```
{fn concat("Hot", "Java")}
```

The following syntax gets the name of the current database user:

```
{fn user()}
```

Scalar functions may be supported by different data sources with slightly different native syntax, and they may not be supported by all drivers. The driver will either map the escaped function call into the native syntax or implement the function directly.

Various `DatabaseMetaData` methods list the functions that are supported. For example, the method `getNumericFunctions` returns a comma-separated list of the Open Group CLI names of numeric functions, the method `getStringFunctions` returns string functions, and so on.

Refer to Appendix C “Scalar Functions” for a list of the scalar functions a driver is expected to support. A driver is required to implement these functions only if the data source supports them, however.

13.4.2 Date and Time Literals

Data sources differ widely in the syntax they use for date, time, and timestamp literals. The JDBC API supports ISO standard format for the syntax of these literals, using an escape clause that the driver translates to native syntax.

The escape syntax for date literals is:

```
{d 'yyyy-mm-dd'}
```

The driver will replace the escape clause with the equivalent native representation. For example, the driver might replace `{d '1999-02-28'}` with `'28-FEB-99'` if that is the appropriate format for the underlying data source.

The escape syntax for `TIME` and `TIMESTAMP` literals are:

```
{t 'hh:mm:ss'}
```

```
{ts 'yyyy-mm-dd hh:mm:ss.f . . .'}  
.
```

The fractional seconds (`.f . . .`) portion of the timestamp can be omitted.

13.4.3 Outer Joins

Outer joins are an advanced feature and are not supported by all data sources. Consult relevant SQL documentation for an explanation of outer joins.

The escape syntax for an outer join is:

```
{oj <outer-join>}
```

where <outer-join> has the form:

```
table {LEFT|RIGHT|FULL} OUTER JOIN {table | <outer-join>} ON search-  
condition
```

(Note that curly braces ({}) in the preceding line indicate that one of the items between them must be used; they are not part of the syntax.) The following SELECT statement uses the escape syntax for an outer join.

```
Statement stmt = con.createStatement();  
stmt.executeQuery("SELECT * FROM {oj TABLE1 " +  
    "LEFT OUTER JOIN TABLE2 ON DEPT_NO = 003420930}");
```

The JDBC API provides three DatabaseMetaData methods for determining the kinds of outer joins a driver supports: supportsOuterJoins, supportsFullOuterJoins, and supportsLimitedOuterJoins.

13.4.4 Stored Procedures

If a database supports stored procedures, they can be invoked using JDBC escape syntax as follows:

```
{call <procedure_name> [(<argument-list>)]}
```

or, where a procedure returns a result parameter:

```
{? = call <procedure_name> [(<argument-list>)]}
```

The square brackets indicate that the (argument-list) portion is optional. Input arguments may be either literals or parameter markers. See “Setting Parameters” on page 103 for information on parameters.

The method DatabaseMetaData.supportsStoredProcedures returns true if the database supports stored procedures.

13.4.5 LIKE Escape Characters

The percent sign (%) and underscore (_) characters are wild card characters in SQL LIKE clauses (% matches zero or more characters, and _ matches exactly one character). In order to interpret them literally, they can be preceded by a backslash (\), which is a special escape character in strings. One can specify which character to use as the escape character by including the following syntax at the end of a query:

```
{escape '<escape-character>'}
```

For example, the following query uses the backslash as an escape character, and finds identifier names that begin with an underscore. Note that the Java compiler will not recognize the backslash as a character unless it is preceded by a backslash.

```
stmt.executeQuery("SELECT name FROM Identifiers " +  
                  "WHERE Id LIKE '\\_%' {escape '\\'}");
```

13.5 Performance Hints

The `Statement` interface has two methods that can be used to provide hints to a JDBC driver: `setFetchDirection` and `setFetchSize`. The values supplied to these methods are applied to each result set produced by the statement. The methods of the same name in the `ResultSet` interface can be used to supply hints for just that result set.

Hints provided to the driver via this interface may be ignored by the driver if they are not appropriate.

The methods `getFetchDirection` and `getFetchSize` return the current value of the hints. If either of these methods is called before the corresponding setter method has been called, the value returned is implementation-defined.

13.6 Retrieving Auto Generated Keys

Many database systems have a mechanism that automatically generates a unique key field when a row is inserted. The method `Statement.getGeneratedKeys`, which can be called to retrieve the value of such a key, returns a `ResultSet` object with a column for each automatically generated key. A flag indicating that any auto

generated columns should be returned is passed to the methods `execute`, `executeUpdate` or `prepareStatement` when the statement is executed or prepared.

```
Statement stmt = conn.createStatement();
// indicate that the key generated is going to be returned
int rows = stmt.executeUpdate("INSERT INTO ORDERS " +
                              "(ISBN, CUSTOMERID) " +
                              "VALUES (195123018, 'BILLG')",
                              Statement.RETURN_GENERATED_KEYS);
ResultSet rs = stmt.getGeneratedKeys();
boolean b = rs.next();
if (b == true) {
    // retrieve the new key value
    ...
}
```

CODE EXAMPLE 13-28 Retrieving auto generated keys

Additional methods allow the ordinals or names of the columns that should be returned to be specified. In **CODE EXAMPLE 13-29** the `Statement` method `executeUpdate` is called with two parameters, the first is the SQL statement to be executed, the second is an array of `String` containing the column name that should be returned when `getGeneratedKeys` is called:

```
String keyColumn[] = {"ORDER_ID"};
...
Statement stmt = conn.createStatement();
int rows = stmt.executeUpdate("INSERT INTO ORDERS " +
                              "(ISBN, CUSTOMERID) " +
                              "VALUES (966431502, 'BILLG')",
                              keyColumn);
ResultSet rs = stmt.getGeneratedKeys();
....
```

CODE EXAMPLE 13-29 Retrieving a named column using `executeUpdate` and `getGeneratedKeys`

See the API Specification for more details.

Calling `ResultSet.getMetaData` on the `ResultSet` object returned by `getGeneratedKeys` will produce a `ResultSetMetaData` object that can be used to determine the number, type and properties of the generated keys.

In some cases, such as in an insert select statement, more than one key may be returned. The `ResultSet` object returned by `getGeneratedKeys` will contain a row for each key that a statement generated. If no keys are generated, an empty result set will be returned.

The concurrency of the `ResultSet` object returned by `getGeneratedKeys` must be `CONCUR_READ_ONLY`. The type of the `ResultSet` object must be either `TYPE_FORWARD_ONLY` or `TYPE_SCROLL_INSENSITIVE`.

The method `DatabaseMetaData.supportsGetGeneratedKeys` returns `true` if a JDBC driver and underlying data source support the retrieval of automatically generated keys.

Result Sets

The `ResultSet` interface provides methods for retrieving and manipulating the results of executed queries.

14.1 Kinds of `ResultSet` Objects

`ResultSet` objects can have different functionality and characteristics. These characteristics are result set type, result set concurrency, and cursor holdability.

14.1.1 `ResultSet` Types

The type of a `ResultSet` object determines the level of its functionality in two main areas: (1) the ways in which the cursor can be manipulated and (2) how concurrent changes made to the underlying data source are reflected by the `ResultSet` object. The latter is called the *sensitivity* of the `ResultSet` object.

The three different `ResultSet` types are described below.

1. `TYPE_FORWARD_ONLY`

- The result set is not scrollable; its cursor moves forward only, from before the first row to after the last row.
- The rows contained in the result set depend on how the underlying database materializes the results. That is, it contains the rows that satisfy the query at either the time the query is executed or as the rows are retrieved.

2. `TYPE_SCROLL_INSENSITIVE`

- The result set is scrollable; its cursor can move both forward and backward relative to the current position, and it can move to an absolute position.

- The result set is insensitive to changes made to the underlying data source while it is open. It contains the rows that satisfy the query at either the time the query is executed or as the rows are retrieved.

3. TYPE_SCROLL_SENSITIVE

- The result set is scrollable; its cursor can move both forward and backward relative to the current position, and it can move to an absolute position.
- The result set reflects changes made to the underlying data source while the result set remains open.

The default `ResultSet` type is `TYPE_FORWARD_ONLY`.

The method `DatabaseMetaData.supportsResultSetType` returns `true` if the specified type is supported by the driver and `false` otherwise.

If the driver does not support the type supplied to the methods `createStatement`, `prepareStatement`, or `prepareCall`, it generates an `SQLWarning` on the `Connection` object that is creating the statement. When the statement is executed, the driver returns a `ResultSet` object of a type that most closely matches the requested type. An application can find out the type of a `ResultSet` object by calling the method `ResultSet.getType`.

14.1.2 ResultSet Concurrency

The concurrency of a `ResultSet` object determines what level of update functionality is supported.

The two concurrency levels are:

- `CONCUR_READ_ONLY`

The `ResultSet` object cannot be updated using the `ResultSet` interface.

- `CONCUR_UPDATABLE`

The `ResultSet` object can be updated using the `ResultSet` interface.

The default `ResultSet` concurrency is `CONCUR_READ_ONLY`.

The method `DatabaseMetaData.supportsResultSetConcurrency` returns `true` if the specified concurrency level is supported by the driver and `false` otherwise.

If the driver does not support the concurrency level supplied to the methods `createStatement`, `prepareStatement`, or `prepareCall`, it generates an `SQLWarning` on the `Connection` object that is creating the statement. An application can find out the concurrency of a `ResultSet` object by calling the method `ResultSet.getConcurrency`.

If the driver cannot return a `ResultSet` object at the requested type and concurrency, it determines the appropriate type before determining the concurrency.

14.1.3 `ResultSet` Holdability

Calling the method `Connection.commit` can close the `ResultSet` objects that have been created during the current transaction. In some cases, however, this may not be the desired behaviour. The `ResultSet` property `holdability` gives the application control over whether `ResultSet` objects (cursors) are closed when a commit operation is implicitly or explicitly performed.

The following `ResultSet` constants may be supplied to the `Connection` methods `createStatement`, `prepareStatement`, and `prepareCall`:

1. `HOLD_CURSORS_OVER_COMMIT`

- `ResultSet` objects (cursors) are not closed; they are held open when a commit operation is implicitly or explicitly performed.

2. `CLOSE_CURSORS_AT_COMMIT`

- `ResultSet` objects (cursors) are closed when a commit operation is implicitly or explicitly performed. Closing cursors at commit can result in better performance for some applications.

The default holdability of `ResultSet` objects is implementation defined. The `DatabaseMetaData` method `getResultSetHoldability` can be called to determine the default holdability of result sets returned by the underlying data source.

14.1.4 Specifying `ResultSet` Type, Concurrency and Holdability

The parameters supplied to the methods `Connection.createStatement`, `Connection.prepareStatement`, and `Connection.prepareCall` determine the type, concurrency, and holdability of `ResultSet` objects that the statement produces. CODE EXAMPLE 14-1 creates a `Statement` object that will return scrollable, read-only `ResultSet` objects that are insensitive to updates made to the data source and that will be closed when the transaction in which they were created is committed.

```
Connection conn = ds.getConnection(user, passwd);
Statement stmt = conn.createStatement(
    ResultSet.TYPE_SCROLL_INSENSITIVE,
```

```
ResultSet.CONCUR_READ_ONLY,  
ResultSet.CLOSE_CURSORS_AT_COMMIT);
```

CODE EXAMPLE 14-1 Creating a scrollable, insensitive, read-only result set with a cursor that is not holdable

The `Statement`, `PreparedStatement` and `CallableStatement` interfaces also provide setter and getter methods for each of these properties.

14.2 Creating and Manipulating `ResultSet` Objects

14.2.1 Creating `ResultSet` Objects

A `ResultSet` object is most often created as the result of executing a `Statement` object. The `Statement` methods `executeQuery` and `getResultSet` both return a `ResultSet` object, as do various `DatabaseMetaData` methods. CODE EXAMPLE 14-2 executes an SQL statement returning a `ResultSet` object.

```
Statement stmt = conn.createStatement();  
ResultSet rs = stmt.executeQuery("select author, title, isbn " +  
                                "from booklist");
```

CODE EXAMPLE 14-2 Executing a query returning a `ResultSet` object

For each book in the table `booklist`, the `ResultSet` object will contain a row consisting of three columns, `author`, `title`, and `isbn`. The following sections detail how these rows and columns can be retrieved.

14.2.2 Cursor Movement

A `ResultSet` object maintains a cursor, which points to its current row of data. When a `ResultSet` object is first created, the cursor is positioned before the first row. The following methods can be used to move the cursor:

- `next()` — moves the cursor forward one row. Returns `true` if the cursor is now positioned on a row and `false` if the cursor is positioned after the last row.

- `previous()` — moves the cursor backwards one row. Returns `true` if the cursor is now positioned on a row and `false` if the cursor is positioned before the first row.
- `first()` — moves the cursor to the first row in the `ResultSet` object. Returns `true` if the cursor is now positioned on the first row and `false` if the `ResultSet` object does not contain any rows.
- `last()` — moves the cursor to the last row in the `ResultSet` object. Returns `true` if the cursor is now positioned on the last row and `false` if the `ResultSet` object does not contain any rows.
- `beforeFirst()` — positions the cursor at the start of the `ResultSet` object, before the first row. If the `ResultSet` object does not contain any rows, this method has no effect.
- `afterLast()` — positions the cursor at the end of the `ResultSet` object, after the last row. If the `ResultSet` object does not contain any rows, this method has no effect.
- `relative(int rows)`— moves the cursor relative to its current position.

If *rows* is 0 (zero), the cursor is unchanged. If *rows* is positive, the cursor is moved forward *rows* rows. If the cursor is less than the specified number of rows from the last row, the cursor is positioned after the last row. If *rows* is negative, the cursor is moved backward *rows* rows. If the cursor is less than *rows* rows from the first row, the cursor is positioned before the first row.

The method `relative` returns `true` if the cursor is positioned on a valid row and `false` otherwise.

If *rows* is 1, `relative` is identical to the method `next`. If *rows* is -1, `relative` is identical to the method `previous`.

- `absolute(int row)`— positions the cursor on the *row*-th row of the `ResultSet` object.

If *row* is positive, the cursor is moved *row* rows from the beginning of the `ResultSet` object. The first row is 1, the second 2, and so on. If *row* is greater than the number of rows in the `ResultSet` object, the cursor is positioned after the last row.

If *row* is negative, the cursor is moved *row* rows from the end of the `ResultSet` object. The last row is -1, the penultimate -2, and so on. If *row* is greater than the number of rows in the `ResultSet` object, the cursor is positioned before the first row.

Calling `absolute(0)` moves the cursor before the first row.

For a `ResultSet` object that is of type `TYPE_FORWARD_ONLY`, the only valid cursor movement method is `next`. All other cursor movement methods throw an `SQLException`.

14.2.3 Retrieving Values

The `ResultSet` interface provides methods for retrieving the values of columns from the row where the cursor is currently positioned.

Two getter methods exist for each JDBC type: one that takes the column index as its first parameter and one that takes the column name or label.

The columns are numbered from left to right, as they appear in the select list of the query, starting at 1.

Column names supplied to getter methods are case insensitive. If a select list contains the same column more than once, the first instance of the column will be returned.

The index of the first instance of a column name can be retrieved using the method `findColumn`. If the specified column is not found, the method `findColumn` throws an `SQLException`.

```
ResultSet rs = stmt.executeQuery(sqlstring);
int colIdx = rs.findColumn("ISBN");
```

CODE EXAMPLE 14-3 Mapping a column name to a column index

14.2.3.1 Data Type Conversions

The recommended `ResultSet` getter method for each JDBC type is shown in TABLE B-6 on page B-182. This table also shows all of the possible conversions that a JDBC driver may support. The method

`DataBaseMetaData.supportsConvert(int fromType, int toType)` returns `true` if the driver supports the given conversion.

14.2.3.2 ResultSet Metadata

When the `ResultSet` method `getMetaData` is called on a `ResultSet` object, it returns a `ResultSetMetaData` object describing the columns of that `ResultSet` object. In cases where the SQL statement being executed is unknown until runtime, the result set metadata can be used to determine which of the getter methods should be used to retrieve the data. In CODE EXAMPLE 14-4, result set metadata is used to determine the type of each column in the result set.

```

ResultSet rs = stmt.executeQuery(sqlString);
ResultSetMetaData rsmd = rs.getMetaData();
int colType [] = new int[rsmd.getColumnCount()];
for (int idx = 0, int col = 1; idx < colType.length; idx++, col++)
    colType[idx] = rsmd.getColumnType(col);

```

CODE EXAMPLE 14-4 Retrieving result set metadata

14.2.3.3 Retrieving NULL values

The method `wasNull` can be called to determine if the last value retrieved was a JDBC NULL in the database.

When the column value in the database is JDBC NULL, it may be returned to the Java application as `null`, `0`, or `false`, depending on the type of the column value. Column values that map to Java Object types are returned as a Java `null`; those that map to numeric types are returned as `0`; those that map to a Java `boolean` are returned as `false`. Therefore, it may be necessary to call the `wasNull` method to determine whether the last value retrieved was a JDBC NULL.

14.2.4 Modifying ResultSet Objects

`ResultSet` objects with concurrency `CONCUR_UPDATABLE` can be updated using `ResultSet` methods. Columns can be updated, new rows can be inserted, and rows can be deleted using methods defined in the `ResultSet` interface.

14.2.4.1 Updating a Row

Updating a row in a `ResultSet` object is a two-phase process. First, the new value for each column being updated is set, and then the change is applied to the row. The row in the underlying data source is not updated until the second phase is completed.

The `ResultSet` interface contains two update methods for each JDBC type, one specifying the column to be updated as an index and one specifying the column name as it appears in the select list.

Column names supplied to updater methods are case insensitive. If a select list contains the same column more than once, the first instance of the column will be updated.

The method `updateRow` is used to apply all column changes to the current row. The changes are not made to the row until `updateRow` has been called. The method `cancelUpdates` can be used to back out changes made to the row before the method `updateRow` is called. CODE EXAMPLE 14-5 shows the current row being updated to change the value of the column “author” to “Zamyatin, Evgenii Ivanovich”:

```
Statement stmt = conn.createStatement(ResultSet.TYPE_FORWARD_ONLY,
                                       ResultSet.CONCUR_UPDATABLE);
ResultSet rs = stmt.executeQuery("select author from booklist " +
                                "where isbn = 140185852");

rs.next();
rs.updateString("author", "Zamyatin, Evgenii Ivanovich");
rs.updateRow();
```

CODE EXAMPLE 14-5 Updating a row in a `ResultSet` object

The method `DatabaseMetaData.ownUpdatesAreVisible(int type)` returns `true` if a `ResultSet` object of the specified type is able to see its own updates and `false` otherwise.

A `ResultSet` object may be able to use the method `rowUpdated` to detect rows that have had the method `updateRow` called on them. The method `DatabaseMetaData.updatesAreDetected(int type)` returns `true` if a `ResultSet` object of the specified type can determine if a row is updated using the method `rowUpdated` and `false` otherwise.

14.2.4.2 Deleting a Row

A row in a `ResultSet` object can be deleted using the method `deleteRow`. CODE EXAMPLE 14-6 shows the fourth row of the `ResultSet` `rs` being deleted.

```
rs.absolute(4);
rs.deleteRow();
```

CODE EXAMPLE 14-6 Deleting a row in a `ResultSet` object

After the method `deleteRow` has been called, the current row is deleted in the underlying data source. This deletion is visible as a change in the open `ResultSet` object if the row is either removed or replaced by an empty or invalid row.

If the deleted row is removed or replaced by an empty row, the method `DatabaseMetaData.ownDeletesAreVisible(int type)` will return `true`. It returns `false` if the `ResultSet` object still contains the deleted row, which means that the deletion is not visible as a change to `ResultSet` objects of the given type.

The method `DatabaseMetaData.othersDeletesAreVisible(int type)` checks whether deletions made by others (another transaction or another `ResultSet` object in the same transaction) are visible to `ResultSet` objects of the specified type. This method returns `true` if a row deleted by others is visible and `false` if it is not.

If a `ResultSet` object can detect deletions, the `ResultSet` method `rowDeleted` returns `true` when the current row has been deleted and `false` when it has not. However, `rowDeleted` also returns `false` if the `ResultSet` object cannot detect deletions. The method `DatabaseMetaData.deletesAreDetected(int type)` can be called to see whether a `ResultSet` object of the specified type can call the method `rowDeleted` to detect a deletion that is visible. The method `deletesAreDetected` returns `false` if a row deleted from the `ResultSet` object is removed from it and `true` if the deleted row is replaced by an empty or invalid row.

In CODE EXAMPLE 14-7, application code uses metadata to process a `ResultSet` object that may contain deleted rows.

```
if (dbmd.ownDeletesAreVisible(ResultSet.TYPE_SCROLL_INSENSITIVE) &&
    dbmd.deletesAreDetected(ResultSet.TYPE_SCROLL_INSENSITIVE)) {
    while (rs.next()) {
        if (rs.rowDeleted()) {
            continue;
        } else {
            // process row
            ...
        }
    }
} else {
    // if up-to-date data is needed, it is better to close this
    // ResultSet object and reexecute the query to get an updated
    // ResultSet object
    ...
    rs.close();
    break;
```

```
}
```

CODE EXAMPLE 14-7 Processing a `ResultSet` object containing deleted rows

Note – CODE EXAMPLE 14-7 does not cover the case where `ownDeletesAreVisible` returns `true` and `deletesAreDetected` returns `false`. This will cause an `SQLException` to be thrown when the cursor is positioned on a deleted row, so an implementation with these characteristics requires that an application handle the exception. Such an implementation does not appear to be a very likely.

After the method `deleteRow` has been called, the cursor will be positioned before the next valid row. If the deleted row is the last row, the cursor will be positioned after the last row.

14.2.4.3 Inserting a Row

New rows may be inserted using the `ResultSet` interface. New rows are constructed in a special *insert row*. The steps to insert a new row are:

1. Move the cursor to the insert row
2. Set the values for the columns of the row using the `ResultSet` interface update methods
3. Insert the new row into the `ResultSet` object

CODE EXAMPLE 14-8 shows the steps necessary to insert a new row into the table `booklist`.

```
// select all the columns from the table booklist
ResultSet rs = stmt.executeQuery("select author, title, isbn " +
                                "from booklist");

rs.moveToInsertRow();
// set values for each column
rs.updateString(1, "Huxley, Aldous");
rs.updateString(2, "Doors of Perception and Heaven and Hell");
rs.updateLong(3, 60900075);
// insert the row
rs.insertRow();
// move the cursor back to its position in the result set
```

```
rs.moveToCurrentRow();
```

CODE EXAMPLE 14-8 Inserting a new row into a `ResultSet` object

Each column in the insert row that does not allow null as a value and does not have a default value must be given a value using the appropriate update method. If this is not the case, the method `insertRow` will throw an `SQLException`.

The method `DatabaseMetaData.ownInsertsAreVisible(int type)` will return `true` if newly inserted rows can be seen in result sets of the specified type.

If the `ResultSet` objects of the specified type can identify newly inserted rows, the method `DatabaseMetaData.insertsAreDetected(int type)` will return `true`. This indicates that the inserted rows are visible to the `ResultSet` object.

14.2.4.4 Positioned Updates and Deletes

JDBC drivers or DBMSs that do not support performing updates via the `ResultSet` interface may support positioned updates and deletes via SQL commands. This method of updating a row relies on using named cursors to allow multiple statements to act on a single result set. CODE EXAMPLE 14-9 shows the use of the method `setCursorName` to associate a cursor with a `Statement` object and then the use of the method `getCursorName` to retrieve the name for use by a second `Statement` object.

```
Statement stmt1 = conn.createStatement();
stmt1.setCursorName("CURSOR1");
ResultSet rs = stmt1.executeQuery("select author, title, isbn " +
    "from booklist for update of author");
// move to the row we want to update
while ( ... ) {
    rs.next()
}
String cursorName = rs.getCursorName();
Statement stmt2 = conn.createStatement();
// now update the row
int updateCount = stmt2.executeUpdate("update booklist " +
    "set author = 'Zamyatin, Evgenii Ivanovich' " +
    "where current of " + cursorName);
```

CODE EXAMPLE 14-9 Updating a row using positioned updates

The syntax of both the select statement and the update statement may vary among driver or DBMS implementations.

The method `DatabaseMetaData.supportsPositionedUpdates` returns `true` if the JDBC driver and DBMS support this facility.

14.2.5 Closing a `ResultSet` Object

A `ResultSet` object is automatically closed when the `Statement` object that produced it is closed. The method `close` can be called explicitly to close a `ResultSet` object, thereby releasing any external resources and making it immediately available for garbage collection.

Batch Updates

The batch update facility allows multiple update operations to be submitted to a data source for processing at once. Submitting multiple updates together, instead of individually, can greatly improve performance. `Statement`, `PreparedStatement`, and `CallableStatement` objects can be used to submit batch updates.

15.1 Description of Batch Updates

15.1.1 Statements

The batch update facility allows a `Statement` object to submit a set of heterogeneous update commands together as a single unit, or batch, to the underlying data source.

Since the JDBC 2.0 API, a `Statement` object has had the ability to keep track of a list of commands—or batch—that can be submitted together for execution. When a `Statement` object is created, its associated batch is empty. An application adds commands to a statement's batch one at a time by calling the method `Statement.addBatch` and providing it with the SQL update command to be added. All of the commands added to a batch must be statements that return an update count.

If an application decides not to submit a batch of updates that has been constructed for a statement, it can call the method `Statement.clearBatch` to clear the batch of all commands.

In CODE EXAMPLE 15-1, all of the update operations required to insert a new employee into a fictitious company database are submitted as a single batch.

```
// turn off autocommit
con.setAutoCommit(false);
Statement stmt = con.createStatement();
stmt.addBatch("INSERT INTO employees VALUES (1000, 'Joe Jones')");
stmt.addBatch("INSERT INTO departments VALUES (260, 'Shoe')");
stmt.addBatch("INSERT INTO emp_dept VALUES (1000, 260)");

// submit a batch of update commands for execution
int[] updateCounts = stmt.executeBatch();
```

CODE EXAMPLE 15-1 Creating and executing a batch of insert statements

In the example, auto-commit mode is disabled to prevent the driver from committing the transaction when `Statement.executeBatch` is called. Disabling auto-commit allows an application to decide whether or not to commit the transaction in the event that an error occurs and some of the commands in a batch cannot be processed successfully. For this reason, auto-commit should always be turned off when batch updates are done. The commit behaviour of `executeBatch` is always implementation-defined when an error occurs and auto-commit is `true`.

It is not possible to set a savepoint "within" a batch of statements to enable partial recovery. If a savepoint is set any time before the method `executeBatch` is called, it is set before any of the statements that have been added to the batch are executed.

Although the focus in this section is on using `Statement` objects to do batch updates, the discussion that follows applies to `PreparedStatement` and `CallableStatement` objects as well.

15.1.2 Successful Execution

The `Statement.executeBatch` method submits a statement's batch to the underlying data source for execution. Batch commands are executed serially (at least logically) in the order in which they were added to the batch. When all of the commands in a batch execute successfully, the method `executeBatch` returns an integer array containing one entry for each command in the batch.

The entries in the array are ordered according to the order in which the commands were processed (which, again, is the same as the order in which the commands were originally added to the batch). When all of the commands in a batch have been executed successfully, an entry in the array of update counts may have the following values :

- 0 or greater — the command was processed successfully and the value is an update count indicating the number of rows in the database that were affected by the command's execution
- `Statement.SUCCESS_NO_INFO` — the command was processed successfully, but the number of rows affected is unknown

Calling the method `executeBatch` closes the calling `Statement` object's current result set if one is open. The statement's batch is reset to empty once `executeBatch` returns. The behaviour of the methods `executeQuery`, `executeUpdate`, and `execute` is implementation-defined when a statement's batch is non-empty.

Only DDL and DML commands that return a simple update count may be executed as part of a batch. The method `executeBatch` throws a `BatchUpdateException` if any of the commands in the batch fail to execute properly or if a command attempts to return a result set. When a `BatchUpdateException` is thrown, an application can call the `BatchUpdateException.getUpdateCounts` method to obtain an integer array of update counts that describes the outcome of the batch execution.

15.1.3 Handling Failures during Execution

A JDBC driver may or may not continue processing the remaining commands in a batch once execution of a command fails. However, a JDBC driver must always provide the same behaviour with a particular data source. For example, a driver cannot continue processing after a failure for one batch and not continue processing for another batch.

If a driver stops processing after the first failure, the array returned by the method `BatchUpdateException.getUpdateCounts` will always contain fewer entries than there were statements in the batch. Since statements are executed in the order that they are added to the batch, if the array contains *N* elements, this means that the first *N* elements in the batch were processed successfully when `executeBatch` was called.

When a driver continues processing in the presence of failures, the number of elements in the array returned by the method `BatchUpdateException.getUpdateCounts` always equals the number of commands in the batch. When a `BatchUpdateException` object is thrown and the driver continues processing after a failure, the array of update counts will contain the following `BatchUpdateException` constant:

- `Statement.EXECUTE_FAILED` — the command failed to execute successfully. This value is also returned for commands that could not be processed for some reason—such commands fail implicitly.

JDBC drivers that do not continue processing after a failure never return `Statement.EXECUTE_FAILED` in an update count array. Drivers of this type simply return a status array containing an entry for each command that was processed successfully.

A JDBC technology-based application can distinguish a JDBC driver that continues processing after a failure from one that does not by examining the size of the array returned by `BatchUpdateException.getUpdateCounts`. A JDBC driver that continues processing always returns an array containing one entry for each element in the batch. A JDBC driver that does not continue processing after a failure will always return an array whose number of entries is less than the number of commands in the batch.

15.1.4 PreparedStatement Objects

When a `PreparedStatement` object is used, a command in a batch consists of a parameterized SQL statement and an associated set of parameters. The batch update facility is used with a `PreparedStatement` object to associate multiple sets of input parameter values with a single `PreparedStatement` object. The sets of parameter values together with their associated parameterized update commands can then be sent to the underlying data source engine for execution as a single unit.

CODE EXAMPLE 15-2 inserts two new employee records into a database as a single batch. The `PreparedStatement` interface setter methods are used to create each parameter set, one for each employee. The `PreparedStatement.addBatch` method adds a set of parameters to the current command.

```
// turn off autocommit
con.setAutoCommit(false);

PreparedStatement stmt = con.prepareStatement(
    "INSERT INTO employees VALUES (?, ?)");

stmt.setInt(1, 2000);
stmt.setString(2, "Kelly Kaufmann");
stmt.addBatch();

stmt.setInt(1, 3000);
stmt.setString(2, "Bill Barnes");
stmt.addBatch();
```

```
// submit the batch for execution
int[] updateCounts = stmt.executeBatch();
```

CODE EXAMPLE 15-2 Creating and executing a batch of prepared statements

Finally, the method `PreparedStatement.executeBatch` is called to submit the updates to the underlying data source. Calling this method clears the statement's associated list of commands. The array returned by `PreparedStatement.executeBatch` contains an element for each set of parameters in the batch, similar to the case for `Statement` objects. Each element contains either an update count or the generic 'success' indicator `SUCCESS_NO_INFO`.

Error handling in the case of `PreparedStatement` objects is the same as error handling in the case of `Statement` objects. Some drivers may stop processing as soon as an error occurs, while others may continue processing the rest of the batch.

As with `Statement` objects, the number of elements in the array returned by `BatchUpdateException.getUpdateCounts` indicates whether or not the driver continues processing after a failure. The same three array element values are possible: 0 or higher, `Statement.SUCCESS_NO_INFO`, or `Statement.EXECUTE_FAILED`. The order of the entries in the array is the same order as the order in which commands were added to the batch.

15.1.5 CallableStatement Objects

The batch update facility works the same with `CallableStatement` objects as it does with `PreparedStatement` objects. Multiple sets of input parameter values may be associated with a `CallableStatement` object and sent to the underlying data source together.

Stored procedures invoked using the batch update facility with a callable statement must return a maximum of one update count. If no update count is returned, the array element value will be `Statement.SUCCESS_NO_INFO`. Additionally, a batchable stored procedure may not have OUT or INOUT parameters. The `CallableStatement.executeBatch` method throws an exception if this restriction is violated. Error handling is analogous to that for `PreparedStatement` objects.

Advanced Data Types

Chapter 16 “Advanced Data Types” and Chapter 17 “Customized Type Mapping” discuss additions to the JDBC API that allow an application written in the Java programming language to access SQL99 data types, such as binary large objects and structured types. If a data source does not support an advanced data type described in these two chapters, a driver for that data source is not required to implement the methods and interfaces associated with that data type.

16.1 Taxonomy of SQL Types

The latest version of the ANSI/ISO SQL standard is commonly referred to as *SQL99*. The JDBC API incorporates a model of the new SQL99 data types that includes only those properties that are essential to exchanging data between a database and an application written in the Java programming language.

SQL99 specifies these data types:

- SQL92 built-in types—the familiar SQL ‘column types’
 - CHAR
 - FLOAT
 - DATE
 - and so on
- New built-in types — new types added by SQL99
 - BOOLEAN — a truth value
 - BLOB — a Binary Large Object
 - CLOB — a Character Large Object
- User Defined Types
 - Structured type — a user-defined type; for example:

```
CREATE TYPE PLANE_POINT AS (X FLOAT, Y FLOAT) NOT FINAL
```

- **DISTINCT type** — a user-defined type based on a built-in type; for example:

```
CREATE TYPE MONEY AS NUMERIC(10,2) FINAL
```

- **Constructed types** — new types based on a given base type
 - **REF(structured-type)** — a pointer that persistently denotes an instance of a structured type that resides in the database
 - **base-type ARRAY[n]** — an array of n base-type elements
- **Locators** — new entities that are logical pointers to data that resides on the database server. A **LOCATOR** exists in the client environment and is a transient, logical pointer to data on the server. A locator typically refers to data that is too large to materialize on the client, such as images or audio. There are operators defined at the SQL level to retrieve random-access pieces of the data denoted by the locator.
 - **LOCATOR(structured-type)** — locator to a structured instance in server
 - **LOCATOR(array)** — locator to an array in server
 - **LOCATOR(blob)** — locator to a Binary Large Object in server
 - **LOCATOR(clob)** — locator to a Character Large Object in server
- **Type for managing data external to the data source**
 - **Datalink** — a reference to data external to the data source that is managed by the data source. At the time of this writing, **Datalink** values are being standardized as part of SQL MED (Management of External Data), a part of the SQL ANSI/ISO standard specification. Having the data source manage the reference to external data has several advantages:
 - i. **Referential integrity** — the referenced data can no longer be deleted or renamed directly through file system APIs
 - ii. **Access control** — access to the data may be configured such that it is controlled by the data source instead of the file system
 - iii. **Coordinated backup and recovery** — data referenced by **Datalink** values may be included in the data source's backup process
 - iv. **Transaction consistency** — changes that affect both relational and external data are executed in a transactional context to preserve the integrity and consistency of the data

The remainder of this chapter discusses the default mechanism provided by the JDBC API for accessing each of the new SQL data types mentioned above. The JDBC API also provides a means of customizing the mapping of SQL **DISTINCT** and structured types into Java classes. This mechanism is discussed in Chapter 17 “Customized Type Mapping”.

16.2 Mapping of SQL99 Types

The JDBC API provides default mappings for the new SQL99 types. Except for the `DISTINCT` and `DATALINK` types, these default mappings take the form of interfaces. The following list gives the SQL99 types and the interfaces to which they are mapped.

- `BLOB` — the `Blob` interface
- `CLOB` — the `Clob` interface
- `ARRAY` — the `Array` interface
- Structured types — the `Struct` interface
- `REF(structured type)` — the `Ref` interface

The other SQL99 data types with default mappings to the Java programming language are:

- `DISTINCT` — the type to which the base type is mapped. For example, a `DISTINCT` value based on an SQL `NUMERIC` type maps to a `java.math.BigDecimal` type because `NUMERIC` maps to `BigDecimal` in the Java programming language.
- `DATALINK` — a `java.net.URL` object.

16.3 Blob and Clob Objects

16.3.1 Retrieving BLOB and CLOB Values

The binary large object (`BLOB`) and character large object (`CLOB`) data types are treated similarly to the more primitive built-in types. Values of these types can be retrieved by calling the `getBlob` and `getClob` methods in the `ResultSet` and `CallableStatement` interfaces. For example, `CODE EXAMPLE 16-1` retrieves a `BLOB` value from the first column of the `ResultSet` `rs` and a `CLOB` value from the second column.

```
Blob blob = rs.getBlob(1);  
Clob clob = rs.getClob(2);
```

CODE EXAMPLE 16-1 Retrieving `BLOB` and `CLOB` values

The `Blob` interface contains operations for returning the length of the BLOB value, a specific range of bytes contained in the BLOB value, and so on. The `Clob` interface contains corresponding operations that are character based. The API documentation gives more details.

An application does not deal directly with the `LOCATOR(blob)` and `LOCATOR(clob)` types that are defined in SQL. By default, a JDBC driver should implement the `Blob` and `Clob` interfaces using the appropriate locator type. Also by default, `Blob` and `Clob` objects remain valid only during the transaction in which they are created.

16.3.2 Storing Blob and Clob Objects

A `Blob` or `Clob` object can be passed as an input parameter to a `PreparedStatement` object just like other data types. The method `setBlob` sets a `PreparedStatement` parameter with a `Blob` object, and the method `setClob` sets a `Clob` object as a parameter. In CODE EXAMPLE 16-2, *authorImage* is an instance of `java.sql.Blob` retrieved from another SQL statement, and *authorBio* is an instance of `java.sql.Clob` retrieved from another SQL statement.

```
PreparedStatement pstmt = conn.prepareStatement(
    "INSERT INTO bio (image, text) VALUES (?, ?)");
pstmt.setBlob(1, authorImage);
pstmt.setClob(2, authorBio);
```

CODE EXAMPLE 16-2 Setting `Blob` and `Clob` objects as parameters to a `PreparedStatement` object

The `setBinaryStream` and `setObject` methods may also be used to set a `Blob` object as a parameter in a `PreparedStatement` object. The `setAsciiStream`, `setCharacterStream`, and `setObject` methods are alternate means of setting a `Clob` object as a parameter.

The `updateBlob` and `updateClob` methods can be used to update a column value in an updatable result set.

16.3.3 Altering Blob and Clob Objects

The `Blob` and `Clob` interfaces provide methods to alter their internal content. In CODE EXAMPLE 16-3, the method `setBytes` is used to write the first five bytes of the `Blob` object retrieved from the column `DATA`.

```

byte[] val = {0,1,2,3,4};
...
Blob data = rs.getBlob("DATA");
int numWritten = data.setBytes(1, val);
if (dbmd.locatorsUpdateCopy() == true) {
    PreparedStatement ps = conn.prepareStatement(
        "UPDATE datatab SET data = ?");
    ps.setBlob("DATA", data);
    ps.executeUpdate();
}

```

CODE EXAMPLE 16-3 Writing bytes to a Blob object

Similarly, the Clob methods `setString` and `truncate` can be used to change the value of a column containing a Clob object.

The semantics of updates made to LOB objects are implementation defined. In some implementations, the changes may be made to a copy of the LOB, and in others the changes are made directly to the LOB. In implementations where the changes are made to a copy of the LOB, a separate update statement must be issued to update the LOB stored in the DBMS.

The method `locatorsUpdateCopy` in the `DatabaseMetaData` interface returns `true` if the implementation updates a copy of the LOB and `false` if updates are made directly to the LOB. CODE EXAMPLE 16-3 shows a typical use of the `locatorsUpdateCopy` method.

16.4 Array Objects

16.4.1 Retrieving Array Objects

Data of type SQL ARRAY can be retrieved by calling the `getArray` method of the `ResultSet` and `CallableStatement` interfaces. For example, the following line of code retrieves an Array value from the first column of the `ResultSet` *rs*.

```
Array a = rs.getArray(1);
```

By default, a JDBC driver should implement the `Array` interface using an SQL `LOCATOR(array)` internally. Also by default, `Array` objects remain valid only during the transaction in which they are created.

The `Array` object returned to an application by the `ResultSet.getArray` and `CallableStatement.getArray` methods is a logical pointer to the SQL `ARRAY` value in the database; it does not contain the contents of the SQL `ARRAY` value. The `Array` interface provides several versions of the methods `getArray` and `getResultSet` that return the contents of an SQL `ARRAY` value to the client as a materialized Java programming language array ("Java array") or as a `ResultSet` object. The API documentation gives complete details.

16.4.2 Storing Array Objects

The `PreparedStatement` methods `setArray` and `setObject` may be called to pass an `Array` value as an input parameter to a `PreparedStatement` object. CODE EXAMPLE 16-4 sets the `Array` object *member_array*, which was retrieved from another table in the database, as the second parameter to the `PreparedStatement pstmt`.

```
PreparedStatement pstmt = conn.prepareStatement(
    "INSERT INTO dept (name, members) VALUES (?, ?)");
pstmt.setString(1, "biology");
pstmt.setArray(2, member_array);
pstmt.executeUpdate();
```

CODE EXAMPLE 16-4 Storing an `Array` object

A Java array may be passed as an input parameter by calling the method `PreparedStatement.setObject`.

16.4.3 Updating Array Objects

The `ResultSet` methods `updateArray` and `updateObject` can be used to update a column value.

CODE EXAMPLE 16-5 uses the method `ResultSet.updateArray` to update the value of the column `LATEST_NUMBERS` in one `ResultSet` object with an `Array` object retrieved from the column `NUMBERS` in another `ResultSet` object.

```
// retrieve a column containing an SQL ARRAY value from ResultSet rs
java.sql.Array num = rs.getArray("NUMBERS");
...
// update the column "LATEST_NUMBERS" in a second ResultSet
// with the value retrieved...
rs2.updateArray("LATEST_NUMBERS", num);
rs2.updateRow();
```

CODE EXAMPLE 16-5 Updating a column with an Array object

16.5 Ref Objects

16.5.1 Retrieving REF Values

An SQL REF(structured type) value can be retrieved as a Ref object by calling the `getRef` method of the `ResultSet` and `CallableStatement` interfaces. For example, in CODE EXAMPLE 16-6, the `ResultSet` *rs* contains a reference to an instance of the SQL structured type `dog` that is stored in the table `dogs`. The code retrieves this REF(`dog`) from the first column of *rs*.

```
ResultSet rs = stmt.executeQuery("SELECT oid FROM dogs WHERE " +
                                "name = rover");

rs.next();
Ref ref = rs.getRef(1);
```

CODE EXAMPLE 16-6 Retrieving a REF value

An SQL REF value is a pointer; therefore, a `Ref` object, which is the mapping of a REF value, is likewise a pointer and does not contain the data of the structured type instance to which it refers. A `Ref` object remains valid while the session or connection on which it is created is open.

An SQL REF value may either be system-generated or user-generated. For user-generated references, the representation of the reference values is based on a predefined data type, such as `INTEGER`, and the reference values can be directly inspected or generated by an end user or application. A vendor supporting user-generated references may, in addition to the functionality based on the `Ref` object, also support the retrieval and storage of reference values by using the getter and setter methods that are appropriate to the representation type of the reference.

16.5.2 Retrieving the Referenced Value

The `Ref` object returned from the method `getRef` is a reference to an instance of a structured type in the underlying data source. The methods `getObject()` and `getObject(Map map)` can be used to retrieve the structured type instance that is referenced. CODE EXAMPLE 16-7 shows how a reference to an instance of the structured type `Address` can be dereferenced to retrieve the instance of `Address`. This example would require that a map, mapping `Address` to its SQL type, had been supplied to the `Connection` interface using the method `setMap`.

```
Ref ref = rs.getRef(1);
Address addr = (Address)ref.getObject();
```

CODE EXAMPLE 16-7 Retrieving the structured type instance referenced by a `Ref` object

16.5.3 Storing Ref Objects

The `PreparedStatement.setRef` method may be called to pass a `Ref` object as an input parameter to a `PreparedStatement` object.

16.5.4 Storing the Referenced Value

An instance of a structured type retrieved with the method `ResultSet.getRef` or `CallableStatement.getRef` is stored using the `Ref.setObject` method. In CODE EXAMPLE 16-8, the table `DOGS` stores instances of the structured type `DOG`. The `SELECT` statement selects the `REF(DOG)` that refers to the instance in which the name is Rover. The referenced instance of the type `DOG` is retrieved using `Ref.getObject`. The parameter `map` describes a mapping from the SQL type `DOG` to the Java class `Dog`, which implements the `SQLData` interface.

```
ResultSet rs = stmt.executeQuery("SELECT OID FROM DOGS " +
                                "WHERE NAME = 'ROVER'");

rs.next();
Ref rover = rs.getRef("OID");
Dog dog = (Dog)rover.getObject(map);
// manipulate instance of Dog
dog.setAge(14);
...
// store updated Dog
rover.setObject((Object)dog);
```

CODE EXAMPLE 16-8 Retrieving and storing the structured type instance referenced by a Ref object

16.5.5 Metadata

The type REF is defined in the class `java.sql.Types`. This value is returned by methods such as `DatabaseMetaData.getTypeInfo` and `DatabaseMetaData.getColumns` when a JDBC driver supports the Ref data type.

16.6 Distinct Types

An SQL DISTINCT type is a new user-defined data type that is based on one of the primitive types. C and C++ programmers can think of it as being similar to a typedef.

16.6.1 Retrieving Distinct Types

By default, a column of SQL type DISTINCT is retrieved by calling any getter method that is appropriate to the type on which it is based. For example, the following type declaration creates the type MONEY, which is based on the SQL type NUMERIC.

```
CREATE TYPE MONEY AS NUMERIC(10,2) FINAL
```

CODE EXAMPLE 16-9 Creating a distinct type

CODE EXAMPLE 16-10 uses the method `getBigDecimal` to retrieve a MONEY value because the underlying SQL NUMERIC type is mapped to the `java.math.BigDecimal` type.

```
java.math.BigDecimal bd = rs.getBigDecimal(1);
```

CODE EXAMPLE 16-10 Retrieving a distinct type

16.6.2 Storing Distinct Types

Any setter method in the `PreparedStatement` interface that is appropriate for the base type of an SQL `DISTINCT` type may be used to pass an input parameter of that distinct type to a prepared statement. For example, given the definition of type `MONEY` in `CODE EXAMPLE 16-9`, the method `PreparedStatement.setBigDecimal` would be used.

16.6.3 Metadata

The type code `DISTINCT` is defined in the class `java.sql.Types`. This value is returned by methods such as `DatabaseMetaData.getTypeInfo` and `DatabaseMetaData.getColumns` when a JDBC driver supports this data type.

An SQL `DISTINCT` type must be defined as part of a particular database schema before it can be used in a schema table definition. Information on schema-specific user-defined types—of which `DISTINCT` types are one particular kind—can be retrieved by calling the `DatabaseMetaData.getUDTs` method. For example, `CODE EXAMPLE 16-11` returns descriptions of all the SQL `DISTINCT` types defined in the `catalog-name.schema-name` schema. If the driver does not support UDTs or no matching UDTs are found, the `getUDTs` method returns an empty result set.

```
int[] types = {Types.DISTINCT};
ResultSet rs = dmd.getUDTs("catalog-name", "schema-name",
                           "%", types);
```

CODE EXAMPLE 16-11 Querying a `DatabaseMetaData` object for distinct types

Each row in the `ResultSet` object returned by the method `getUDTs` describes a UDT. Each row contains the following columns:

TYPE_CAT	String => the type's catalog (may be null)
TYPE_SCHEM	String => the type's schema (may be null)
TYPE_NAME	String => the SQL type name
CLASS_NAME	String => a Java class name
DATA_TYPE	short => value defined in <code>java.sql.Types</code> , such as <code>DISTINCT</code>
REMARKS	String => explanatory comment on the type
BASE_TYPE	short => value defined in <code>java.sql.Types</code> , for <code>DISTINCT</code> or reference types (may be null)

Most of the columns above should be self-explanatory. The `TYPE_NAME` is the SQL type name given to the `DISTINCT` type—`MONEY` in the example above. This is the name used in a `CREATE TABLE` statement to specify a column of this type.

When `DATA_TYPE` is `Types.DISTINCT`, the `CLASS_NAME` column contains a fully qualified Java class name. Instances of this class will be created if `getObject` is called on a column of this `DISTINCT` type. For example, `CLASS_NAME` would default to `java.math.BigDecimal` in the case of `MONEY` above. The JDBC API does not prohibit a driver from returning a subtype of the class named by `CLASS_NAME`. The `CLASS_NAME` value reflects a custom type mapping when one is used. See Chapter 17 “Customized Type Mapping” for details.

16.7 Structured Types

16.7.1 Retrieving Structured Types

An SQL structured type value is always retrieved by calling the method `getObject`. By default, `getObject` returns a value of type `Struct` for a structured type. For example, the following line of code retrieves a `Struct` value from the first column of the current row of the `ResultSet` object `rs`.

```
Struct struct = (Struct)rs.getObject(1);
```

The `Struct` interface contains methods for retrieving the attributes of a structured type as an array of `java.lang.Object` values. By default, a JDBC driver materializes the contents of a `Struct` prior to returning a reference to it to the application. Also, by default a `Struct` object is considered valid as long as the Java application maintains a reference to it.

16.7.2 Storing Structured Types

The `PreparedStatement.setObject` method may be called to pass a `Struct` object as an input parameter to a prepared statement.

16.7.3 Metadata

The type code `STRUCT` is defined in the class `java.sql.Types`. This value is returned by methods such as `DatabaseMetaData.getTypeInfo` and `DatabaseMetaData.getColumns` when a JDBC driver supports structured data types.

An SQL structured type must be defined as part of a particular database schema before it can be used in a schema table definition. Information on schema-specific user-defined types—of which `STRUCT` types are one particular kind—can be retrieved by calling the `DatabaseMetaData.getUDTs` method. For example, `CODE EXAMPLE 16-1` returns descriptions of all the SQL structured types defined in the `catalog-name.schema-name` schema.

```
int[] types = {Types.STRUCT};  
ResultSet rs = dmd.getUDTs("catalog-name", "schema-name",  
                           "%", types);
```

CODE EXAMPLE 16-12 Querying a `DatabaseMetaData` object for structured types

If the driver does not support UDTs or no matching UDTs are found, an empty result set is returned. See section 16.6.3 for a description of the result set returned by the method `getUDTs`.

When the `DATA_TYPE` returned by `getUDTs` is `Types.STRUCT`, the `CLASS_NAME` column contains the fully qualified Java class name of a Java class. Instances of this class are manufactured by the JDBC driver when `getObject` is called on a column of this `STRUCT` type. Thus, `CLASS_NAME` defaults to `java.sql.Struct` for structured types. If there is a custom mapping for the `STRUCT` type, `CLASS_NAME` will be the implementation of the interface `SQLData` that specifies the mapping. The JDBC API does not prohibit a driver from returning a subtype of the class named by `CLASS_NAME`. Chapter 17 “Customized Type Mapping” provides more information about implementations of the `SQLData` interface.

16.8 Datalinks

A `DATALINK` value references a file outside of the underlying data source that the data source manages.

16.8.1 Retrieving References to External Data

A reference to external data being managed by the data source can be retrieved using the method `ResultSet.getURL`. The `java.net.URL` object that is returned can be used to manipulate the data.

```
java.net.URL url = rs.getURL(1);
```

CODE EXAMPLE 16-13 Retrieving a reference to an external data object

In cases where the type of URL returned by the methods `getObject` or `getURL` is not supported by the Java platform, the URL can be retrieved as a `String` by calling the method `getString`.

16.8.2 Storing References to External Data

The method `PreparedStatement.setURL` can be used to pass a `java.net.URL` object to a prepared statement. In cases where the type of URL being set is not supported by the Java platform, the URL can be stored using the `setString` method.

16.8.3 Metadata

The type code `DATALINK` is defined in the class `java.sql.Types`. This value is returned by methods such as `DatabaseMetaData.getTypeInfo` and `DatabaseMetaData.getColumns` when a JDBC driver supports the `Datalink` data type or references to external files.

Customized Type Mapping

This chapter describes the support that the JDBC API provides for mapping SQL structured and distinct types to classes in the Java programming language. Because the mechanism for this custom mapping is an extension of the existing `getObject` and `setObject` mechanism, it involves minimal extensions to the JDBC API from the user's point of view.

17.1 The Type Mapping

The SQL user-defined types (UDTs), structured types and `DISTINCT` types, can be given a custom mapping to a class in the Java programming language. The default is for a driver to use the default mappings between SQL data types and types in the Java programming language. The default mapping for an SQL structured type is to the interface `Struct`; the default mapping for an SQL `DISTINCT` type is to the type to which the underlying type is mapped. If a custom mapping has been set up for a UDT, the driver will use the custom mapping instead of the default mapping when an application calls the `getObject` or `setObject` methods on that UDT.

Setting up a custom mapping requires two things:

1. Writing an implementation of the `SQLData` interface for the UDT. This class typically maps the attribute(s) of an SQL structured type (or the single attribute of a `DISTINCT` type) to fields. There is, however, great latitude allowed in how a UDT is custom mapped. It is expected that most `SQLData` implementations will be created using a tool.
2. Putting an entry in a `java.util.Map` object. The entry must contain the following two items:
 - a. The fully qualified name of the SQL UDT that is to be mapped.

- b. The `Class` object for the `SQLData` implementation. It is an error if the class listed in a type map entry does not implement the `SQLData` interface.

For example, if the UDT is named `mySchemaName.AUTHORS` and the `SQLData` implementation is the class `Authors`, the entry for the type map associated with the `Connection` object `conn` would look like CODE EXAMPLE 17-1.

```
java.util.Map map = conn.getTypeMap();
map.put("mySchemaName.AUTHORS", Class.forName("Authors"));
conn.setTypeMap(map);
```

CODE EXAMPLE 17-1 Putting an entry in a connection's type map

The method `Connection.getTypeMap` returns the type map associated with the `Connection` object `conn`; the method `Connection.setTypeMap` sets the given `java.util.Map` object as the type map for `conn`.

When an SQL value with a custom mapping is being retrieved (by the method `ResultSet.getObject`, `CallableStatement.getObject`, or any of the other methods that materialize an SQL value's data on the client), the driver will check to see if there is an entry in the connection's type map for the SQL value that is to be retrieved. If there is, the driver will map the SQL UDT to the class specified in the type map. If there is no entry for the UDT in the connection's type map, the UDT is mapped to the default mapping.

Certain methods may take a type map as a parameter. A type map supplied as a parameter supersedes the type map associated with the connection. A UDT that does not have an entry in the type map supplied as a parameter will be mapped to the default mapping. When a type map is explicitly supplied to a method, the connection's type map is never used.

17.2 Class Conventions

A class that appears in a type map entry must do the following:

1. Implement the interface `java.sql.SQLData`
2. Provide a no-arg constructor, that is, a constructor that takes no parameters

The `SQLData` interface contains methods that convert instances of SQL UDTs to Java class instances and that convert Java class instances back to SQL UDTs. For example, the method `SQLData.readSQL` reads a stream of data values and builds a Java

object, while the method `SQLData.writeSQL` writes a sequence of values from a Java object to a stream. These methods will typically be generated by a tool that understands the database schema.

This stream-based approach for exchanging data between SQL and the Java programming language is conceptually similar to Java object serialization. The data are read from and written to an SQL data stream provided by the JDBC driver. The SQL data stream may be implemented on various network protocols and data formats. It may be implemented on any logical data representation in which the leaf SQL data items (of which SQL structured types are composed) can be read from (written to) the data stream in a "depth-first" traversal of the structured types. That is, each attribute value, which may itself be a structured type, appears fully (its structure recursively elaborated) in the stream before the next attribute. In addition, the attributes of an SQL structured type must appear in the stream in the order in which they are declared in the type definition. For data of SQL structured types that use inheritance, the attributes must appear in the stream in the order that they are inherited. That is, the attributes of a supertype must appear before attributes of a subtype.

If multiple inheritance is used, then the attributes of supertypes should appear in the stream in the order in which the supertypes are listed in the type declaration. This protocol does not require the database server to have any knowledge of the Java programming language. However, as there is no support for multiple inheritance in the SQL99 specification, this issue should not arise.

17.3 Streams of SQL Data

This section describes the stream interfaces, `SQLInput` and `SQLOutput`, which support customization of the mapping of SQL UDTs to Java data types.

17.3.1 Retrieving Data

In a custom mapping, when data of SQL structured and distinct types are retrieved from the database, they "arrive" in a stream implementing the `SQLInput` interface. The `SQLInput` interface contains methods for reading individual data values sequentially from the stream. CODE EXAMPLE 17-2 illustrates how a driver can use an `SQLInput` stream to provide values for the fields of an `SQLData` object. The `SQLData` object—the `this` object in the example—contains three persistent fields: the `String` `str`, the `Blob` object `blob`, and the `Employee` object `emp`.

```
SQLInput sqlin;  
...
```

```
this.str = sqlin.readString();
this.blob = sqlin.readBlob();
this.emp = (Employee)sqlin.readObject();
```

CODE EXAMPLE 17-2 Retrieving data using the `SQLInput` interface

The `SQLInput.readString` method reads a `String` value from the stream; the `SQLInput.readBlob` method reads a `Blob` value from the stream. By default, the `Blob` interface is implemented using an SQL locator, so calling the method `readBlob` doesn't materialize the SQL BLOB contents on the client. The `SQLInput.readObject` method retrieves an object reference from the stream. In the example, the `Object` returned is narrowed to an `Employee` object.

There are a number of additional methods defined on the `SQLInput` interface for reading each of the types (`readLong`, `readBytes`, and so on). The `SQLInput.wasNull` method can be called to check whether the last value read was SQL NULL in the database.

17.3.2 Storing Data

When an instance of a class that implements `SQLData` is passed to a driver as an input parameter via the `setObject` method, the JDBC driver calls the object's `SQLData.writeSQL` method. It also creates an `SQLOutput` stream to which the method `writeSQL` writes the attributes of the custom mapped UDT. The method `writeSQL` will typically have been generated by a tool from an SQL type definition. **CODE EXAMPLE 17-3** illustrates the use of the `SQLOutput` object `sqlout`.

```
sqlout.writeString(this.str);
sqlout.writeBlob(this.blob);
sqlout.writeObject(this.emp);
```

CODE EXAMPLE 17-3 Storing data using the `SQLOutput` interface

The example shows how the contents of an `SQLData` object can be written to an `SQLOutput` stream. The `SQLData` object—the `this` object in the example—contains three persistent fields: the `String` `str`, the `Blob` object `blob`, and the `Employee` object `emp`. Each field is written in turn to the `SQLOutput` stream, `sqlout`. The `SQLOutput` interface contains methods for writing each of the types defined in the JDBC API.

17.4 Examples

This section gives examples of SQL code as well as code in the Java programming language. SQL code is used for creating structured types, creating tables for instances of those types, populating the tables with instances of the structured types, and creating an SQL DISTINCT type. This code sets up the SQL values that will be mapped to classes in the Java programming language.

The examples of code in the Java programming language create implementations of the `SQLData` interface for the newly created SQL UDTs and also show how a class in the Java programming language can mirror SQL inheritance for structured types.

17.4.1 An SQL Structured Type

CODE EXAMPLE 17-4, which defines the structured types `PERSON`, `FULLNAME`, and `RESIDENCE`, shows that it is possible for an attribute to be a REF value or another structured type. `PERSON` and `RESIDENCE` each have an attribute that is a REF value, and the REF value in one structured type references the other structured type. Note also that `FULLNAME` is used as an attribute of `PERSON`.

```
CREATE TYPE RESIDENCE AS
(
    DOOR NUMERIC(6),
    STREET VARCHAR(100),
    CITY VARCHAR(50),
    OCCUPANT REF(PERSON)
) NOT FINAL
```

```
CREATE TYPE FULLNAME AS
(
    FIRST VARCHAR(50),
    LAST VARCHAR(50)
) NOT FINAL
```

```
CREATE TYPE PERSON AS
(
    NAME FULLNAME,
```

```

        HEIGHT NUMERIC,
        WEIGHT NUMERIC,
        HOME REF(RESIDENCE)
    ) NOT FINAL

```

CODE EXAMPLE 17-4 Creating SQL structured types

The types created in CODE EXAMPLE 17-4 are presumed to be created in the current schema for the following examples.

CODE EXAMPLE 17-5 creates two tables that are maintained by the DBMS automatically. The CREATE statements do two things:

1. Create tables that store instances of the structured types named in the **OF** part of the statement (**RESIDENCE** in the first one, **PERSON** in the second). Each of the subsequent **INSERT INTO** statements adds a new row representing an instance of the UDT.
2. Create a **REF** value that is a pointer to each instance that is inserted into the table. As indicated in the **CREATE** statement, the **REF** value is generated by the system, which is done implicitly. Because **REF** values are stored in the table, they are persistent pointers. This contrasts with **LOCATOR** types, which are logical pointers but exist only as long as the transactions in which they are created.

```

CREATE TABLE HOMES OF RESIDENCE
    (REF IS OID SYSTEM GENERATED,
     OCCUPANT WITH OPTIONS SCOPE PEOPLE)
CREATE TABLE PEOPLE OF PERSON
    (REF IS OID SYSTEM GENERATED,
     OCCUPANT WITH OPTIONS SCOPE HOMES)

```

CODE EXAMPLE 17-5 Creating tables to store instances of a structured type

CODE EXAMPLE 17-6 uses **INSERT INTO** statements to populate the tables created in CODE EXAMPLE 17-5. For example, the **INSERT INTO PEOPLE** statement inserts an instance of the UDT **PERSON** into the table **PEOPLE**. When this command is executed, the DBMS will also automatically generate a **REF** value that is a pointer to this instance of **PERSON** and store it in the column **OID** (the column name specified in the **CREATE** statement that created the table **PEOPLE**).

Each column value in these special tables is an attribute of the UDT, which may itself be a UDT. For example, the first attribute of the UDT **PERSON** is the value in the column **NAME**, which must be an instance of the UDT **FULLNAME**. The example assumes that the UDT **FULLNAME** has an additional two-parameter constructor.

A column value may also be a reference to an SQL structured type. For example, the attribute `OCCUPANT` of the UDT `RESIDENCE` is of type `REF(PERSON)`. It takes an SQL `SELECT` statement to retrieve the `REF` value from the table `HOMES` and use it as the value for `OCCUPANT`, which is shown at the end of `CODE EXAMPLE 17-6`.

```
INSERT INTO PEOPLE (NAME, HEIGHT, WEIGHT) VALUES
(
    NEW FULLNAME('DAFFY', 'DUCK'),
    4,
    58
);

INSERT INTO HOMES (DOOR, STREET, CITY, OCCUPANT) VALUES
(
    1234,
    'CARTOON LANE',
    'LOS ANGELES',
    (SELECT OID FROM PEOPLE P WHERE P.NAME.FIRST = 'DAFFY')
)

UPDATE PEOPLE SET HOME = (SELECT OID FROM HOMES H WHERE
    H.OCCUPANT->NAME.FIRST = 'DAFFY') WHERE
    FULLNAME.FIRST = 'DAFFY'
```

CODE EXAMPLE 17-6 Populating and updating tables that store instances of structured types

17.4.2 SQLData Implementations

The Java classes defined in `CODE EXAMPLE 17-7` are mappings of the SQL structured types used in the examples in Section 17.4.1. We expect that such classes will typically be generated by a tool that reads the definitions of those structured types from the catalog tables and, subject to customizations that a user of the tool may provide for name mappings and type mappings of primitive fields, will generate Java classes like those in the example.

In each implementation of `SQLData`, the method `SQLData.readSQL` reads the attributes in the order in which they appear in the SQL definition of the structured type. Attributes are also read in "row order, depth-first" order, where the complete

structure of each attribute is read recursively before the next attribute is read. The method `SQLData.writeSQL` writes each attribute to the output stream in the same order.

```
public class Residence implements SQLData {
    public int door;
    public String street;
    public String city;
    public Ref occupant;

    private String sql_type;
    public String getSQLTypeName() { return sql_type; }

    public void readSQL (SQLInput stream, String type)
        throws SQLException {
        sql_type = type;
        door = stream.readInt();
        street = stream.readString();
        city = stream.readString();
        occupant = stream.readRef();
    }

    public void writeSQL (SQLOutput stream) throws SQLException {
        stream.writeInt(door);
        stream.writeString(street);
        stream.writeString(city);
        stream.writeRef(occupant);
    }
}

public class Fullname implements SQLData {
    public String first;
    public String last;

    private String sql_type;
    public String getSQLTypeName() { return sql_type; }
```

```

        public void readSQL (SQLInput stream, String type)
            throws SQLException {
            sql_type = type;
            first = stream.readString();
            last = stream.readString();
        }

        public void writeSQL (SQLOutput stream) throws SQLException {
            stream.writeString(first);
            stream.writeString(last);
        }
    }

    public class Person implements SQLData {
        Fullname name;
        float height;
        float weight;
        Ref home;

        private String sql_type;
        public String getSQLTypeName() { return sql_type; }

        public void readSQL (SQLInput stream, String type)
            throws SQLException {
            sql_type = type;
            name = (Fullname)stream.readObject();
            height = stream.readFloat();
            weight = stream.readFloat();
            home = stream.readRef();
        }

        public void writeSQL (SQLOutput stream)
            throws SQLException {
            stream.writeObject(name);
            stream.writeFloat(height);
            stream.writeFloat(weight);

```

```

        stream.writeRef(home);
    }
}

```

CODE EXAMPLE 17-7 Classes implementing the `SQLData` interface

CODE EXAMPLE 17-8 puts entries for custom mappings in the connection's type map. Then it retrieves the `Ref` instance stored in the `OCCUPANT` column of the table `HOMES`. This `Ref` instance is set as a parameter in the `where` clause of the query to get the name of the occupant. When the method `getObject` is called to retrieve an instance of `FULLNAME`, the driver looks in the connections type map and uses the `SQLData` implementation, `Fullname`, to custom map the `FULLNAME` value.

```

// set up mappings for the connection
try {
    java.util.Map map = con.getTypeMap();
    map.put("S.RESIDENCE", Class.forName("Residence"));
    map.put("S.FULLNAME", Class.forName("Fullname"));
    map.put("S.PERSON", Class.forName("Person"));
}
catch (ClassNotFoundException ex) {}

PreparedStatement pstmt;
ResultSet rs;
pstmt = con.prepareStatement("SELECT OCCUPANT FROM HOMES");
rs = pstmt.executeQuery();
rs.next();
Ref ref = rs.getRef(1);

pstmt = con.prepareStatement(
    "SELECT FULLNAME FROM PEOPLE WHERE OID = ?");
pstmt.setRef(1, ref);
rs = pstmt.executeQuery(); rs.next();
Fullname who = (Fullname)rs.getObject(1);

// prints "Daffy Duck"
System.out.println(who.first + " " + who.last);

```

CODE EXAMPLE 17-8 Retrieving a custom mapping

17.4.3 Mirroring SQL Inheritance in the Java Programming Language

SQL structured types may be defined to form an inheritance hierarchy. For example, consider SQL type `STUDENT`, which inherits from `PERSON`:

```
CREATE TYPE PERSON AS
    (NAME VARCHAR(20),
     BIRTH DATE)
    NOT FINAL;
CREATE TYPE STUDENT UNDER PERSON AS
    (GPA NUMERIC(4,2))
    NOT FINAL;
```

CODE EXAMPLE 17-9 Creating a hierarchy of SQL types

The following Java classes can represent data of those SQL types. Class `Student` extends `Person`, mirroring the SQL type hierarchy. Methods `SQLData.readSQL` and `SQLData.writeSQL` of the subclass cascade each call to the corresponding method in its superclass in order to read or write the superclass attributes before reading or writing the subclass attributes.

```
import java.sql.*;

...

public class Person implements SQLData {
    public String name;
    public Date birth;

    private String sql_type;
    public String getSQLTypeName() { return sql_type; }

    public void readSQL (SQLInput data, String type)
        throws SQLException {
        sql_type = type;
        name = data.readString();
        birth = data.readDate();
    }

    public void writeSQL (SQLOutput data)
```

```

        throws SQLException {
            data.writeString(name);
            data.writeDate(birth);
        }
    }

    public class Student extends Person {
        public float GPA;

        private String sql_type;
        public String getSQLTypeName() { return sql_type; }

        public void readSQL (SQLInput data, String type)
            throws SQLException {
            sql_type = type;
            super.readSQL(data, type);
            GPA = data.readFloat();
        }

        public void writeSQL (SQLOutput data)
            throws SQLException {
            super.writeSQL(data);
            data.writeFloat(GPA);
        }
    }
}

```

CODE EXAMPLE 17-10 Mirroring SQL type hierarchies in Java classes

The Java class hierarchy need not mirror the SQL inheritance hierarchy. For example, the class `Student` above could have been declared without a superclass. In this case, `Student` could contain fields to hold the inherited attributes of the SQL type `STUDENT` as well as the attributes declared by `STUDENT` itself.

17.4.4 Example Mapping of SQL DISTINCT Type

CODE EXAMPLE 17-11 illustrates creating an SQL DISTINCT type, `MONEY`, and CODE EXAMPLE 17-12 illustrates a Java class, `Money`, that represents it.

```
CREATE TYPE MONEY AS NUMERIC(10,2) FINAL;
```

CODE EXAMPLE 17-11 Creating an SQL DISTINCT type

```
public class Money implements SQLData {
    public java.math.BigDecimal value;

    private String sql_type;
    public String getSQLTypeName() { return sql_type; }

    public void readSQL (SQLInput stream, String type)
        throws SQLException {
        sql_type = type;
        value = stream.readBigDecimal();
    }

    public void writeSQL (SQLOutput stream) throws SQLException {
        stream.writeBigDecimal(value);
    }
}
```

CODE EXAMPLE 17-12 Java class that represents a DISTINCT type

17.5 Effect of Transform Groups

Transform groups (SQL99) can be used to convert a user-defined SQL type into predefined SQL types. This transformation is performed by the underlying data source before it is returned to the JDBC driver.

If transform groups are used for a user-defined type, and the application has not defined a mapping for that type to a Java class, then the `ResultSetMetaData` method `getColumnClass` should return the Java class corresponding to the data type produced by the transformation function (that is, `String` for a `VARCHAR`).

Note – This is consistent with the behaviour for `DISTINCT` types.

If transform groups are used for a UDT, and the application has defined a mapping for that type to a Java class, then the `SQLInput` stream delivered by the JDBC driver during an invocation of the method `readSQL` contains only a single value, that is, the result produced by the transformation function. The same model holds for the method `writesQL`.

17.6 Generality of the Approach

Users have great flexibility in customizing the Java classes that represent SQL structured and `DISTINCT` types. They control the mappings of built-in SQL attribute types to Java field types. They control the mappings of SQL names (of types and attributes) to Java names (of classes and fields). Users may add (to Java classes that represent SQL types) fields and methods that implement domain-specific functionality. Users can generate JavaBeans components as the classes that represent SQL types.

A user can even map a single SQL type to different Java classes, depending on arbitrary conditions. To do that, the user must customize the implementation of `SQLData.readSQL` to construct and return objects of different classes under different conditions.

Similarly, the user can map a single SQL value to a graph of Java objects. Again, that is accomplished by customizing the implementation of the method `SQLData.readSQL` to construct multiple objects and distribute the SQL attributes into fields of those objects.

A customization of the `SQLData.readSQL` method could populate a connection's type map incrementally. This flexibility will allow users to map SQL types appropriately for different kinds of applications.

17.7 NULL Data

An application uses the existing `getObject` and `setObject` mechanism to retrieve and store `SQLData` values. We note that when the second parameter, `x`, of method `PreparedStatement.setObject` has the value `null`, the driver executes the SQL statement as if the SQL literal `NULL` had appeared in its place.

```
void setObject (int i, Object x) throws SQLException;
```

When parameter `x` is `null`, there is no enforcement that the corresponding argument expression is of a Java type that could successfully be passed to that SQL statement if its value were not `null`. The Java programming language `null` carries no type information. For example, a `null` Java programming language variable of class `AntiMatter` could be passed as an argument to an SQL statement that requires a value of SQL type `MATTER`, and no error would result, even though the relevant type map object did not permit the translation of `MATTER` to `AntiMatter`.

Rowsets

A `javax.sql.RowSet` object encapsulates a set of rows that have been retrieved from a tabular data source. Because the `RowSet` interface includes an event notification mechanism and supports getting and setting properties, every `RowSet` object is a JavaBeans™ component. This means, for example, that a rowset can be used as a JavaBeans component in a visual JavaBeans development environment. As a result, a `RowSet` instance can be created and configured at design time, and its methods can be executed at run time.

18.1 Rowsets at Design Time

18.1.1 Properties

The `RowSet` interface provides a set of JavaBeans properties that allow a `RowSet` instance to be configured to connect to a data source and retrieve a set of rows.

CODE EXAMPLE 18-1 sets some properties for the `RowSet` object *rset*.

```
rset.setDataSourceName("jdbc/SomeDataSourceName");
rset.setTransactionIsolation(
    Connection.TRANSACTION_READ_COMMITTED);
rset.setCommand("SELECT NAME, BREED, AGE FROM CANINE");
```

CODE EXAMPLE 18-1 Setting properties for a `RowSet` object

The data source name property is used by a `RowSet` object to look up a `DataSource` object in a JNDI naming service. (Relational databases are the most common type of data source used by rowsets.) The `DataSource` object is used to create a connection to the physical data source that it represents. The transaction

isolation property specifies that only data that was produced by committed transactions may be read by the rowset. Lastly, the command property specifies the command that will be executed to retrieve a set of rows. In this case, the NAME, BREED, and AGE columns for all rows in the CANINE table are retrieved.

18.1.2 Events

`RowSet` components support JavaBeans events, which allows other JavaBeans components in an application to be notified when an event on a rowset occurs. A component that wishes to register for `RowSet` events must implement the `RowSetListener` interface. Event listeners are registered with a rowset by calling the `addRowSetListener` method as shown below. Any number of listeners may be registered with an individual `RowSet` object. CODE EXAMPLE 18-2 adds one listener to the `RowSet` object *rset*.

```
RowSetListener listener = ...;  
rset.addRowSetListener(listener);
```

CODE EXAMPLE 18-2 Adding a listener to a `RowSet` object

Rowsets can generate three different types of events:

1. Cursor movement events — indicate that the rowset's cursor has moved
2. Row change events — indicate that a particular row has been inserted, updated or deleted
3. Rowset change events — indicate that the entire contents of a rowset have changed, which may happen, for example, when the method `RowSet.execute` is called.

When an event occurs, the appropriate listener method is called behind the scenes to notify the registered listener(s). If a listener is not interested in a particular kind of event, it may implement the method for that event so that it does nothing. Listener methods take a `RowSetEvent` object, which identifies the `RowSet` object that is the source of the event.

18.2 Rowsets at Run Time

18.2.1 Parameters

The command property in CODE EXAMPLE 18-1 was set with a simple SQL command that takes no input parameters, but it could also have been set with a command that accepts input parameters. The `RowSet` interface provides a group of setter methods for setting these input parameters.

CODE EXAMPLE 18-3 shows a command that takes a `String` input parameter. The `RowSet.setString` method is used to pass the input parameter value to the `RowSet` object *rset*. Typically, the command property is specified at design time, whereas parameters are not set until run time when their values are known.

```
rset.setCommand("SELECT NAME, BREED, AGE FROM CANINE WHERE NAME = ?");  
rset.setString(1, "spot");
```

CODE EXAMPLE 18-3 Setting parameters in a `RowSet` object's command

18.2.2 Command Execution

A rowset may be filled with data by calling the `RowSet.execute` method. This method uses the appropriate property values internally to connect to a data source and retrieve some data. The `RowSet` interface includes the properties that are needed to connect to a data source. The exact properties that must be set may vary between `RowSet` implementations, so developers need to check the documentation for the particular rowset they are using. The method `execute` throws an `SQLException` if the necessary properties have not been set. The current contents of a rowset, if any, are lost when the method `execute` is called.

18.2.3 Traversing a Rowset

The `javax.sql.RowSet` interface extends the `java.sql.ResultSet` interface, so in many ways a rowset behaves just like a result set. In fact, most components that make use of a `RowSet` component will likely treat it as a `ResultSet` object. A `RowSet` object is simply a `ResultSet` object that can function as a `JavaBeans`

component. The code below shows how to iterate forward through a rowset. Notice that since a rowset is a result set, this code is identical to the code that would be used to iterate forward through a result set.

```
// iterate forward through the rowset
rset.beforeFirst();
while (rset.next()) {
    System.out.println(rset.getString(1) + " " + rset.getFloat(2));
}
```

CODE EXAMPLE 18-4 Printing all the rows in a two-column RowSet object

Other cursor movements, such as iterating backward through the rowset and positioning the cursor on a specific row, are also done the same way they are for result sets.

Relationship to Connectors

The J2EE Connector Architecture 1.0 Specification defines a set of contracts that allow a resource adapter to extend a container in a pluggable way. A resource adapter provides connectivity to an external system from the application server. The resource adapter's functionality is similar to that provided by the JDBC interfaces used in the J2EE platform to establish a connection with a data source. These interfaces, which the Connector specification refers to as the *service provider interface* (SPI), are the following:

- `DataSource`
- `ConnectionPoolDataSource`
- `XADataSource`

Additionally, the Connector Architecture defines a packaging format to allow a resource adapter to be deployed into a J2EE compliant application server.

19.1 System Contracts

The system contracts defined in the Connector specification describe the interface between an application server and one or more resource adapters. This interface allows a resource adapter to be bundled in such a way that it can be used by any application server that supports the system contracts.

The following standard contracts are defined between an application server and a back end system:

- A connection management contract that enables application components to connect to a back end system.

The connection management contract is equivalent to the services described by the JDBC interfaces `DataSource` and `ConnectionPoolDataSource`.

- A transaction management contract between the transaction manager and a back end system supporting transactional access to its resources.

The transaction contract is equivalent to the services described by the JDBC interface `XDataSource`.

- A security contract that enables secure access to a back end system.

The security contract does not have an equivalent in the JDBC API. Authentication in the JDBC API always consists of providing a user name and a password.

The JDBC specification does not make a distinction between its application programming interface (API) and the SPI. However, a driver can map the JDBC interfaces in the SPI to the Connector system contracts.

19.2 Mapping Connector System Contracts to JDBC Interfaces

Driver vendors who want to supply JDBC drivers that use the Connector system contracts have several options:

1. To write a set of classes that wrap a JDBC driver and implement the Connector system contracts. Constructing these wrappers is fairly straightforward and should allow JDBC driver vendors to provide resource adapters quickly enough so that they are available when application server vendors have implemented the Connector contracts.
2. To implement the Connector system contracts natively. This approach avoids the overhead of wrapper classes, but the implementation effort may be more involved and time-consuming. This alternative is a more long-term option.

Either approach will allow JDBC driver vendors to package their drivers as resource adapters and get all the benefits of pluggability, packaging, and deployment.

Note – There are no plans to deprecate or remove the current JDBC interfaces, `DataSource`, `ConnectionPoolDataSource` and `XDataSource`.

19.3 Packaging JDBC Drivers in Connector RAR File Format

Resource adapters can be packaged, along with a deployment descriptor, into a Resource adapter ARchive, or RAR file. The RAR file contains the Java classes / interfaces, native libraries, deployment descriptor, and other resources needed to deploy the adapter.

The deployment descriptor maps the classes in the resource adapter to the specific roles that they perform. The descriptor also details the capabilities of the resource adapter in terms of what level of transactional support it provides, the kind of security it supports, and so on.

CODE EXAMPLE 19-1 is an example of a deployment descriptor for a JDBC driver. The class `com.acme.JdbcManagedConnectionFactory` could be supported by an implementation of `javax.sql.XADataSource`. The resource adapter section contains information on the level of transaction support, the mechanism used for authentication, and configuration information for deploying the data source in the JNDI namespace.

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE connector PUBLIC "-//Sun Microsystems, Inc.//DTD Connector 1.0//EN"
'http://java.sun.com/j2ee/dtds/connector_1_0.dtd'>
<connector>
  <display-name>Acme JDBC Adapter</display-name>
  <vendor-name>Acme Software Inc</vendor-name>
  <spec-version>1.0</spec-version>
  <version>1.0</version>
  <eis-type>JDBC Database</eis-type>
  <resourceadapter>
    <managedconnectionfactory-class>com.acme.JdbcManagedConnectionFactory</
managedconnectionfactory-class>
    <connectionfactory-interface>javax.sql.DataSource<connectionfactory-
interface>
    <connectionfactory-impl-class>com.acme.JdbcDataSource<connectionfactory-impl-
class>
    <connection-interface>java.sql.Connection</connection-interface>
    <connection-impl-class>com.acme.JdbcConnection</connection-impl-class>
    <transaction-support>xa_transaction</transaction-support>
    <config-property>
      <config-property-name>XADataSourceName</config-property-name>
      <config-property-type>java.lang.String</config-property-type>
```

```
        <config-property-value>jdbc/XAAcme</config-property-value>
    </config-property>
    <auth-mechanism>
        <auth-mech-type>basic-password</auth-mech-type>
        <credential-interface>javax.resource.security.PasswordCredential</
credential-interface>
    </auth-mechanism>
    <reauthentication-support>false</reauthentication-support>
</resourceadapter>
</connector>
```

CODE EXAMPLE 19-1 Example of a resource adapter deployment descriptor.

See the Connector specification for more details.

Revision History

Appendix TABLE A-1 presents a summary of the revisions made to this specification.

TABLE A-1 Revision History

Revision	Dash	Date	Comments
Expert Draft	01	June 2000	First expert draft
Expert Draft	02	July 2000	Second expert draft. Addressed expert group comment and major editorial updates.
Community Draft		July 2000	First community draft. Minor editorial changes and clarifications from Expert Draft 02.

TABLE A-1 Revision History

Revision	Dash	Date	Comments
First Public Draft		Sept 2000	Continuing editorial changes and clarifications. • updated section 6.2 to clarify reference to SQL92. Made note that SQL99 behaviour is to be used for new features • updated section 8.1 to allow for SQL99 SQLStates to be returned. Added new DatabaseMetaData method to indicate if SQLStates being returned are X/Open or SQL99 • updated section 14.2.2 to align the ResultSet methods relative and absolute with SQL99 behaviour. • updated section 14.1.3 making the default holdability implementation-defined and adding new database metadata method getResultSetHoldability to report the default. • updated section 13.6 removing setGenerateKeys(boolean). Now passing flag into execute/executeUpdate/prepareStatement. • Added BOOLEAN data type. Logically equivalent to BIT. Updated data type table to reflect new type.

TABLE A-1 Revision History

Revision	Dash	Date	Comments
Proposed Final Draft	1	Oct 2000	Continued minor editorial changes and clarification. • added <code>getJDBCMajorVersion</code> and <code>getJDBCMinorVersion</code> to the <code>DatabaseMetaData</code> interface. • updated section 6.5 adding new <code>DatabaseMetaData</code> methods • updated section 6.7 adding missing constructor <code>java.sql.Time(int, int, int)</code> to the list of deprecated methods • further updated SQL syntax used in Chapters 16 & 17. • renamed the methods <code>getValue</code> and <code>setValue</code> in the <code>Ref</code> interface to <code>getObject</code> and <code>setObject</code> respectively. • added new example to section 13.6 to clarify process of retrieving named columns. • Added new section 9.3.1 on <code>SQLPermission</code>. • Added missing methods to register parameters and retrieve parameters by name.
Proposed Final Draft	2	Jan 2001	Minor editorial changes and clarification. • fixed errors in Chapter 5, some classes/interfaces were misclassified. • added new sections 9.2.1 and 9.2.2 on URL syntax and subprotocol registration. • added new wording 16.5.5 allowing user-defined references to be retrieved and set using the appropriate getter and setter methods.
Proposed Final Draft	3	April 2001	Removed method <code>PooledConnection.CloseAll()</code>. Changes to the <code>javax.sql</code> interfaces in this release break J2EE compatibility rules.

TABLE A-1 Revision History

Revision	Dash	Date	Comments
Proposed Final Draft	4	October 2001	• clarified how updates affect LOBs. Added new method <code>locatorsUpdateCopy</code> to <code>DatabaseMetaData</code> interface. • noted further limitations of XA and transactional properties • removed checked exception <code>MalformedURLException</code> from interface definition of <code>getURL</code> methods in the <code>ResultSet</code>, <code>CallableStatement</code>, <code>SQLInput</code> interfaces • updated <code>JavaDoc</code> to reflect the definition of the <code>ResultSet</code> method <code>relative</code>

Data Type Conversion Tables

The tables provided here describe the various mappings and conversions that drivers must support.

TABLE B-1 JDBC Types Mapped to Java Types

This table shows the conceptual correspondence between JDBC types and Java types. A programmer should write code with this mapping in mind. For example, if a value in the database is a `SMALLINT`, a `short` should be the data type used in a JDBC application.

All `CallableStatement` getter methods except for `getObject` use this mapping. The `getObject` methods for both the `CallableStatement` and `ResultSet` interfaces use the mapping in TABLE B-3.

TABLE B-2 Java Types Mapped to JDBC Types

This table shows the mapping a driver should use for the updater methods in the `ResultSet` interface and for `IN` parameters. `PreparedStatement` setter methods, `RowSet` setter methods and `SQLOutput` writer methods use this table for mapping an `IN` parameter, which is a Java type, to the JDBC type that will be sent to the database. Note that the `setObject` methods for `PreparedStatement` and `Rowset` use the mapping shown in TABLE B-4.

TABLE B-3 JDBC Types Mapped to Java Object Types

`ResultSet.getObject` and `CallableStatement.getObject` use the mapping shown in this table for standard mappings.

TABLE B-4 Java Object Types Mapped to JDBC Types

`PreparedStatement.setObject` and `RowSet.setObject` use the mapping shown in this table when no parameter specifying a target JDBC type is provided.

TABLE B-5 Conversions by `setObject` from Java Obejct Types to JDBC Types

This table shows which JDBC types may be specified as the target JDBC type to the methods `PreparedStatement.setObject` and `RowSet.setObject`.

In addition, for user-generated REF types and for DISTINCT types, the setter methods that are appropriate for the representation type of the REF type or for the source type of the DISTINCT type can be used.

TABLE B-6 Type Conversions Supported by ResultSet getter Methods

This table shows which JDBC types may be returned by `ResultSet` getter methods. A bold **X** indicates the method recommended for retrieving a JDBC type. A plain x indicates for which JDBC types it is possible to use a getter method.

This table also shows the conversions used by the `SQLInput` reader methods, except that they use only the recommended conversions. For user generated REF types, the reader methods that are appropriate for the representation type of the REF type may be used.

JDBC Type	Java Type
CHAR	String
VARCHAR	String
LONGVARCHAR	String
NUMERIC	java.math.BigDecimal
DECIMAL	java.math.BigDecimal
BIT	boolean
BOOLEAN	boolean
TINYINT	byte
SMALLINT	short
INTEGER	int
BIGINT	long
REAL	float
FLOAT	double
DOUBLE	double
BINARY	byte[]
VARBINARY	byte[]
LONGVARBINARY	byte[]
DATE	java.sql.Date
TIME	java.sql.Time
TIMESTAMP	java.sql.Timestamp
CLOB	Clob
BLOB	Blob
ARRAY	Array
DISTINCT	mapping of underlying type
STRUCT	Struct
REF	Ref
DATALINK	java.net.URL
JAVA_OBJECT	underlying Java class

TABLE B-1 JDBC Types Mapped to Java Types

Java Type	JDBC Type
String	CHAR, VARCHAR, or LONGVARCHAR
java.math.BigDecimal	NUMERIC
boolean	BIT or BOOLEAN
byte	TINYINT
short	SMALLINT
int	INTEGER
long	BIGINT
float	REAL
double	DOUBLE
byte[]	BINARY, VARBINARY, or LONGVARBINARY
java.sql.Date	DATE
java.sql.Time	TIME
java.sql.Timestamp	TIMESTAMP
Clob	CLOB
Blob	BLOB
Array	ARRAY
Struct	STRUCT
Ref	REF
java.net.URL	DATALINK
Java class	JAVA_OBJECT

TABLE B-2 Standard Mapping from Java Types to JDBC Types

JDBC Type	Java Object Type
CHAR	String
VARCHAR	String
LONGVARCHAR	String
NUMERIC	java.math.BigDecimal
DECIMAL	java.math.BigDecimal
BIT	Boolean
BOOLEAN	Boolean
TINYINT	Integer
SMALLINT	Integer
INTEGER	Integer
BIGINT	Long
REAL	Float
FLOAT	Double
DOUBLE	Double
BINARY	byte[]
VARBINARY	byte[]
LONGVARBINARY	byte[]
DATE	java.sql.Date
TIME	java.sql.Time
TIMESTAMP	java.sql.Timestamp
DISTINCT	Object type of underlying type
CLOB	Clob
BLOB	Blob
ARRAY	Array
STRUCT	Struct or SQLData
REF	Ref
DATALINK	java.net.URL
JAVA_OBJECT	underlying Java class

TABLE B-3 Mapping from JDBC Types to Java Object Types

Java Object Type	JDBC Type
String	CHAR, VARCHAR, or LONGVARCHAR
java.math.BigDecimal	NUMERIC
Boolean	BIT or BOOLEAN
Integer	INTEGER
Long	BIGINT
Float	REAL
Double	DOUBLE
byte[]	BINARY, VARBINARY, or LONGVARBINARY
java.sql.Date	DATE
java.sql.Time	TIME
java.sql.Timestamp	TIMESTAMP
Clob	CLOB
Blob	BLOB
Array	ARRAY
Struct	STRUCT
Ref	REF
java.net.URL	DATALINK
Java class	JAVA_OBJECT

TABLE B-4 Mapping from Java Object Types to JDBC Types

	TINYINT	SMALLINT	INTEGER	BIGINT	REAL	FLOAT	DOUBLE	DECIMAL	NUMERIC	BIT	BOOLEAN	CHAR	VARCHAR	LONGVARCHAR	BINARY	VARBINARY	LONGVARBINARY	DATE	TIME	TIMESTAMP	ARRAY	BLOB	CLOB	STRUCT	REF	DATALINK	JAVA_OBJECT
String	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x							
java.math.BigDecimal	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
Boolean	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
Integer	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
Long	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
Float	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
Double	x	x	x	x	x	x	x	x	x	x	x	x	x	x													
byte[]															x	x	x										
java.sql.Date												x	x	x				x		x							
java.sql.Time												x	x	x					x								
java.sql.Timestamp												x	x	x				x	x	x							
Array																					x						
Blob																						x					
Clob																							x				
Struct																								x			
Ref																									x		
java.net.URL																										x	
Java class																											x

TABLE B-5 Conversions Performed by `setObject` Between Java Object Types and Target JDBC Types

	TINYINT	SMALLINT	INTEGER	BIGINT	REAL	FLOAT	DOUBLE	DECIMAL	NUMERIC	BIT	BOOLEAN	CHAR	VARCHAR	LONGVARCHAR	BINARY	VARBINARY	LONGVARBINARY	DATE	TIME	TIMESTAMP	CLOB	BLOB	ARRAY	REF	DATALINK	STRUCT	JAVA_OBJECT
getBytes															X	X	x										
getBoolean	x	x	x	x	x	x	x	x	x	X	X	x	x	x													
getByte	X	x	x	x	x	x	x	x	x	x	x	x	x	x													
getCharacterStream												x	x	X	x	x	X										
getDouble	x	x	x	x	x	X	X	x	x	x	x	x	x	x													
getFloat	x	x	x	x	X	x	x	x	x	x	x	x	x	x													
getLong	x	x	x	X	x	x	x	x	x	x	x	x	x	x													
getNumeric																											
getShort	x	X	x	x	x	x	x	x	x	x	x	x	x	x													
getString	x	x	x	x	x	x	x	x	x	x	x	X	X	x	x	x	x	x	x	x					x		
getTime												x	x	x				X	x								
getTimestamp												x	x	x				x	x	X							
getURL																									X		
getArray																							X				
getBlob																						X					
getClob																					X						
getObject	x	x	x	x	x	x	x	x	x	x		x	x	x	x	x	x	x	x	x	x	x	x	x	x	X	X

TABLE B-6 Use of ResultSet getter Methods to Retrieve JDBC Data Types

Scalar Functions

The JDBC API supports escape syntax for numeric, string, time, date, system, and conversion functions on scalar values. These scalar functions may be used in SQL strings as described in Section 13.4.1 “Scalar Functions” on page 13-109. The Open Group CLI specification provides more information on the semantics of the scalar functions. The scalar functions are listed below for reference.

If a DBMS supports a scalar function, the driver should also. Because scalar functions are supported by different DBMSs with slightly different syntax, it is the driver’s job either to map them into the appropriate syntax or to implement the functions directly in the driver.

A user should be able to find out which functions are supported by calling metadata methods. For example, the method `DatabaseMetaData.getNumericFunctions` returns a comma separated list of the Open Group CLI names of the numeric functions supported. Similarly, the method `DatabaseMetaData.getStringFunctions` returns a list of string functions supported, and so on.

The scalar functions are listed by category:

C.1 Numeric Functions

<u>Function Name</u>	<u>Function Returns</u>
<code>ABS(number)</code>	Absolute value of number
<code>ACOS(float)</code>	Arccosine, in radians, of float
<code>ASIN(float)</code>	Arcsine, in radians, of float
<code>ATAN(float)</code>	Arctangent, in radians, of float
<code>ATAN2(float1, float2)</code>	Arctangent, in radians, of float2 / float1
<code>CEILING(number)</code>	Smallest integer >= number
<code>COS(float)</code>	Cosine of float radians
<code>COT(float)</code>	Cotangent of float radians
<code>DEGREES(number)</code>	Degrees in number radians

EXP(float)	Exponential function of float
FLOOR(number)	Largest integer <= number
LOG(float)	Base e logarithm of float
LOG10(float)	Base 10 logarithm of float
MOD(integer1, integer2)	Remainder for integer1 / integer2
PI()	The constant pi
POWER(number, power)	number raised to (integer) power
RADIANS(number)	Radians in number degrees
RAND(integer)	Random floating point for seed integer
ROUND(number, places)	number rounded to places places
SIGN(number)	-1 to indicate number is < 0; 0 to indicate number is = 0; 1 to indicate number is > 0
SIN(float)	Sine of float radians
SQRT(float)	Square root of float
TAN(float)	Tangent of float radians
TRUNCATE(number, places)	number truncated to places places

C.2 String Functions

Function Name	Function Returns
ASCII(string)	Integer representing the ASCII code value of the leftmost character in string
CHAR(code)	Character with ASCII code value code, where code is between 0 and 255
CONCAT(string1, string2)	Character string formed by appending string2 to string1; if a string is null, the result is DBMS-dependent
DIFFERENCE(string1, string2)	Integer indicating the difference between the values returned by the function SOUNDEX for string1 and string2
INSERT(string1, start, length, string2)	A character string formed by deleting length characters from string1 beginning at start, and inserting string2 into string1 at start
LCASE(string)	Converts all uppercase characters in string to lowercase
LEFT(string, count)	The count leftmost characters from string
LENGTH(string)	Number of characters in string, excluding trailing blanks
LOCATE(string1, string2[, start])	Position in string2 of the first occurrence of string1, searching from the beginning of string2; if start is specified, the search begins from position start. 0 is returned if string2 does not contain string1. Position 1 is the first character in string2.
LTRIM(string)	Characters of string with leading blank spaces removed
REPEAT(string, count)	A character string formed by repeating string count times
REPLACE(string1, string2, string3)	Replaces all occurrences of string2 in string1 with string3
RIGHT(string, count)	The count rightmost characters in string
RTRIM(string)	The characters of string with no trailing blanks
SOUNDEX(string)	A character string, which is data source-dependent, representing the sound of the words in string; this could be a four-digit SOUNDEX code, a phonetic representation of each word, etc.
SPACE(count)	A character string consisting of count spaces

SUBSTRING(string, start, length)	A character string formed by extracting length characters from string beginning at start
UCASE(string)	Converts all lowercase characters in string to uppercase

C.3 Time and Date Functions

<u>Function Name</u>	<u>Function Returns</u>
CURDATE()	The current date as a date value
CURTIME()	The current local time as a time value
DAYNAME(date)	A character string representing the day component of date; the name for the day is specific to the data source
DAYOFMONTH(date)	An integer from 1 to 31 representing the day of the month in date
DAYOFWEEK(date)	An integer from 1 to 7 representing the day of the week in date; 1 represents Sunday
DAYOFYEAR(date)	An integer from 1 to 366 representing the day of the year in date
HOUR(time)	An integer from 0 to 23 representing the hour component of time
MINUTE(time)	An integer from 0 to 59 representing the minute component of time
MONTH(date)	An integer from 1 to 12 representing the month component of date
MONTHNAME(date)	A character string representing the month component of date; the name for the month is specific to the data source
NOW()	A timestamp value representing the current date and time
QUARTER(date)	An integer from 1 to 4 representing the quarter in date; 1 represents January 1 through March 31
SECOND(time)	An integer from 0 to 59 representing the second component of time
TIMESTAMPADD(interval, count, timestamp)	A timestamp calculated by adding COUNT number of interval(s) to timestamp; interval may be one of the following: SQL_TSI_FRAC_SECOND, SQL_TSI_SECOND, SQL_TSI_MINUTE, SQL_TSI_HOUR, SQL_TSI_DAY, SQL_TSI_WEEK, SQL_TSI_MONTH, SQL_TSI_QUARTER, or SQL_TSI_YEAR
TIMESTAMPDIFF(interval, timestamp1, timestamp2)	An integer representing the number of interval by which timestamp2 is greater than timestamp1; interval may be one of the following: SQL_TSI_FRAC_SECOND, SQL_TSI_SECOND, SQL_TSI_MINUTE, SQL_TSI_HOUR, SQL_TSI_DAY, SQL_TSI_WEEK, SQL_TSI_MONTH, SQL_TSI_QUARTER, or SQL_TSI_YEAR
WEEK(date)	An integer from 1 to 53 representing the week of the year in date
YEAR(date)	An integer representing the year component of date

C.4 System Functions

<u>Function Name</u>	<u>Function Returns</u>
DATABASE()	Name of the database
IFNULL(expression, value)	value if expression is null; expression if expression is not null

USER()

User name in the DBMS

C.5 Conversion Functions

<u>Function Name</u>	<u>Function Returns</u>
CONVERT(value, SQLtype)	value converted to SQLtype where SQLtype may be one of the following SQL types: BIGINT, BINARY, BIT, CHAR, DATE, DECIMAL, DOUBLE, FLOAT, INTEGER, LONGVARBINARY, LONGVARCHAR, REAL, SMALLINT, TIME, TIMESTAMP, TINYINT, VARBINARY, or VARCHAR

Related Documents

This specification makes reference to the following documents.

Data Management: SQL Call Level Interface (X/Open SQL CLI) Available at <http://www.opengroup.org>.

Distributed Transaction Processes: The XA Specification (X/Open CAE) Available at <http://www.opengroup.org>.

JDBC 2.1 API (JDBC 2.1). Copyright 1998, Sun Microsystems, Inc. Available at <http://java.sun.com/products/jdbc>.

JDBC 2.0 Standard Extension API (JDBC extension specification). Copyright 1998, 1999, Sun Microsystems, Inc. Available at <http://java.sun.com/products/jdbc>.

JDBC 1.22 API (JDBC 1.22). Copyright 1998, Sun Microsystems, Inc. Available at <http://java.sun.com/products/jdbc>.

JavaBeans 1.01 Specification (JavaBeans specification). Copyright 1996, 1997, Sun Microsystems, Inc. Available at <http://java.sun.com/beans>.

Java Transaction API, Version 1.0.1 (JTA Specification). Copyright 1998, 1999, Sun Microsystems, Inc. Available at <http://java.sun.com/products/jta>.

Java Naming and Directory Interface 1.2 Specification (JNDI specification). Copyright 1998, 1999, Sun Microsystems, Inc. Available at <http://java.sun.com/products/jndi>.

Enterprise Java Beans, Version 1.1 (EJB). Copyright 1998, 1999, Sun Microsystems, Inc. Available at <http://java.sun.com/products/ejb>.

J2EE Connector Architecture (JCX1.0) Copyright 1999, 2000, Sun Microsystems, Inc. Available at <http://java.sun.com/j2ee>.

The following documents are collectively referred to as SQL99:

ISO/IEC 9075-1:1999, Information technology - Database languages - SQL - Part 1: Framework (SQL / Framework).

ISO/IEC 9075-2:1999, Information technology - Database languages - SQL - Part 2: Foundation (SQL / Foundation).

ISO/IEC 9075-3:1999, Information technology - Database languages - SQL - Part 3: Call-Level Interface (SQL / CLI).

ISO/IEC 9075-4:1999, Information technology - Database languages - SQL - Part 4: Persistent Stored Modules (SQL / PSM).

ISO/IEC 9075-5:1999, Information technology - Database languages - SQL - Part 5: Host Language Bindings (SQL / Bindings).

The following document is a reference for SQL MED:

ISO/IEC 9075-9:2000 Information technology - Database languages - SQL - Part 9: Management of External Data (SQL / MED)

The following document is a reference for SQLJ:

ISO/IEC 9075-10:2000, Information technology - Database Languages SQL - Part 10: Object Language Bindings (SQL / OLB)



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